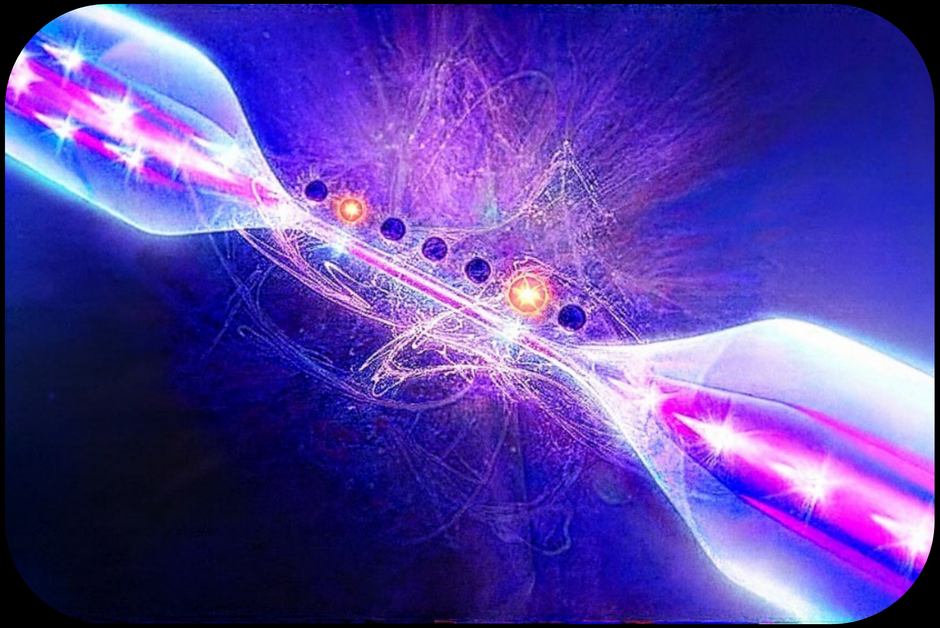
Quantum Interfaces

Transferring quantum information between light and matter



Course materials and homework:
www.iamsquantum.github.io/TIGP.html

Shayne Bennetts

Quantum Technology Based on Atomic, Molecular, and Optical Systems

25 November 2025

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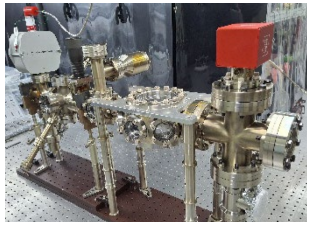
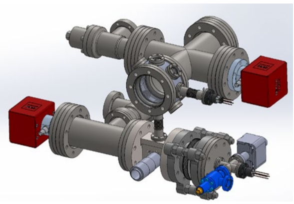
IAMS Ytterbium Lab

Platforms



"Pathfinder"

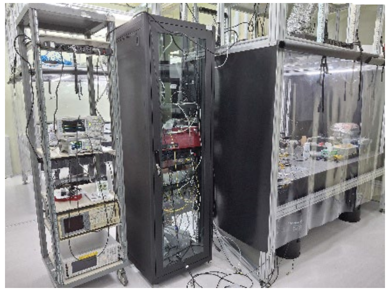


"Nanofiber"

Join us

Projects available
Email us, visit us

Laser Systems



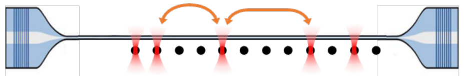
Lab Infrastructure

Research

Hybrid Atom-Nanophotonic Quantum Technology

Shayne Bennetts
s.p.bennetts@g.iams.edu.tw

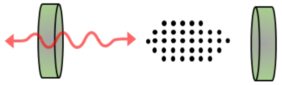
Hybrid atom-nanophotonics



- Any-to-any interaction and gates
- Individual tweezer atom control
- Arbitrary defect free arrays
- Telecom wavelength operation
- Scalability

Tapered nanofiber cavities

Cavity photon – atom interfaces

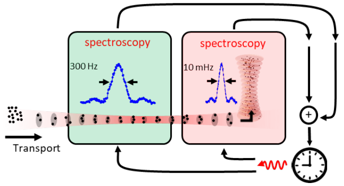


Collective atom-cavity light-matter interfaces

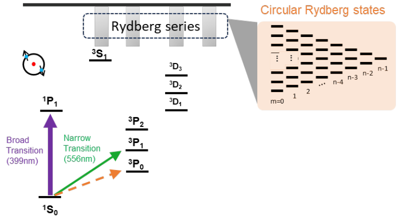
Quantum Metrology Laboratory

Chun-Chia Chen (陳俊嘉)
chunchia.chen@g.iams.sinica.edu.tw

Continuous Passive optical clocks



Ytterbium Rydberg spectroscopy



Funding

NSTC 國家科學及技術委員會 National Science and Technology Council
iMATE
IAMS

Part I: Quantum Interfaces Introduction

This lecture is about how to transfer quantum information between stationary and flying qubits.

Matter qubits ⟷ Flying qubits (photons)

This is a key tool for:

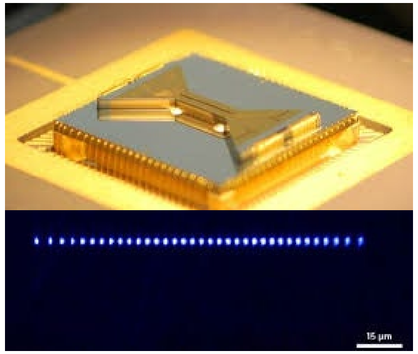
1. Long-distance quantum communication
2. Entanglement distribution
3. Quantum repeaters
4. Modular architectures
   (multiple quantum processors linked optically)

The challenge: Linking different quantum platforms

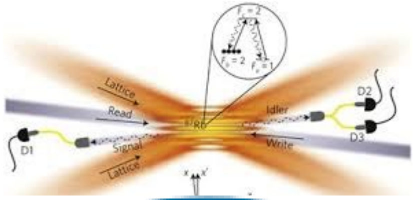
Optical interface

Microwave interface

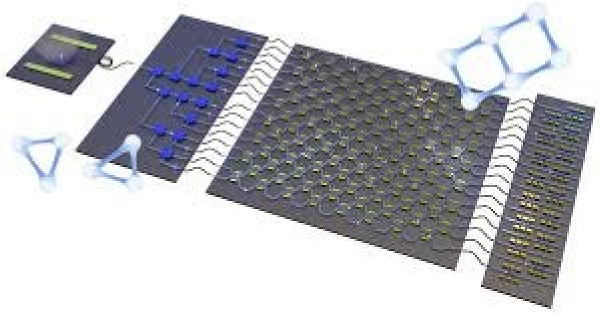
Ions

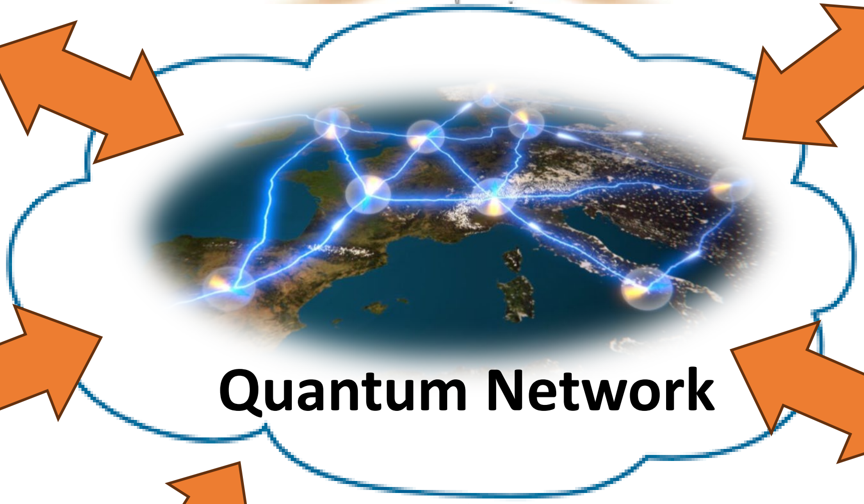


Neutral atom clouds



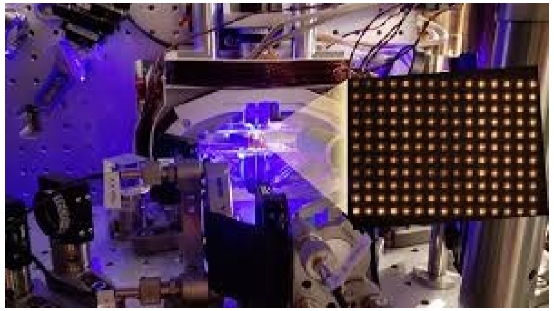
Photonics



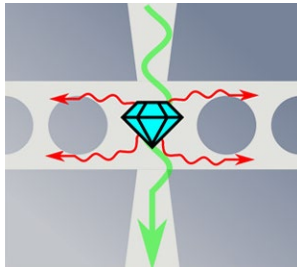


Quantum Network

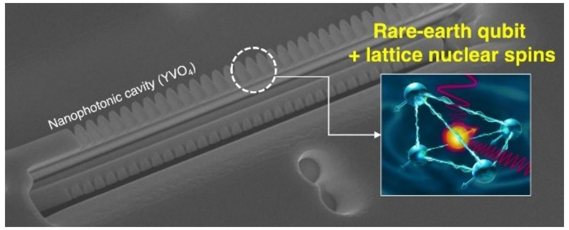
Neutral atoms



NV centers



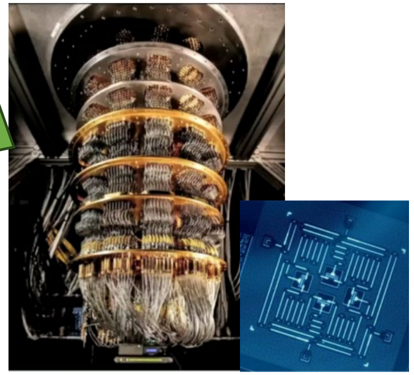
RE ions



Rydberg Atoms

Mechanical

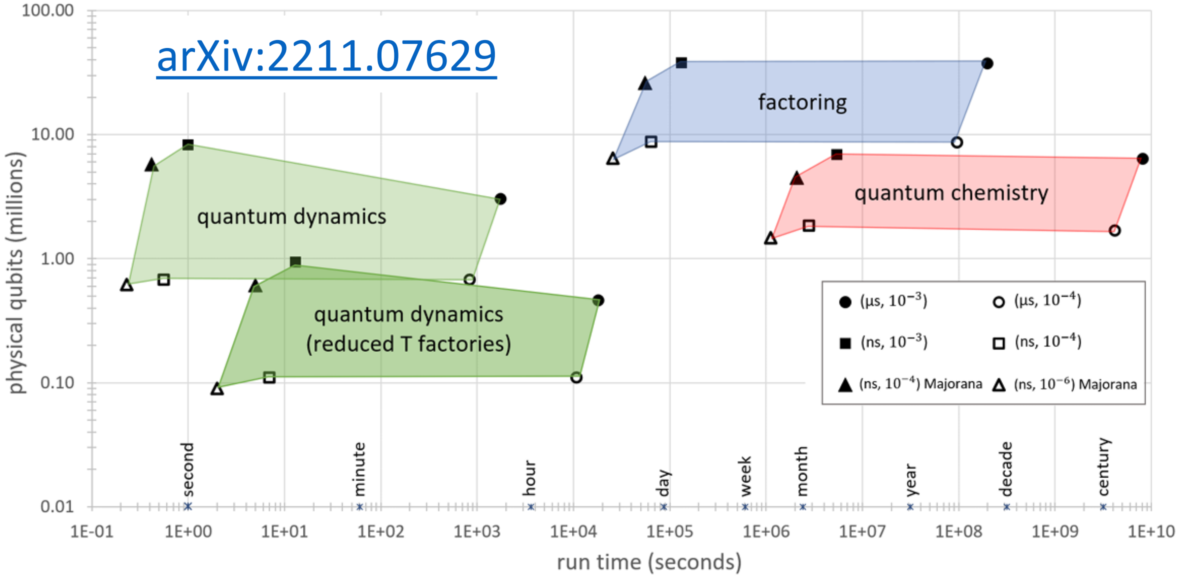
Superconducting



Today there is no "winner" for quantum computing

	Superconducting	Photonic	Neutral atom (cavity)	Neutral atom (Rydberg)	Ions
Speed	~ 50 ns	~100 ps	~1 ns	~1 us/100us	~ 100us
Qbits	1e3	Emerging	Emerging	1e3/1e4	1e1/1e2
Memory		0			
Errors	1e-3	Emerging	Concept	1e-2/1e-3	1e-4
Network	Concept	Native	Advanced	Emerging	Advanced

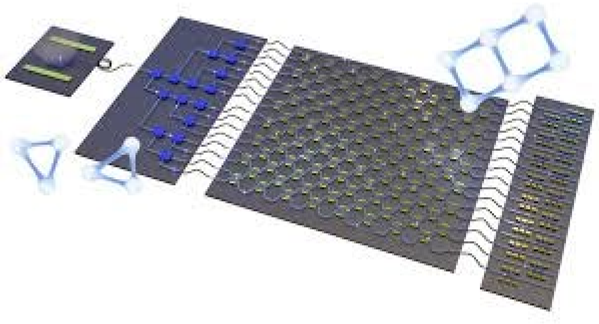
Quantum resource requirements:



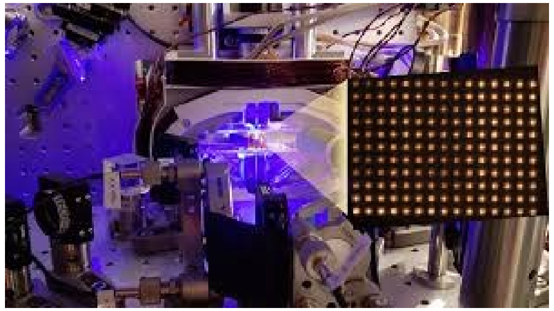
There is no winner: Quantum interfaces allow combining the advantages of different technologies.

Potential benefits from combining quantum platforms

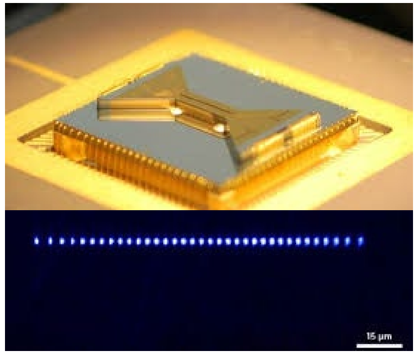
Photonics



Neutral atoms



Ions



Memory: Atoms and Ions can provide long coherence, quantum memories with robust, deterministic error correction.

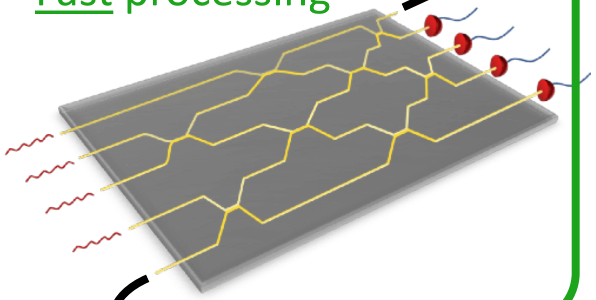
Deep circuits: Atoms/ions can enable deep, deterministic, high fidelity complex circuits with error correction.

Scaling/Distributed Quantum: Photonics allows linking of multiple processors enabling distributed quantum systems and scaling.

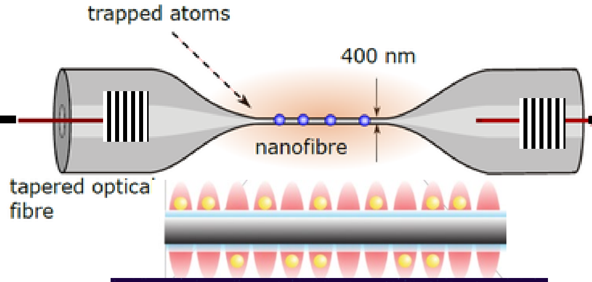
Fast: Photonic quantum coprocessors can execute specific circuits/algorithms with high speed.

The challenge: Interfacing

Photonic QPU
- Fast processing

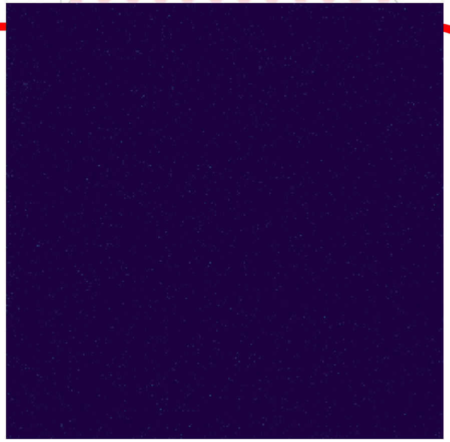


Optical nanofiber cavity
- Interfacing light and matter
- 10s-100s of qubits Q-RAM



Distributed computing and quantum resource distribution

Neutral atom QPU + memory
- 10,000-100,000 qubits/QPU
- Coherent, high fidelity, persistent
- Logical qubits
- High fidelity (slow/deep) processing

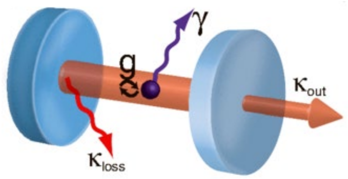


Future quantum systems will combine strengths from multiple technology platforms

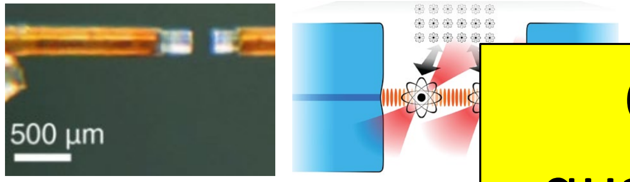
A “zoo” of photonic-atom interfaces

Bulk cavities

Free space cavities



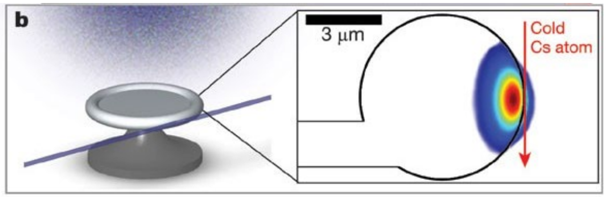
Rempe: Rev. Mod. Phys. 87, 1379 (2015).



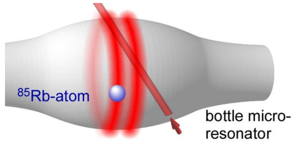
Blatt/Northup: Rev. Sci. Instr. 84, 12310

Lukin/Vuletic groups, Science **387**, 6740, 1

Evanescently coupled cavities



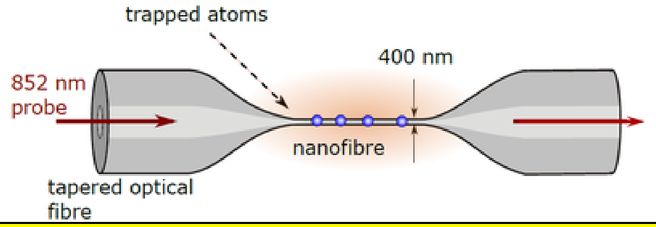
Kimble, Aoki, Nature **443**, 671–674 (2006)



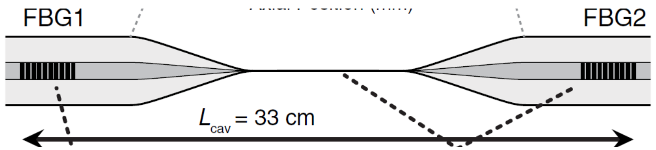
Rauschenbeutel group:
PRL **104**, 203603 (2010).

Nanofibers

Atoms evanescently coupled to optical nanofibers



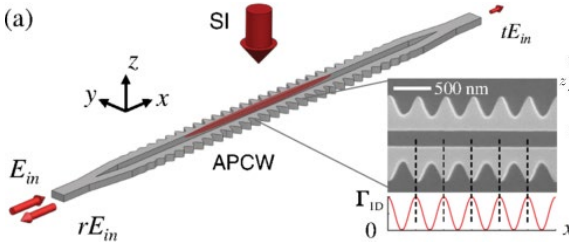
Atoms evanescently coupled to nanofiber cavities

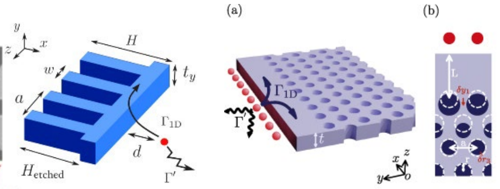


Aoki group/nanoQT: PRL **115**, 093603 (2015)
Optics Lett. **50**, 17 5294 (2025)

Chips

Atoms evanescently coupled to chip waveguides

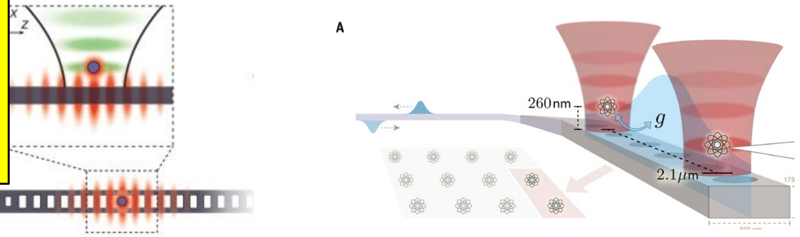




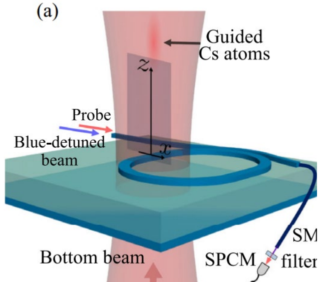
Kimble group: Phys. Rev. Lett. **115**, 063601 (2015)
PNAS 117 (47) 29422 (2020).

Laurat group:
Optics Express, 30, 25, 45093 (2022)
NJP 117 (47) 29422 (2024).

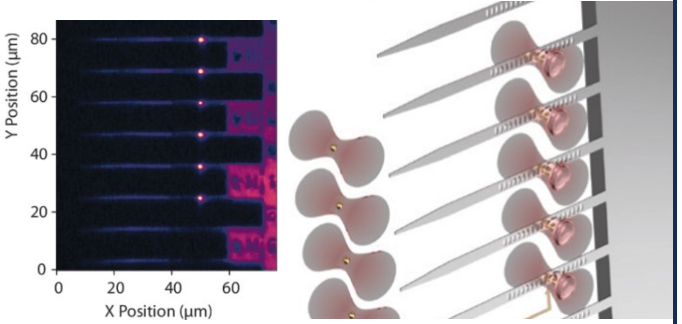
…escently coupled to chip based cavities



Lukin/Vuletic groups, Nature **508**, 241 (2014), Science **373**, 6562, 1511 (2021).



Hung group: PRL **130**, 103601 (2023)



Bernien group: Nature Comms **15**, 6156 (2024)

Course objective: Understand quantum interfacing requirements, solutions and challenges.

Part II: Foundations and Theory

1. Atom–Light Interaction

   (recap of standard equations)

2. 1D/Waveguide atom-photon interactions

3. Cavity QED

Atom – Light Interaction

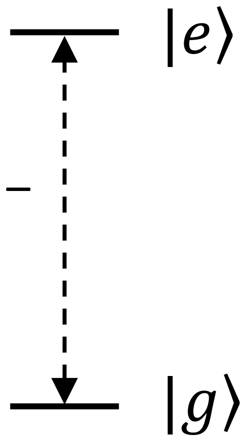
(recap)

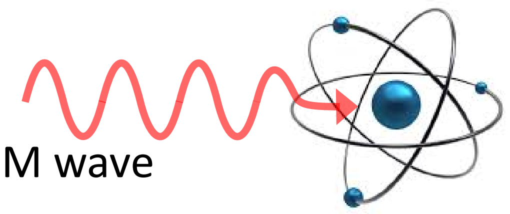
References:
MIT Wolfgang Ketterle's course
Eg Lecture 24-25: Coherence IV-V
https://youtu.be/Y7UsD2SNIIw?list=PLUl4u3cNGP62FPGcyFJkzhqq9c5cHCK32

Quantum Optics (LMU Immanual Bloch's course)
https://youtu.be/pCa1uAvjRN4?list=PLNMgVqt8MREzCQbgPxBiHKlFbV7Xszkds
Atomic Physics, Foot
Quantum Optics, Scully and Zubary
https://hal.science/jpa-00247657v1/document

Atom-light interaction

Consider an atom with a ground state |g⟩ and an excited state |e⟩:





Dipole moment d_{eg}

$$\mathbf{E}(t) = \mathcal{E}_0 \cos(\omega_L t - kz)$$

When light with frequency $\omega_L = (\omega_0 + \Delta)$, Δ the detuning, illuminates the atom, the light interacts with the atomic dipole d:

$$H_{\text{int}} = -\mathbf{d} \cdot \mathbf{E}$$

$$H_{\text{int}} = -\frac{\hbar\Omega_R}{2}\left(e^{-i\omega_L t}|e\rangle\langle g| + h.c.\right)$$ (in the RWA)

The Rabi frequency, Ω_R quantifies strength of the coupling to the field

$$\Omega_R = \frac{\mathbf{d}_{eg} \cdot \mathcal{E}_\mathbf{0}}{\hbar}$$

See Chapter 5 of Quantum Optics by Scully and Zubairy

Optical Bloch Equations

For a 2 level atom the wavefunction is given by:

$$|\Psi(t)\rangle = c_a(t)|g\rangle + c_b(t)|e\rangle$$

The solutions for c_a and c_b are then given by:

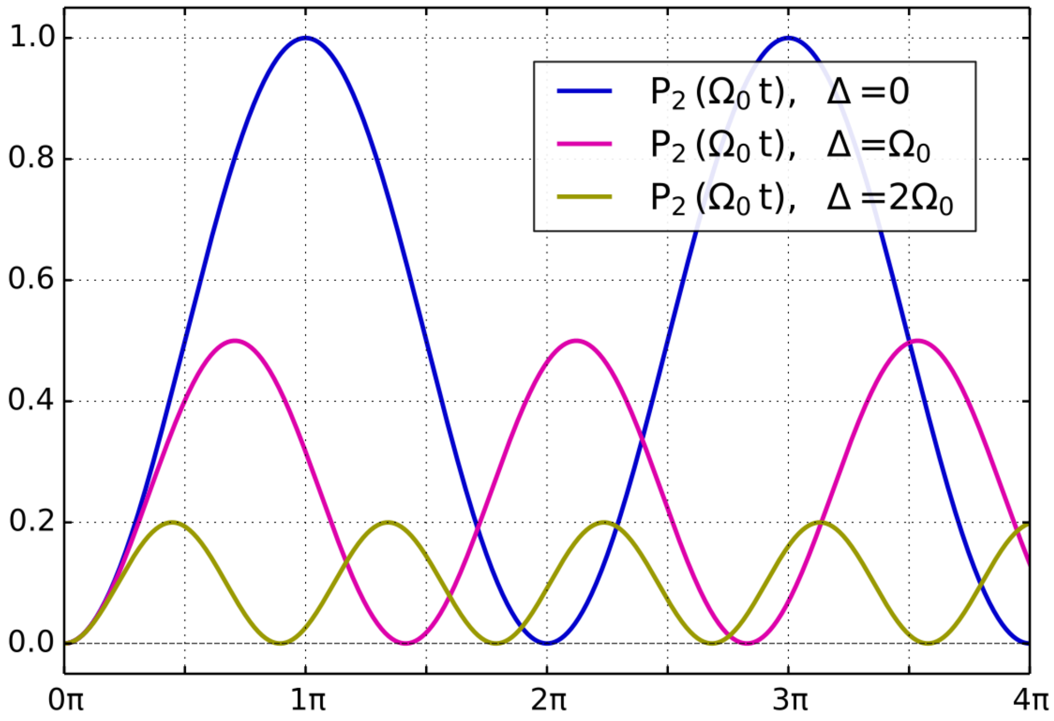
$$c_a(t) = \left(a_1 e^{i\Omega t/2} + a_2 e^{-i\Omega t/2}\right)e^{i\Delta t/2}$$

$$c_b(t) = \left(b_1 e^{i\Omega t/2} + b_2 e^{-i\Omega t/2}\right)e^{-i\Delta t/2}$$

$$\Omega = \sqrt{\Omega_R^2 + \Delta^2}$$

The Optical Bloch Equations

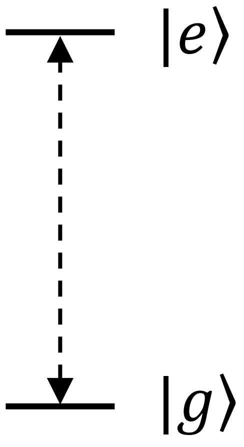
(constants a, b defined by the initial conditions)



*wikipedia

See Chapter 5 of Quantum Optics by Scully and Zubairy

Dissipation: Spontaneous emission



Due to vacuum fluctuations an atom in an excited state |e> can spontaneously decay to the ground state |g> with a natural linewidth/population decay rate Γ_0, an excited state lifetime τ and a dipole coherence decay rate γ.

$$\Gamma_0 = \frac{1}{\tau} = 2\gamma = A_{eg} = \frac{\omega_0^3 \left|\mathbf{d}_{eg}\right|^2}{3\pi\epsilon_0 \hbar c^3}$$

Atom – Photon Interaction in free space

Resonant scattering cross section for a dipole in a beam of light is given by:

$$\sigma_0 = \frac{3\lambda_0^2}{2\pi}$$

The area of the "shadow" caused by a dipole in a beam of light

If our light is detuned by Δ and the transition rate is γ the scattering cross section becomes:

$$\sigma(\Delta) = \frac{\sigma_0}{1 + 4\Delta^2/\gamma^2}$$

Many atom scattering (Beer's law)

Assume

- Optical beam of area $A=(\pi/4)W_0^2$, (W_0 the gaussian waist radius)
- Number of scatterers per unit volume n
- Probability of a single scatterer scattering a photon σ/A

If we have many dipoles we will attenuate light according to Beer's law

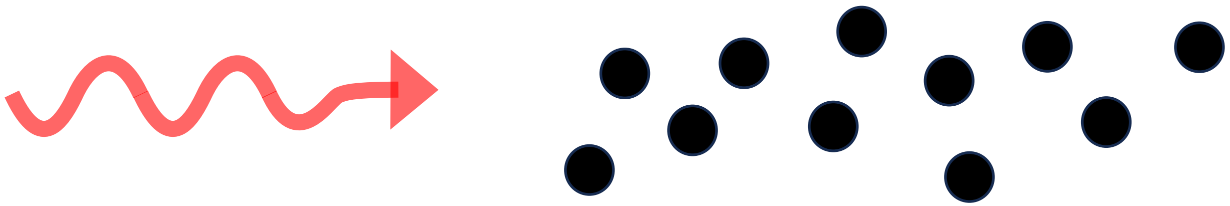
$$dI(z) = -\frac{\sigma}{A}\, n\, A\, I(z)dz$$

$$I(z) = I(0)\exp(-\alpha\, z) \qquad \alpha = \sigma n$$

α – optical depth
$I(z)$ - light intensity vs position
n - density of absorbers
σ - cross section
$A = (\pi/4)W_0^2$ – mode area where W_0 is the gaussian waist radius

Fidelity limit for storing a single photon?

Single photon carrying quantum information



$$\text{Fidelity} \leq 1 - \frac{5.8}{OD}$$

A. V. Gorshkov *et al.*, PRL **98**, 123601 (2007)

OD = Optical Density

Example: Quantum memory fidelity limits

Rb 2DMOT based quantum memory

- Number density n~1e10-1e11 at/cm^3
- Cloud waist radius 1mm, 1cm long

Gives an atom number in cloud ~1e9 atoms

Atoms in the optical mode:

- Waist~200um, L~5mm
- Gives an atom number in optical mode ~3e7 atoms

Optical depth:

- Resonant cross section $\sigma=3\lambda^2/(2\pi)$=3e-9 cm^2
- OD=nσL~70 (used 3e7 atoms)
- Fidelity <= 1-5.8/OD=91.7%

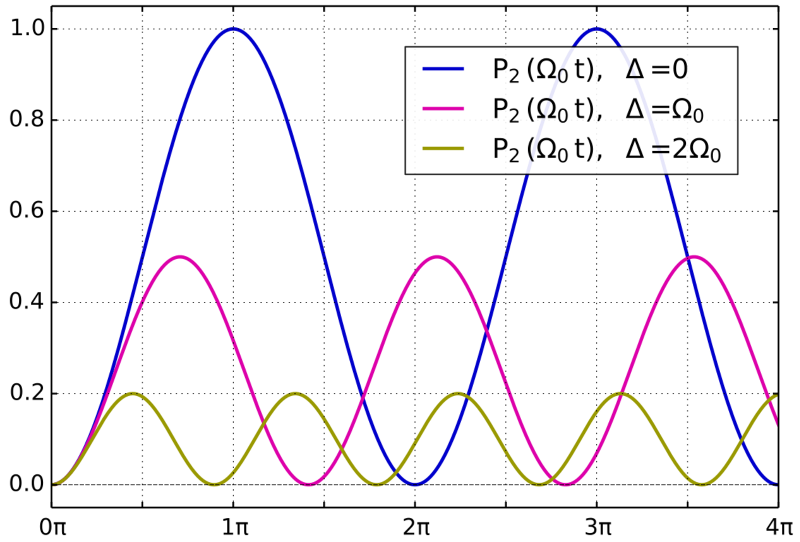
	2D MOT quantum memory
N	$\boxed{\text{3e7}}$
OD	70
β	1e-6 (effective)
Fidelity	91.7%

Atom – Photon Interaction regimes

Coherent interactions (strong)

Closed system, information is not lost to the environment

$$\Omega_R >> \gamma$$



Absorption/scattering

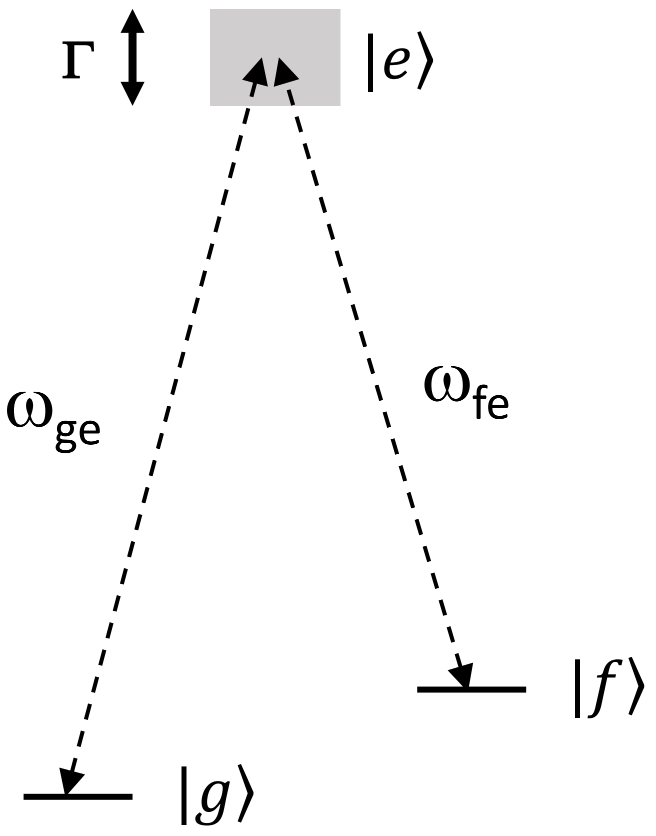
Open system, information rapidly lost to the environment

$$\Omega_R << \gamma$$

3 level atom, dark states

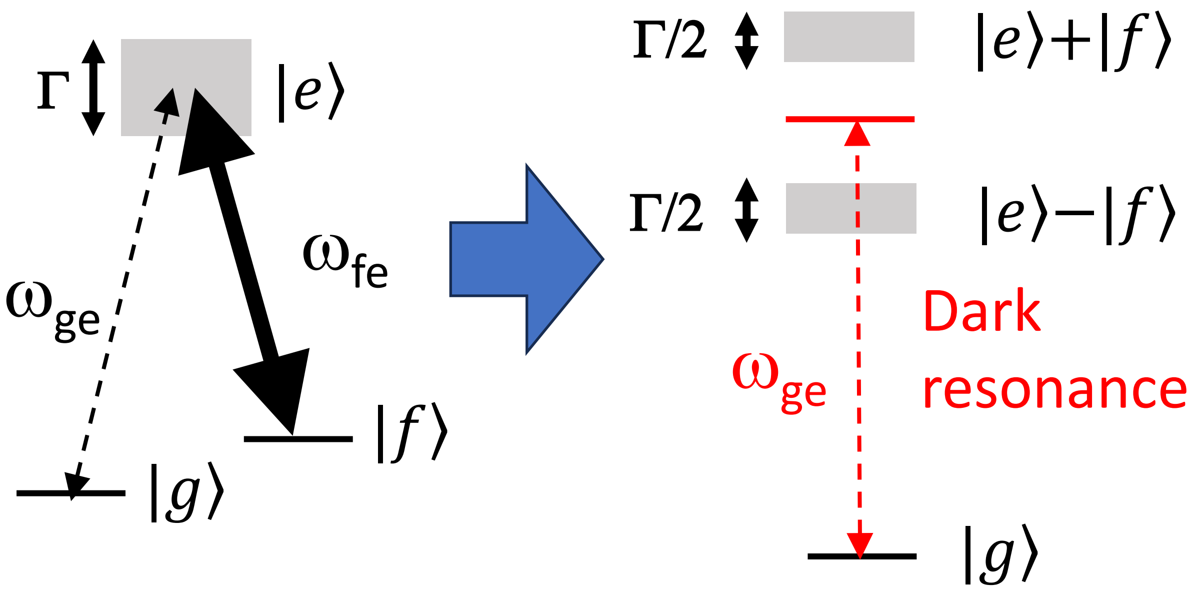
In a 3 level (Λ) system interference between the 2 coherent couplings ω_{ge} and ω_{fe} means there always exists a **dark state**.

The dark state is a coherent superposition of the 3 levels with amplitudes that depend on the strength of the couplings applied.



Electromagnetically induced transparency (EIT)

A **strong coupling** laser with Rabi frequency Ω_C will **mix** the coupled states (coupling rate must be stronger than the spontaneous emission/decoherence rate of the excited state).



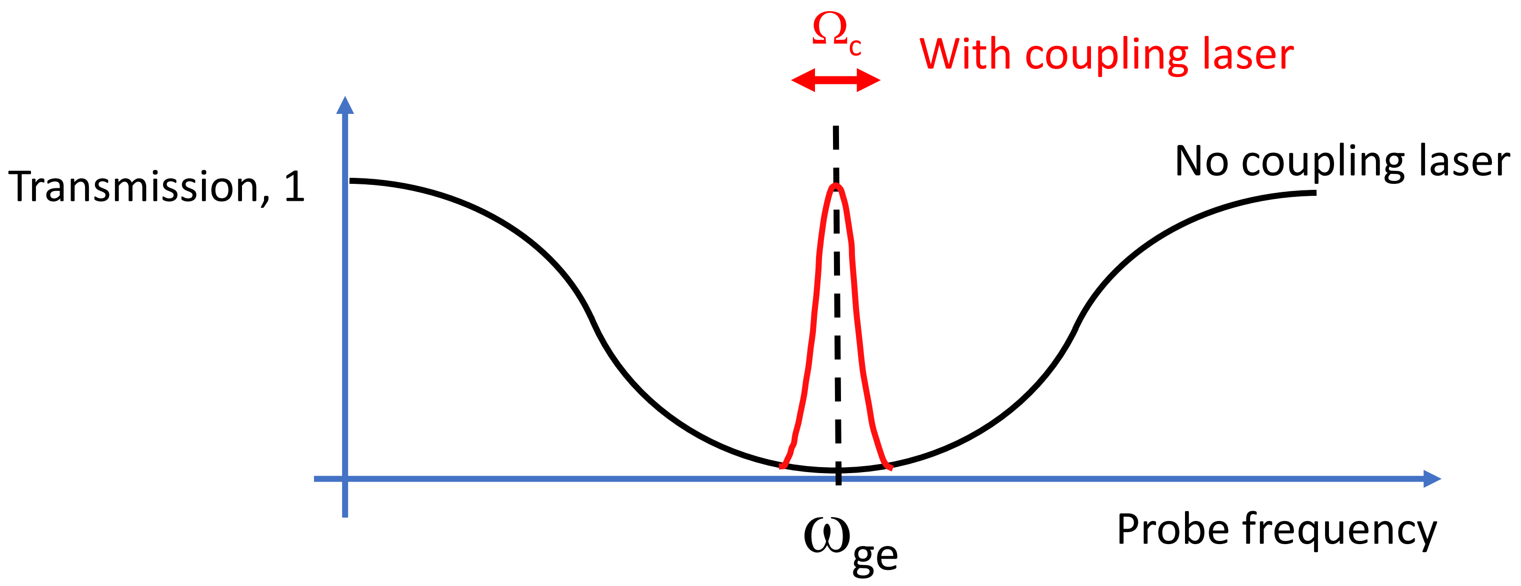
$$|D\rangle = \cos\theta\,|g\rangle - \sin\theta\,|f\rangle$$

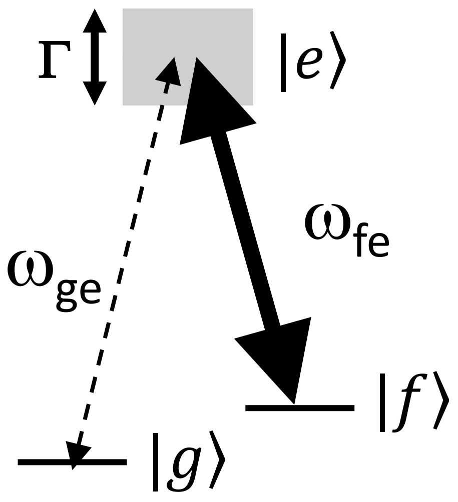
$$\tan\theta = \frac{g\sqrt{N}}{\Omega_c}$$

θ is the mixing angle

Electromagnetically induced transparency (EIT)

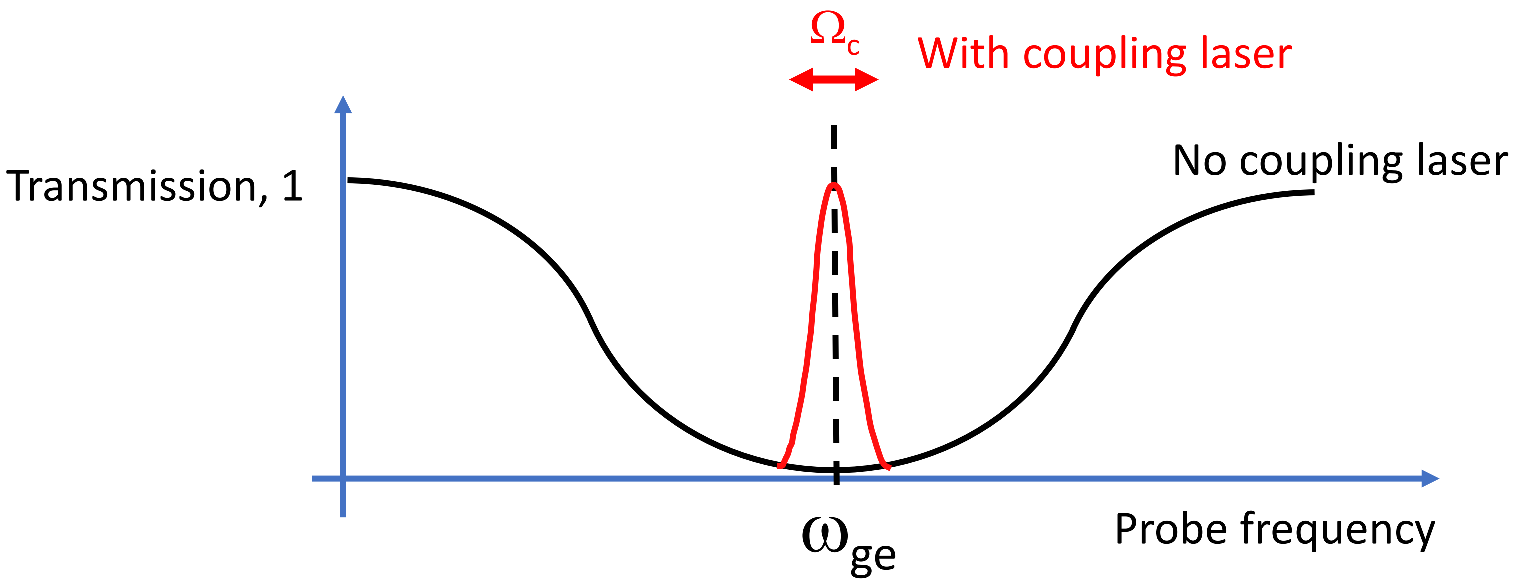
If a probe is applied at a frequency exactly in between the split state interference results in a **dark resonance** with no absorption **(electromagnetically induced transparency).**
The width of the transparency feature is the Rabi frequency of the coupling laser

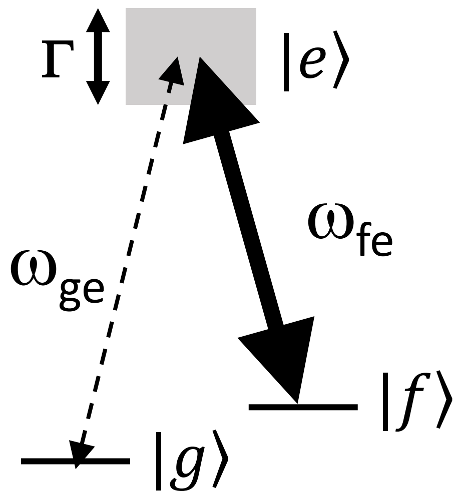




The sharp dispersive feature also slows light with the resulting group velocity: $v_g \approx c\,\frac{\Omega_c^2}{g^2 N}$

Frozen light/stopped light/quantum memory





The sharp dispersive feature also slows light with the resulting group velocity: $v_g \approx c\,\dfrac{\Omega_c^2}{g^2 N}$

If the control laser is switched off while the probe is passing Ω_C (and group velocity) gets smaller and smaller until it becomes 0 and the photon is coherently stored in the levels |g> and |f>

Foundations and Theory

Atom-photon interaction in a waveguide (waveguide QED)

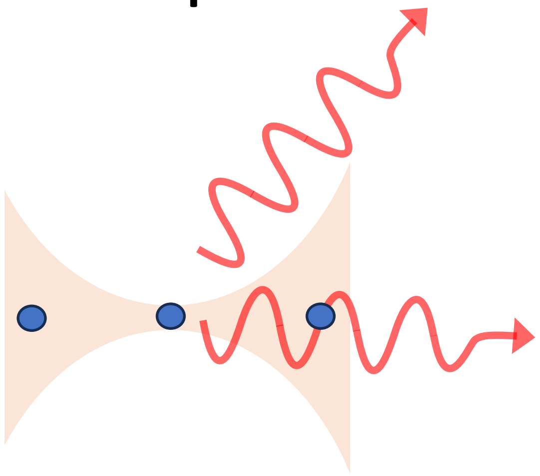
Hsiang-Hua Jen:
arXiv:2410.20665 Photon-mediated dipole–dipole interactions as a resource for quantum science and technology in cold atoms https://sites.google.com/view/hsianghuajen/home

References:

arXiv:2103.06824 "Waveguide quantum electrodynamics: collective radiance and photon-photon correlations"
Nature **541**, 473 (2017), "Chiral quantum optics"
arXiv:1703.10533, "Optical Nanofibers: a new platform for quantum optics"
arXiv:1912.06234, "Atomic-waveguide quantum electrodynamics"
Nature **583**, 775 (2020), "Waveguide quantum electrodynamics with superconducting artificial giant atoms"
Arno Rauschenbeutel Trapping and interfacing cold neutral atoms using optical nanofibers'
https://www.youtube.com/watch?v=AZbv9eHnJIM
Photonic technologies for quantum computing - Peter Lodahl - QCHS 2022
https://www.youtube.com/watch?v=56VktXJTWhI

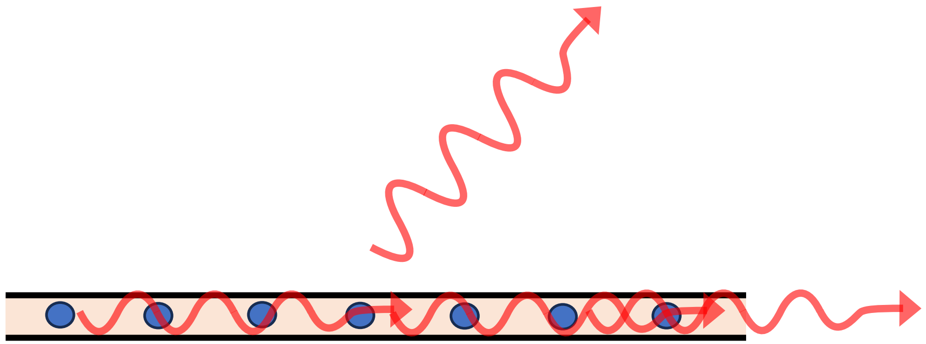
Scattering in a waveguide

Free space:



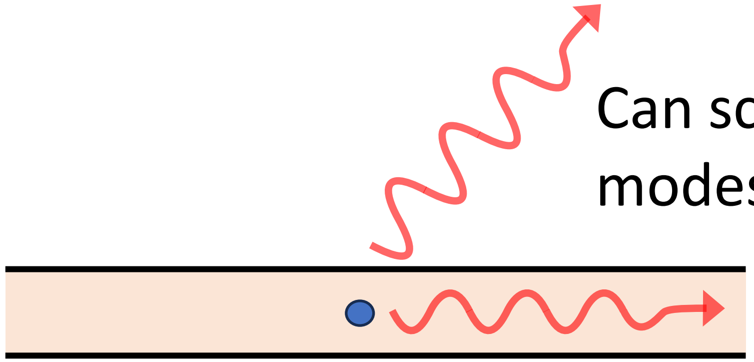
If you focus light to $A \sim \lambda^2$ you get a strong interaction but diffraction means it rapidly diverges

Waveguide:



In a waveguide you can have $A \sim \lambda^2$ and you can get a **strong interaction without diffraction**

Scattering in a waveguide



Can scatter into other non-guided modes (eg free space) Γ'

Can scatter into the waveguide guided mode Γ_{1D}

Total scattering $\Gamma_{TOT} = \Gamma' + \Gamma_{1D}$

β: Proportion of emission that is coupled to the waveguide mode

$$\beta = \frac{\Gamma_{1D}}{\Gamma_{1D} + \Gamma'}$$

Tricks to increase β:

1. Make waveguide **mode very small**
2. Use **subradiant states** that reduce coupling to free space, make $\Gamma' \ll \Gamma_0$

Cooperativity in a waveguide

Other important metrics are:

Single atom cooperativity in a waveguide:

$$\boxed{C = \frac{1}{2}\frac{\beta}{1-\beta}}$$

Purcell factor:

$$\boxed{\text{Purcell factor} = \frac{\Gamma_{1D}}{\Gamma_0}}$$

arXiv:2103.06824

Optical Density in Waveguides

Transmission=exp(-OD)=exp(-2NΓ_{1D}/Γ_0)

$\boxed{OD=2N\beta}$

$$\boxed{\text{Fidelity} \leq 1 - \frac{5.8}{OD} = 1 - \frac{5.8}{2N\beta}}$$

The power of waveguides:

	2D MOT quantum memory	Nanofiber waveguide
N	$\boxed{3e7}$	$\boxed{350}$
OD	70	70
β	1e-6 (effective)	0.1
Fidelity	91.7%	91.7%

A dramatic improvement!

Γ_{1D} : Maximizing coupling to the waveguide

Optimize the overlap between the atomic dipole field pattern and the transverse electric field of the waveguide mode.

The coupling rate is proportional to:

$$\Gamma_{1D} \propto \left| \int \mathbf{E}^*_{\text{guide}}(\mathbf{r}) \cdot \mathbf{u}_{\text{dip}}(\mathbf{r})\, d^3 r \right|^2$$

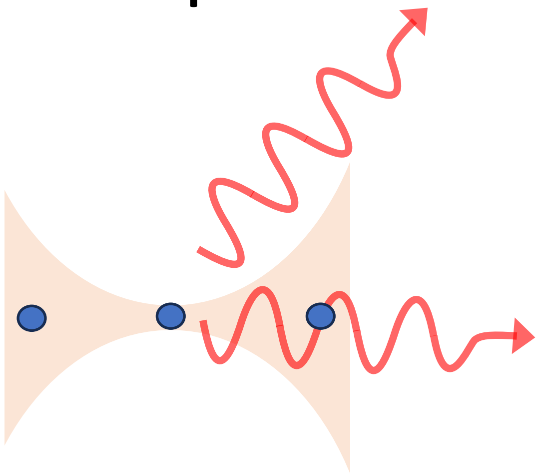
Strategies:

- Nanofibers
- Photonic crystal waveguides

💡 Tip: this will help you estimate beta, it is proportional to the normalized field intensity at the atom.

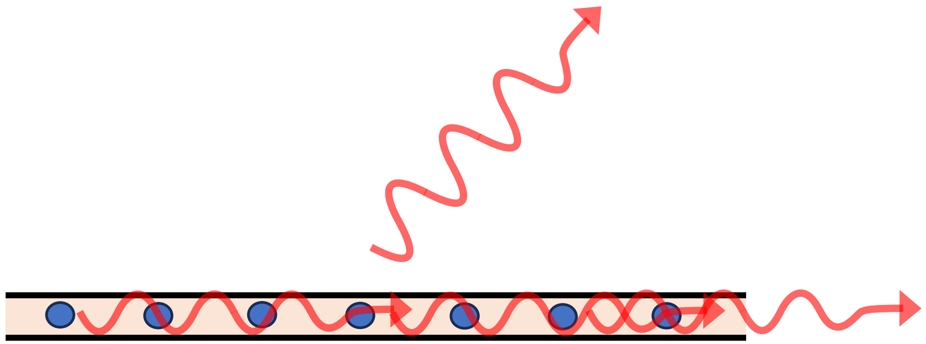
Free Space vs Waveguide: Why Waveguides Win

Free space:



- emission spread over 4π steradians
- $\beta \approx 10^{-4}$–10^{-6} typically
- Realistic few atom light–matter interfaces need optical cavities to enhance coupling

Waveguide:



- emission confined to 2 directions → effectively 1D
- β can exceed 0.3 without cavities
- β ≈ 0.9–0.99 in microwave waveguides
- deterministic scattering without finesse or mirrors

“Perfect” waveguide QED (Superconducting qubits) β=1

Superconducting qubits (eg transmons) are nonlinear LC oscillators with:

- huge dipole moments (>> 1000 e a_0 compared with ~1 e a_0 for atoms)
- transition frequencies in the microwave regime (4-10 GHz)
- engineered strong coupling to a microwave waveguide
- has negligible coupling to free space (due to grounding and geometry)

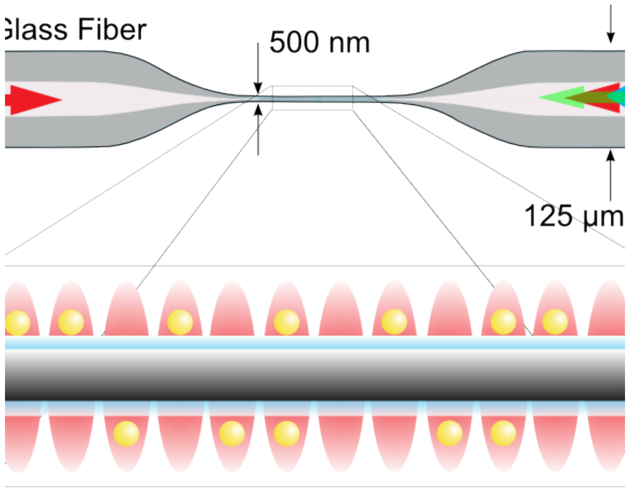
$$\boxed{\beta \sim 1}$$

BUT…

- Thermal noise dominates at room temperature **k_BT~1200 h f**
- Requires superconducting wires, temperatures <1K
- Challenging to scale!

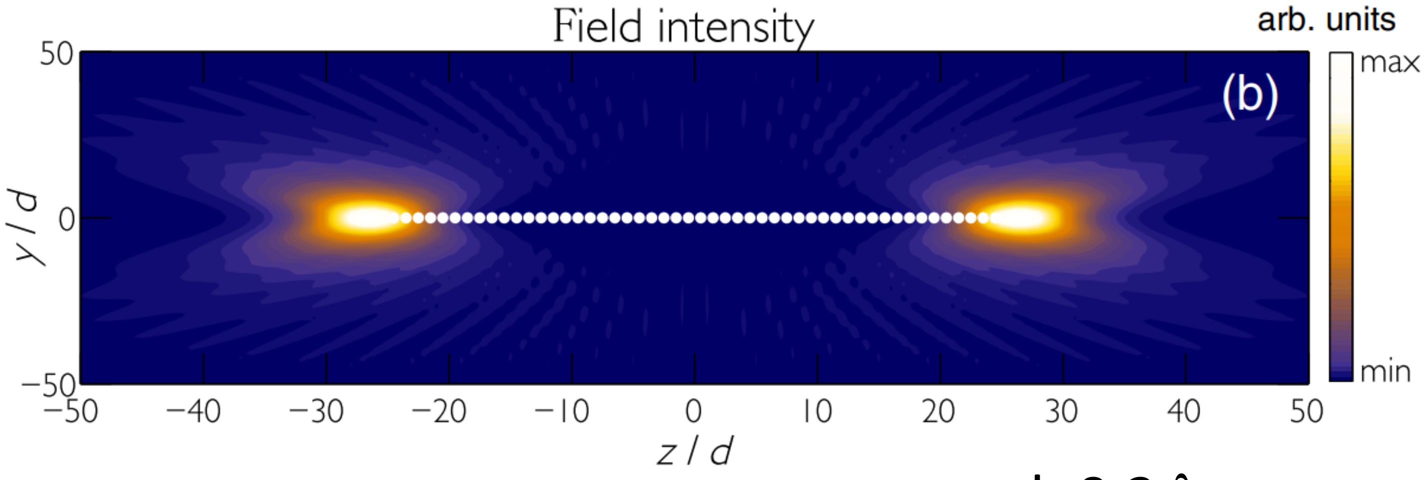
Collective waveguide QED

Random arrays



arXiv:0912.1179

Defect free subwavelength arrays



d=0.3 λ_0

Subwavelength defect free arrays (“phased arrays”) can create collective **superradiant and subradiant** modes

“Exponential Improvement in Photon Storage Fidelities Using Subradiance and “Selective Radiance” in Atomic Arrays”, PRX 7 031024, (2017)

arXiv:1703.03382

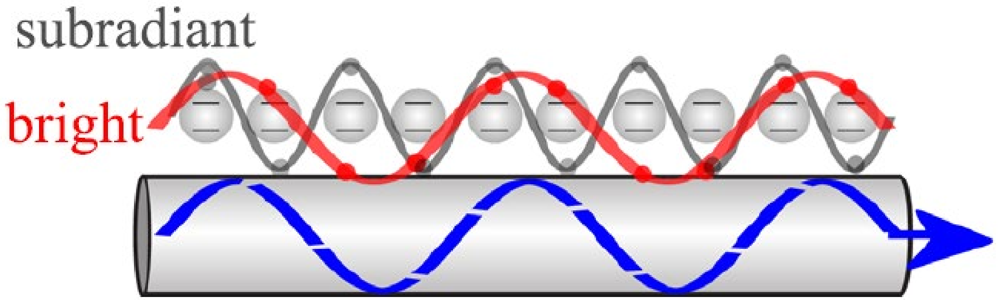
Collective waveguide QED

Superradiance

All atoms oscillate **in phase**
-> bright, enhanced coupling to guided modes

$$|B\rangle = \frac{1}{\sqrt{N}} \sum_j \sigma_j^- |g \dots g\rangle$$

$$\Gamma_{\text{super}} \approx \Gamma' + N\Gamma_{1D}$$



Subradiance

Atoms oscillate **out of phase**
destructive interference of
coupling to guided modes

$$\Gamma_{\text{sub}} \ll \Gamma'$$

arXiv:2103.06824

arXiv:1703.03382

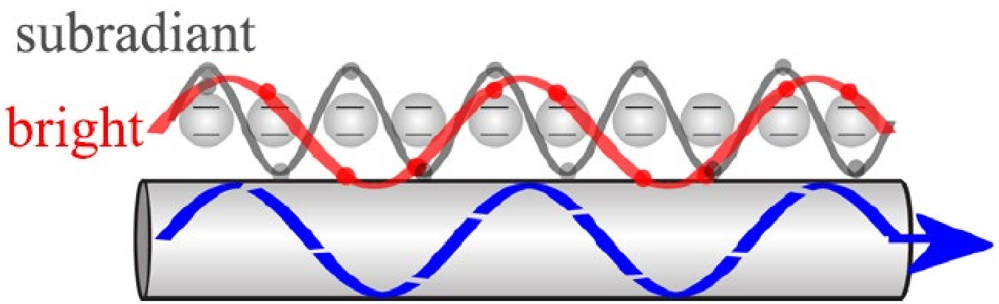
Collective waveguide QED

For atoms equally spaced with period d, excitations form **Bloch states:**

$$|k_B\rangle = \frac{1}{\sqrt{N}} \sum_{j=1}^{N} e^{ik_B z_j} \sigma_j^- |g \dots g\rangle$$

The decay rate of the collective mode is then:

$$\Gamma(k_B) = \Gamma' + \Gamma_{1D} \cos(k_B d)$$



Rich band structure leads to rich physics:

- Photon transport control
- Switching
- Photon localization
- …

arXiv:2103.06824

arXiv:1703.03382

Fidelity in Collective Waveguides

Enhanced coupling for the same OD leads to exponentially better fidelities storing quantum information

Random arrays:

$$\text{Fidelity} \leq 1 - \frac{5.8}{OD} = 1 - \frac{5.8}{2N\beta}$$

Subwavelength defect free arrays:

$$\text{Fidelity} \leq 1 - \frac{15}{N.OD} = 1 - \frac{15}{2N^2\beta}$$

The power of waveguides:

	2D MOT quantum memory	Nanofiber waveguide	Nanofiber waveguide (collective)	
N	3e7	350	30	350
OD	70	70	70	70
β	1e-6 (effective)	0.1	0.1	0.1
Fidelity	91.7%	91.7%	91.7%	99.94%

Foundations and Theory

Atom-photon interaction in a cavity

(cavity QED)

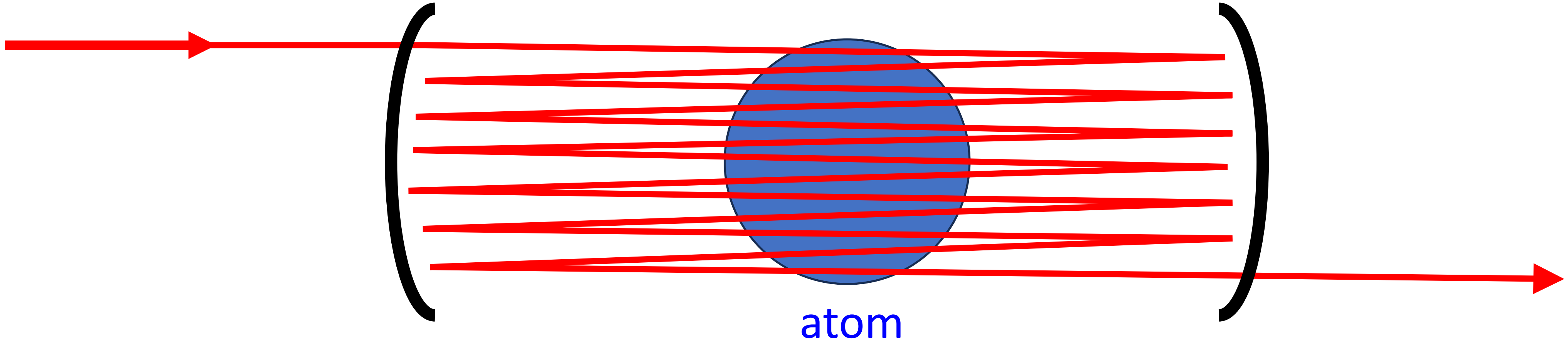
References:

arXiv:1412.2889 “Cavity-based quantum networks with single atoms and optical photons”

Fundamentals of Photonics

Optical Cavities

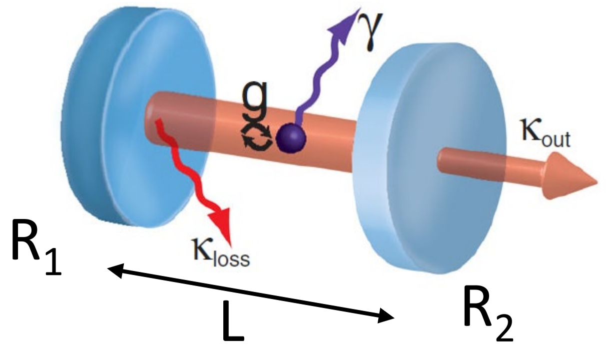
*A cavity bounces a photon so that it **interacts many times** with each atom.*



This increases the effective cross section by the number of bounces F/π (F is called the cavity Finesse)

Cavities help by:
- Minimizing mode volume
- High Finesse: many passes

Optical Cavities (recap)



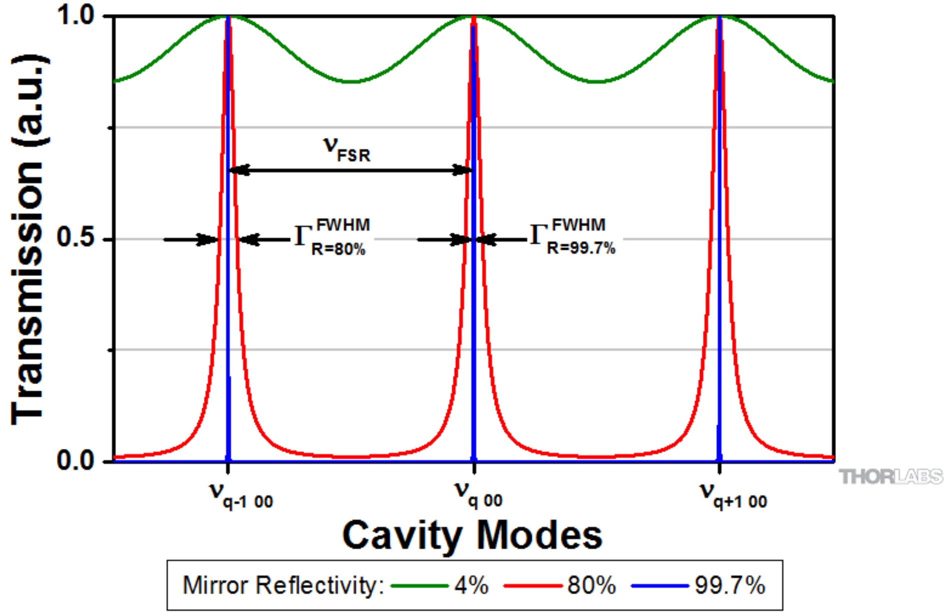
A Fabry–Pérot cavity of length L and mirror reflectivity R_1 and R_2.

Free spectral range: $$\mathrm{FSR} = \frac{c}{2L}$$

Finesse: $$\mathcal{F} = \frac{\pi (R_1 R_2)^{1/4}}{1 - \sqrt{R_1 R_2}}$$

Cavity decay rate: $$\kappa = \frac{\mathrm{FSR}}{\mathcal{F}}$$

Mode volume: $$V = \int \epsilon_0 |\mathbf{E}|^2 d^3 r \ / \ \max |\mathbf{E}|^2$$



https://www.thorlabs.com/newgrouppage9.cfm?objectgroup_id=9021

Cooperativity

Deterministic/guaranteed interaction requires:
$$\underbrace{\frac{3\lambda^2}{2\pi}}_{\sigma_{\text{abs}}} \times \underbrace{\frac{\mathcal{F}}{\pi}}_{\text{bounces}} \gg \underbrace{\frac{\pi}{4} w_0^2}_{\text{beam area}}$$ A

$$\frac{3\lambda^2}{2\pi}\frac{\mathcal{F}}{\pi}\frac{1}{A} \gg 1$$

This can be rewritten as:
single atom **cooperativity**:

$$\boxed{C_1 = \frac{g^2}{2\kappa\gamma} \gg 1}$$

For N atoms in the cavity we care about the **collective cooperativity, C_N**:

$$\boxed{C_N = N C_1 = N\frac{g^2}{2\kappa\gamma} \gg 1}$$

Atom-cavity coupling constant

$$\boxed{g = \sqrt{\frac{d_{ge}^2 \omega}{\epsilon_0 \hbar V}}}$$

Cavity decay rate:

$$\boxed{\kappa = \frac{\pi c}{2L\mathcal{F}}}$$

Atom decay rate:

$$\boxed{\gamma = \frac{d_{ge}^2 \omega^3}{6\pi\epsilon_0 \hbar c^3}}$$

Cavity mode volume:

$$V = \int u^2(\vec{\mathbf{r}}) d^3x$$

u the dimensionless electric field mode normalize to 1 at the field maximum.

**Note some papers/theory uses $C_1 = g^2/\kappa\gamma$, both are common!

Definitions

$$C_1 = \frac{g^2}{2\kappa\gamma} \gg 1$$

$$g = \sqrt{\frac{d_{ge}^2\omega}{\epsilon_0\hbar V}}$$

$$\kappa = \frac{\pi c}{2L\mathcal{F}}$$

$$\gamma = \frac{d_{ge}^2\omega^3}{6\pi\epsilon_0\hbar c^3}$$

- Note: C_1 doesn't depend on L! ($\kappa \propto 1/L$, $g \propto 1/L^{0.5}$)
- Critical atom number (number of atoms needed in the cavity to achieve strong interaction with a photon C_N~1):

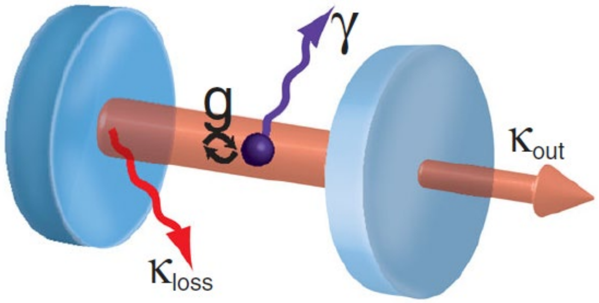
$$N_c = 1/C_1$$

- Critical photon number (number of photons needed saturate the atomic transition):

$$n_c = \frac{\gamma^2}{2g^2}$$

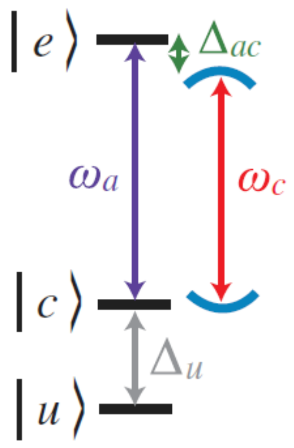
Jaynes–Cummings model

A Fabry–Pérot cavity of length L and mirror reflectivity R has:



$$\mathcal{H}_I = -\vec{d}\,\vec{E} \rightarrow dE(\sigma_{ce}^{\dagger} + \sigma_{ce})(a^{\dagger} + a)$$

$$= (\sigma_{ce}^{\dagger} a^{\dagger} + \sigma_{ce}^{\dagger} a + \sigma_{ce} a^{\dagger} + \sigma_{ce} a)$$



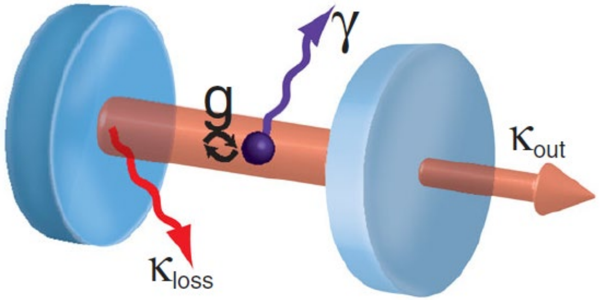
$$\mathcal{H}_{JC} = \hbar\omega_a \sigma_{ce}^{\dagger}\sigma_{ce} + \hbar\omega_c a^{\dagger} a + \hbar g(\underbrace{\sigma_{ec} a + \sigma_{ce} a^{\dagger}}_{\text{ole coupling}})$$

$$E_0 = 0,$$

$$E_{N,\pm} = N\hbar\omega_c + \hbar\frac{\Delta_{ac}}{2} \pm \frac{\hbar}{2}\sqrt{\Omega_N^2 + \Delta_{ac}^2}$$

$$\Omega_N = 2g\sqrt{N} \qquad \Delta_{ac} = \omega_a - \omega_c$$

Jaynes–Cummings solutions



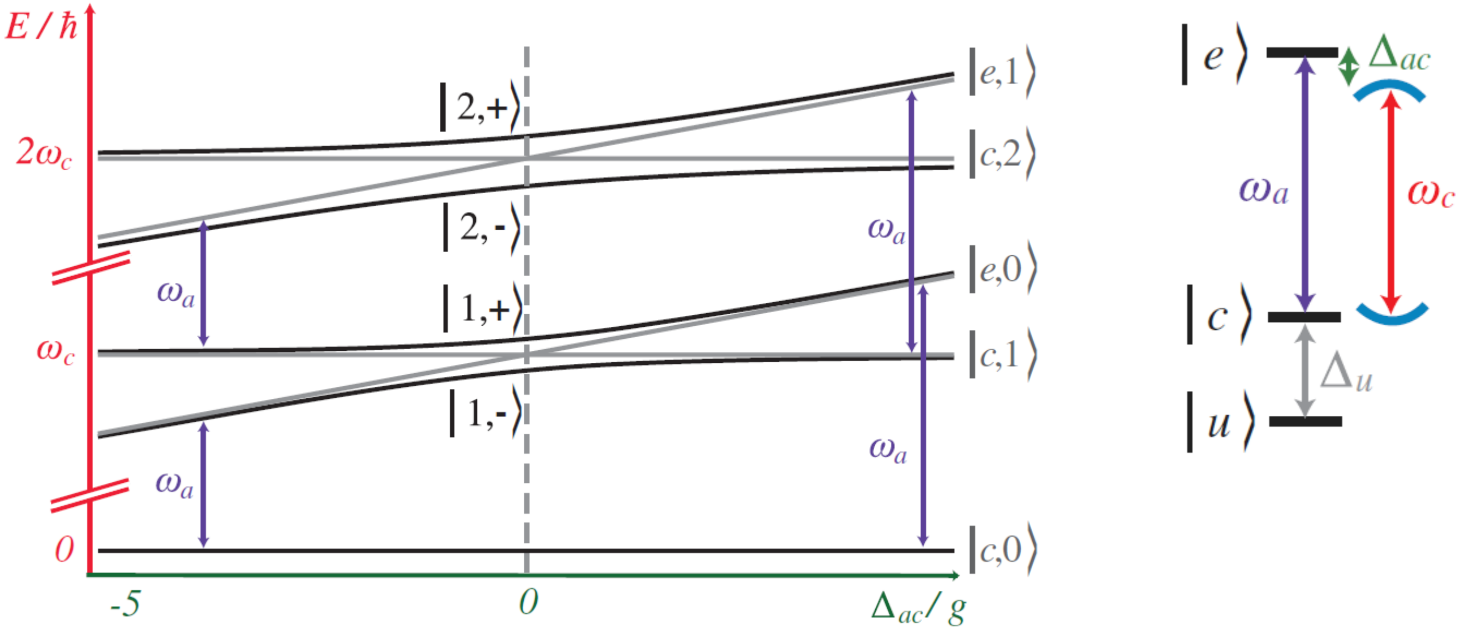
$$E_0 = 0,$$

$$E_{N,\pm} = N\hbar\omega_c + \hbar\frac{\Delta_{ac}}{2} \pm \frac{\hbar}{2}\sqrt{\Omega_N^2 + \Delta_{ac}^2}$$

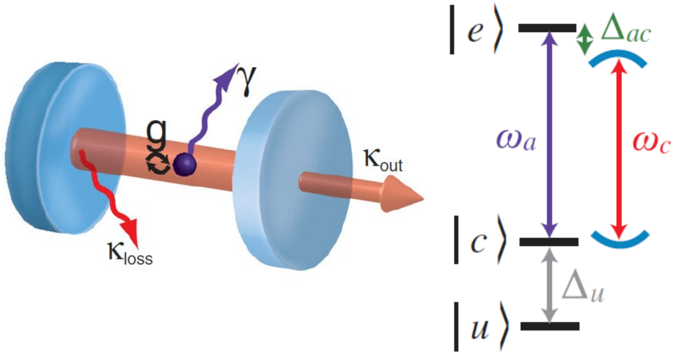
$$\Omega_N = 2g\sqrt{N} \qquad \Delta_{ac} = \omega_a - \omega_c$$

For zero detuning the levels are separated by Ω_N

For zero initial energy (N-1=0) and zero detuning Ω_1=2 g



Vacuum Rabi frequency



$$E_0 = 0,$$
$$E_{N,\pm} = N\hbar\omega_c + \hbar\frac{\Delta_{ac}}{2} \pm \frac{\hbar}{2}\sqrt{\Omega_N^2 + \Delta_{ac}^2}$$

$$\Omega_N = 2g\sqrt{N}$$
$$\Delta_{ac} = \omega_a - \omega_c$$

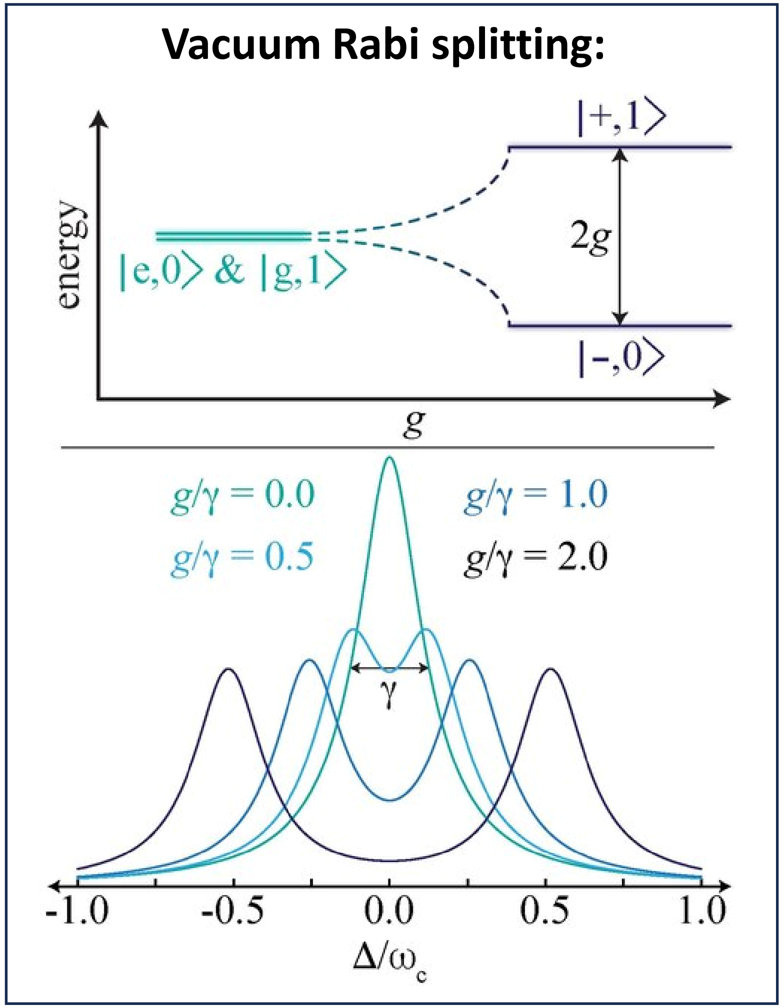
For zero initial energy (N-1=0) and zero detuning:

Ω_1=2 g

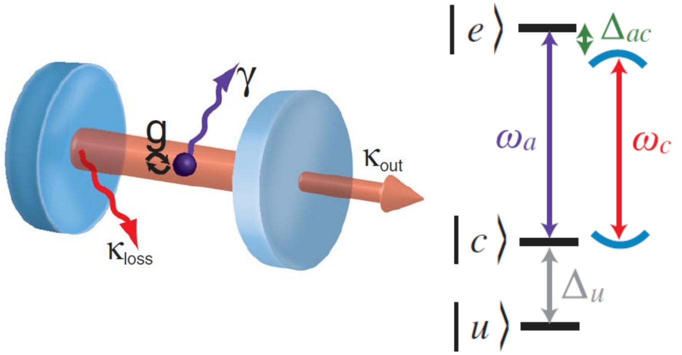
Probability of the atom being in |e> as a function of time:

$$P_{|e\rangle}(t) = \cos^2(\Omega_N t/2)$$

Vacuum Rabi frequency: $\boxed{\Omega_1 = 2\ g}$



Strong coupling



Strong coupling regime:

$$\boxed{g \gg (\kappa, \gamma)}$$

- Reversible atom-photon coupling is faster than all irreversible processes
- Coherent processes
- Vacuum-Rabi splitting observed for weak probes

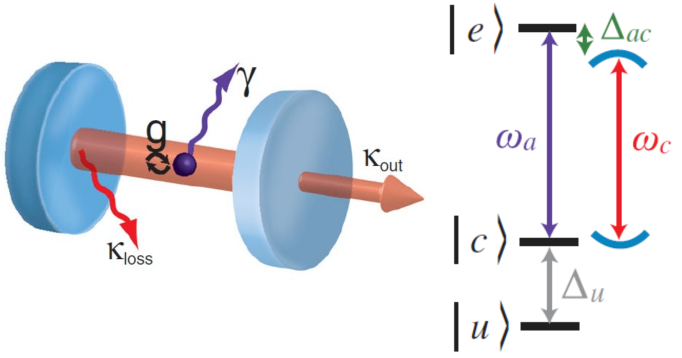
$$g = \sqrt{\frac{d_{ge}^2 \omega}{\epsilon_0 \hbar V}}$$

$$\kappa = \frac{\pi c}{2L\mathcal{F}}$$

Atom – Cavity exchange energy at a rate of: $$\boxed{\Omega_{N=1,\Delta_{ac}} = \sqrt{\Delta_{ac}^2 + 4g^2}}$$

$$\gamma = \frac{d_{ge}^2 \omega^3}{6\pi \epsilon_0 \hbar c^3}$$

Purcell/fast cavity regime



Purcell regime: $\kappa \gg g \gg \gamma$

- Vacuum-Rabi oscillations cannot be observed (photon lost from the resonator before reabsorption)
- Decay into cavity near atomic frequency can be **strongly increased** (Purcell, 1946) **or decreased** (Kleppner, 1981)

Decay rate into the cavity mode from an atom:

$$\gamma_c = \frac{g^2\kappa}{\kappa^2 + \Delta_{ac}^2}$$

Δ_{ac}= detuning between atom and cavity

On resonance decay into the cavity mode can be much larger than natural decay rate γ:

$$\gamma_c = 2C\gamma$$

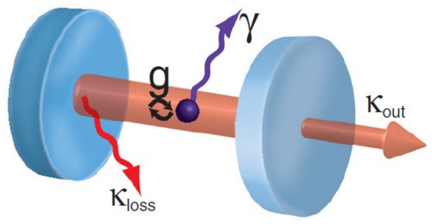
If cavity mode covers a large fraction of the solid angle for an atom in a cavity the spontaneous emission rate into free space is affected:

$$\gamma = (1 - \zeta/4\pi)\frac{\Gamma}{2}$$

ζ = cavity solid angle at the atom

Superradiant lasers

Superradiant laser regime: $\kappa \gg g\sqrt{N} \gg \gamma$

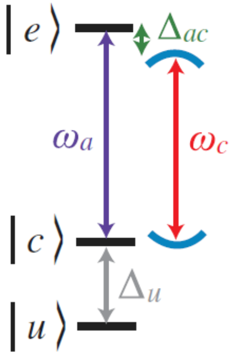


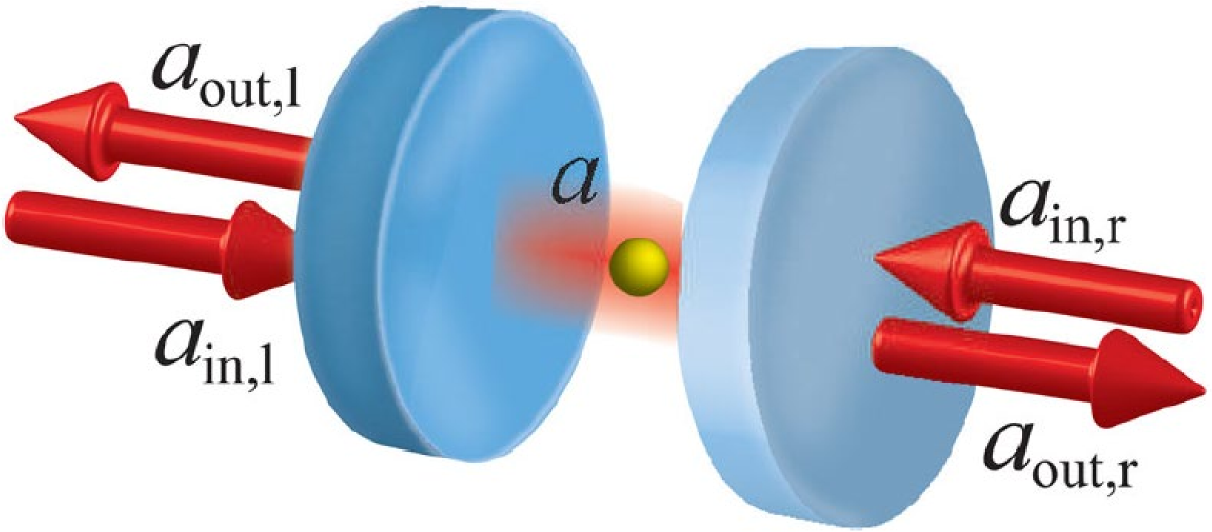
Superradiant lasers are potential next generation atomic clocks.

- Cavity noise is strongly supressed (eg 1e6)
- Enhanced emission rates gain $\propto N^2$
- Allows creation μHz lasers on forbidden transitions

Purcell regime	**Superradiant laser regime**
Cavity controls atomic emission $C_1 \gg 1$	Cavity has no influence on emission $C_1 \ll 1$
Short atomic coherence vs cavity lifetime	Long atomic coherence vs cavity lifetime
Phase defined by cavity	Phase defined by atoms $C_N \gg 1$
Cavity noise NOT suppressed	Cavity noise **strongly** suppressed

Transmission and reflection





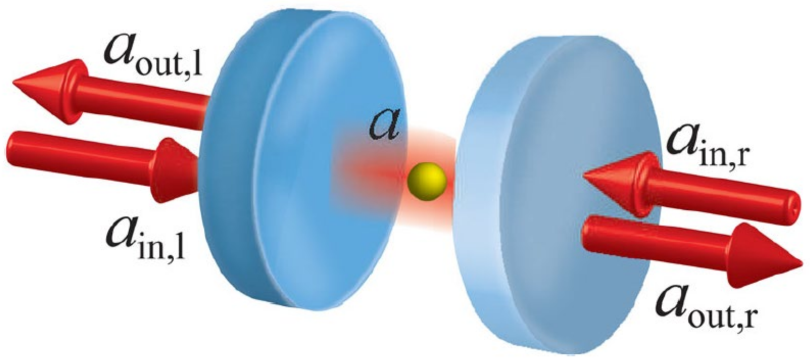
$$r(\omega) = 1 - \frac{2\kappa_l(i\Delta_a + \gamma)}{(i\Delta_c + \kappa)(i\Delta_a + \gamma) + g^2}$$

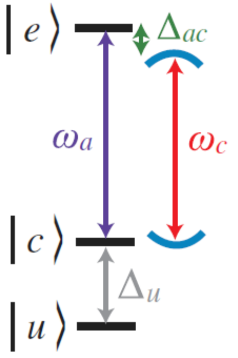
$$t(\omega) = \frac{2\sqrt{\kappa_l\kappa_r}(i\Delta_a + \gamma)}{(i\Delta_c + \kappa)(i\Delta_a + \gamma) + g^2}$$

r, t the **amplitude** reflection and transmission coefficients
Intensity $R=|r|^2$, $T=|t|^2$, reflected phase = arg(r)

Important practical note: Any imperfection in modematching between the incoming light and the cavity

Mode-matching





Practical note: Imperfect mode-matching spatial and temporal between the incoming light mode and the cavity mode will be reflected by the cavity.

For free space cavities this can dominate errors.

$R_{\text{experiment}} = (1-\xi) + \xi R$

ξ = mode matching efficiency

**Optical fiber based cavities can have much better spatial mode matching efficiency compared with free space cavities

Estimating possible fidelities using cavities

For an optical cavity $\boxed{OD \sim 4C_N}$

Using our earlier expression for memory fidelity:

$$\boxed{\text{Fidelity} \leq 1 - \frac{5.8}{OD} = 1 - \frac{5.8}{4NC_1}}$$

	2D MOT quantum memory	Nanofiber waveguide	Nanofiber waveguide (collective)		Cavity	Cavity with waveguide	Cavity with waveguide (collective)
N	3e7	350	30	350	1	100	15
OD	70	70	70	70	100	10000	1500
β	1e-6 (effective)	0.1	0.1	0.1			
C_1	6e-7	6e-2	6e-2	6e-2	25	25	25
Fidelity	91.7%	91.7%	91.7%	99.94%	94.2%	99.94%	99.94%

Realistically other factor will probably start to dominate.

Key Requirements for High-Fidelity Interfaces

High β-factor/cooperativity

(high Rabi frequency compared with dissipation/decoherence rate)

+ good mode-matching/impedance matching

Part III: Experimental Progress

1. Foundational Tools
2. Quantum interfaces to ensembles (recap)
3. Interfacing to trapped ions
4. Interfacing to neutral atoms
5. Interfacing to NV/SiV centers and quantum dots
6. Interfacing to superconducting qubits
7. Summary

Foundational Tools: Entanglement Swapping

Long-distance quantum communication with atomic ensembles and linear optics

L.-M. Duan, M. D. Lukin, J. I. Cirac & P. Zoller

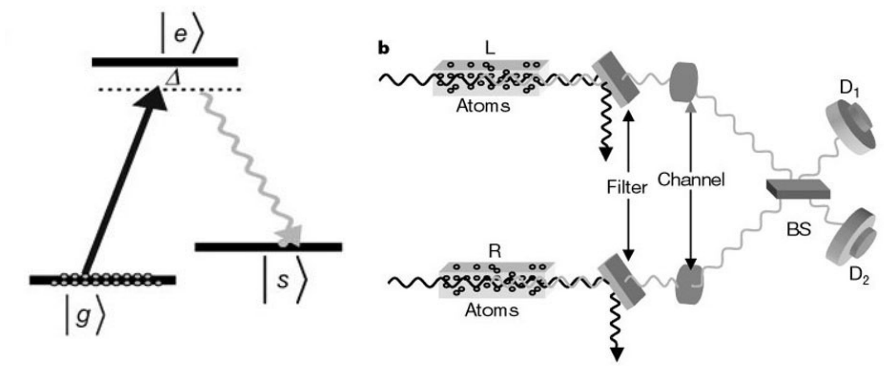
Nature **414**, 413–418 (2001) | Cite this article

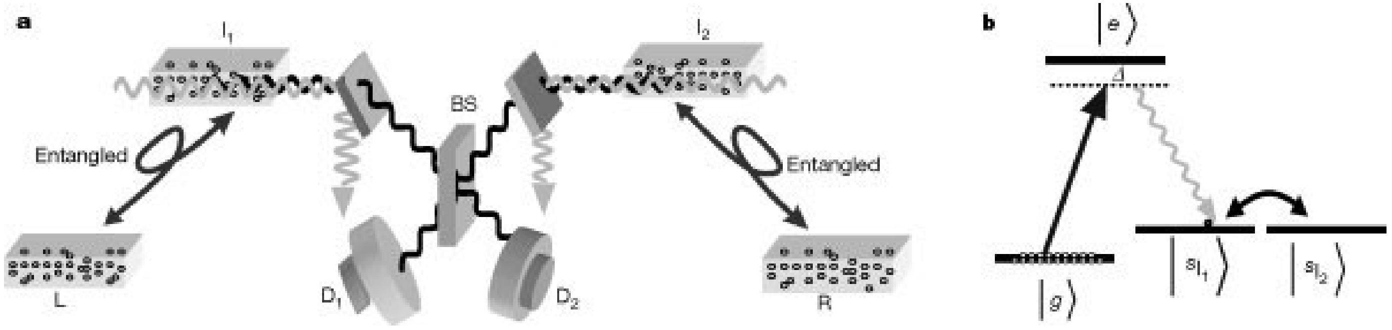
The basis for virtually all quantum communication

DLCZ protocol / Entanglement swapping

Invented heralded entanglement generation

Step 1: Entangle two remote ensembles





Entanglement protocol

1. Prepare two atomic ensembles in |g>
2. Apply a weak write raman pulse to both
3. Interfere stokes output on a beam splitter
4. If only 1 detector clicks one of the ensembles is in |g> and the other is in |s>
5. If no clicks repeat.

Entanglement swapping protocol

1. Apply a read pulse (|s>->|e>) to both I1 and I2 ensembles simultaneously and interfere on beamsplitter
2. If one detector clicks ensembles L and R are entangled
3. Otherwise repeat both entanglement generation and swapping protocols

Foundational Tools: Quantum Non-demolition Measurement

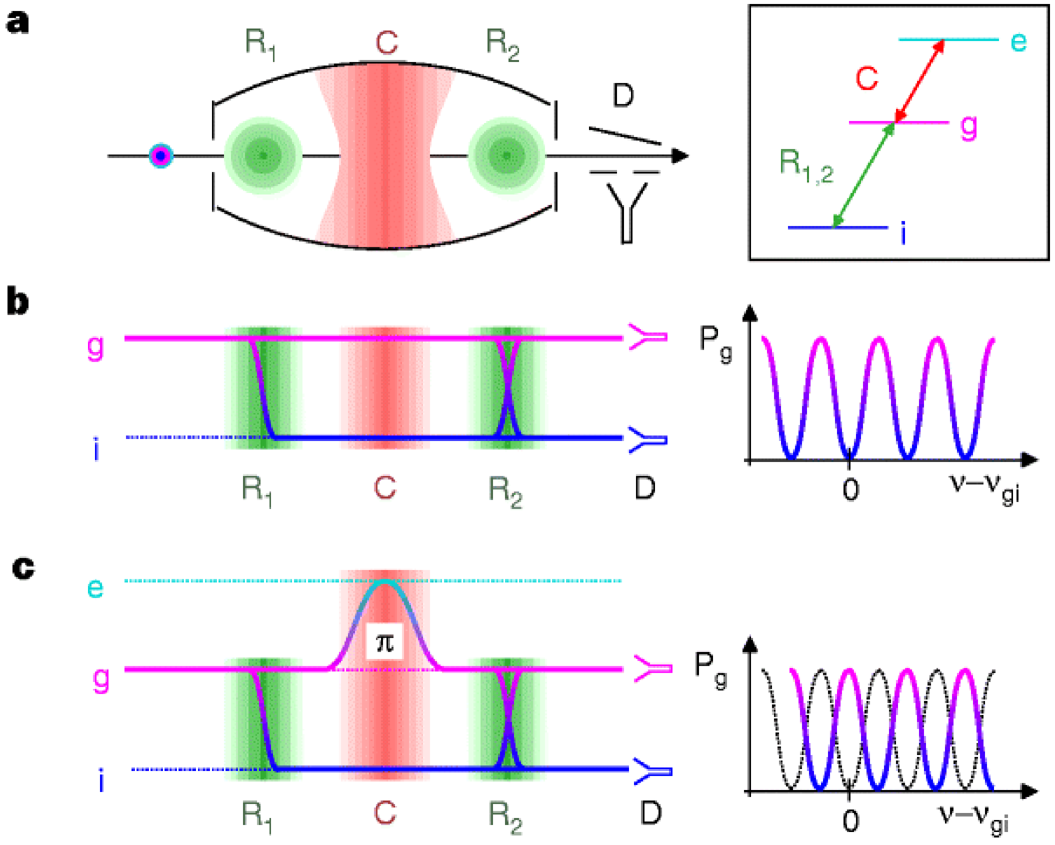
Seeing a single photon without destroying it

G. Nogues, A. Rauschenbeutel, S. Osnaghi, M. Brune, J. M. Raimond & S. Haroche

Nature **400**, 239–242 (1999) | Cite this article

First non-destructive, repeatable detection of a single photon using cavity QED

Basis for phase gates/switches



Experiment:

1. Prepare a single photon in the microwave cavity (lifetime 1ms) (use a Rydberg atom as a photon source).
2. Prepare a Rydberg atom in a superposition via Ramsey zone R_1.
3. Atom flies through cavity → picks up π phase only if photon present.
4. Second Ramsey zone R_2 converts phase into population difference.
5. Detect atom → reveals whether a photon is in the cavity.
6. Photon remains intact → repeat with another atom for confirmation.

2012 Nobel prize – visiting in Jan 2026

Experimental Progress

Quantum interfaces to ensembles

(recap)

References:
arXiv:1002.4659 Optical quantum memory
arXiv:0807.3358 Quantum interface between light and atomic ensembles
arXiv:0806.4195 The Quantum Internet
https://doi.org/10.1126/science.aam9288 Quantum internet: A vision for the road ahead
https://doi.org/10.6028/jres.125.002 Optical Quantum Memory and its Applications in Quantum Communication Systems

arXiv:1605.08519 Highly Efficient Coherent Optical Memory Based on Electromagnetically Induced Transparency

Quantum interfaces to ensembles (recap)

REVIEWS OF MODERN PHYSICS, VOLUME 82, APRIL–JUNE 2010

Quantum interface between light and atomic ensembles

Klemens Hammerer

Institute for Theoretical Physics, University of Innsbruck, and Institute for Quantum Optics and Quantum Information, Austrian Academy of Science, Technikerstrasse 25, 6020 Innsbruck, Austria

Anders S. Sørensen and Eugene S. Polzik

Niels Bohr Institute, Copenhagen University, Blegdamsvej 17, Copenhagen 2100, Denmark

Optical quantum memory

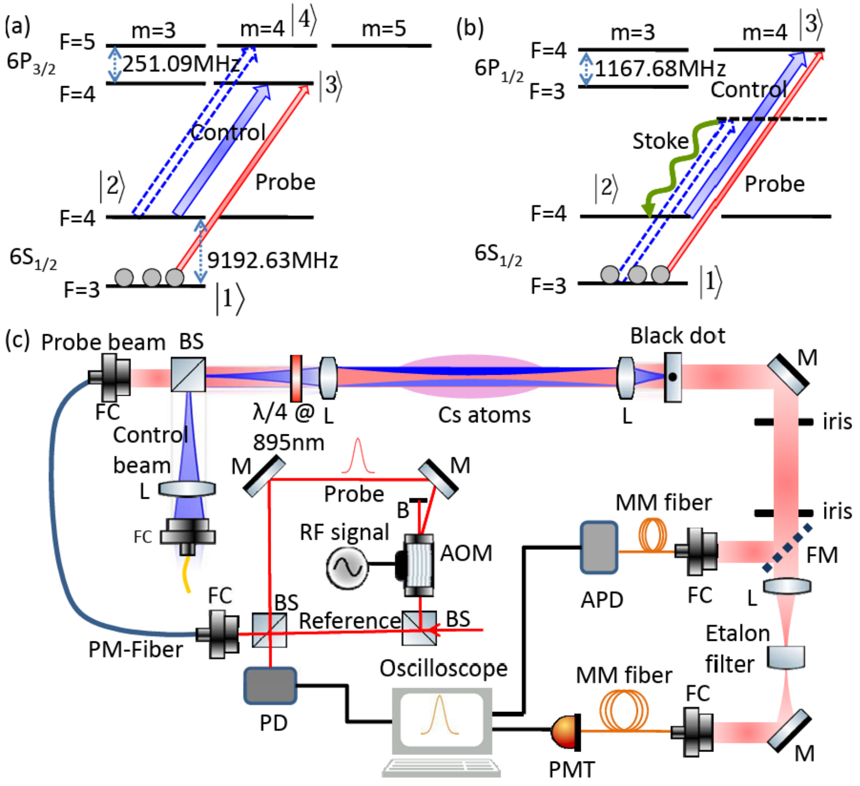
Alexander I. Lvovsky, Barry C. Sanders & Wolfgang Tittel

Nature Photonics **3**, 706–714 (2009) | Cite this article

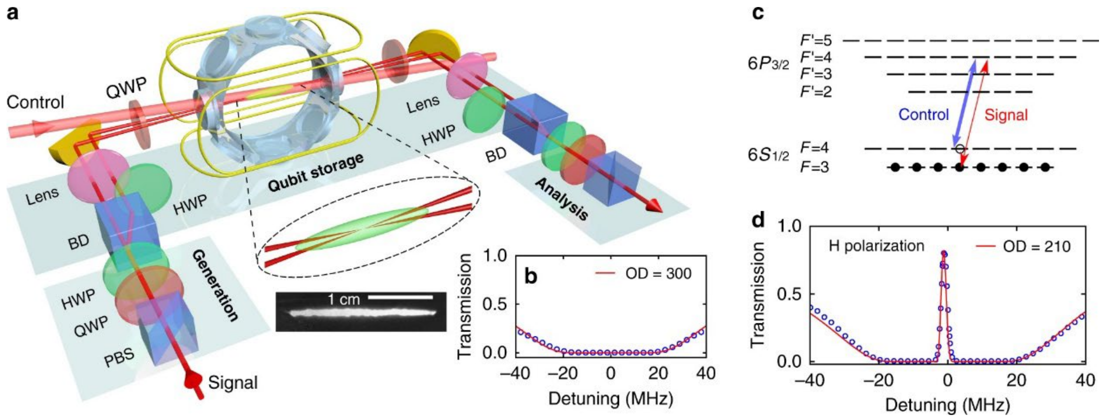
Approaches used for quantum memory in large ensembles

- EIT
- Gradient Echo Memory (GEM)
- …

Quantum interfaces to ensembles: EIT (recap)



92% storage efficiency
OD: 800
Time-bandwidth product: 1200
N atoms: 5e9



- Storing and retrieving single photons in cold atom ensembles is well established with world leading performance at IAMS/NTHU (Ying-Cheng Chen)
- This is a leading technology for quantum communication/entanglement distribution networks
- This technology was covered in detail in earlier lectures

…BUT it requires >1e8 atoms to efficiently store a single qubit
⇒ unsuitable for interfacing to single qubits in quantum processors.

arXiv:1605.08519 Highly Efficient Coherent Optical Memory Based on Electromagnetically Induced Transparency, PRL 120, 183602, (2018)
Highly-efficient quantum memory for polarization qubits in a spatially-multiplexed cold atomic ensemble, Nature Communications **9**, 363 (2018)

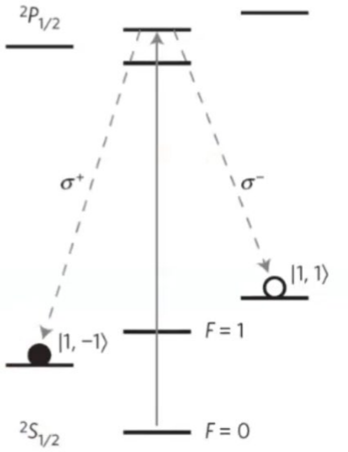
Experimental Progress

Interfacing to trapped ions

References:

Colloquium: Quantum networks with trapped ions, Rev. Mod. Phys. 82, 1209 (2010)
[Quantum interfaces based on trapped ions | Tracy Northup (University of Innsbruck)](https://www.youtube.com/watch?v=PIiWZ8RvmkE)
https://www.youtube.com/watch?v=PIiWZ8RvmkE

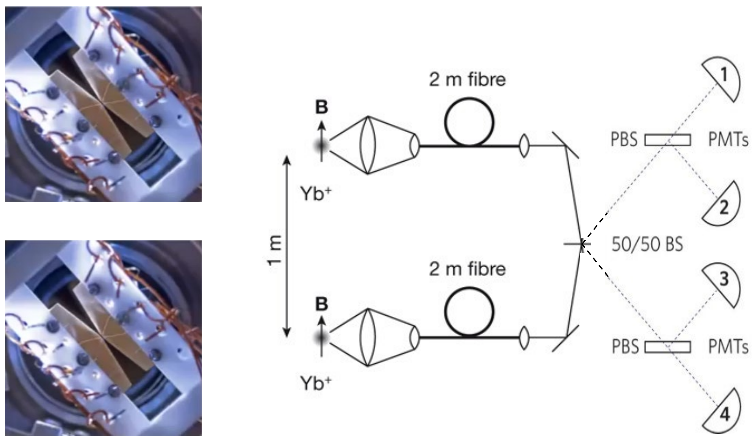
Quantum interfaces for trapped ions



Entanglement of single-atom quantum bits at a distance

D. L. Moehring, P. Maunz, S. Olmschenk, K. C. Younge, D. N. Matsukevich, L.-M. Duan & C. Monroe

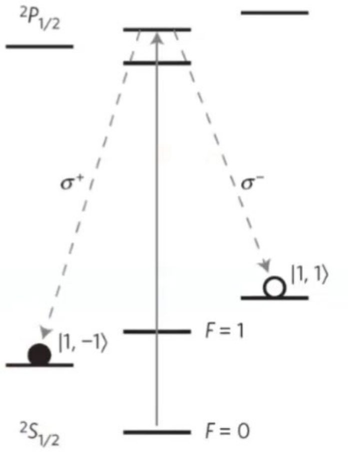
Nature **449**, 68–71 (2007) | Cite this article



1. Excite ions in both traps simultaneously to $^2P_{1/2}$
2. Ions decay emitting a $\sigma+/\sigma-$ polarization photon
3. Both photons simultaneously cross on a 50:50 beamsplitter erasing which way information
4. **If photons have opposite polarizations** and click PMTs simultaneously the ions are entangled
5. If photons have same polarization both ions are in a known state and no entanglement is generated **(fails 50% of the time)**.

Heralding method allows a deterministic outcome (entanglement) with some (50% in the case) probability

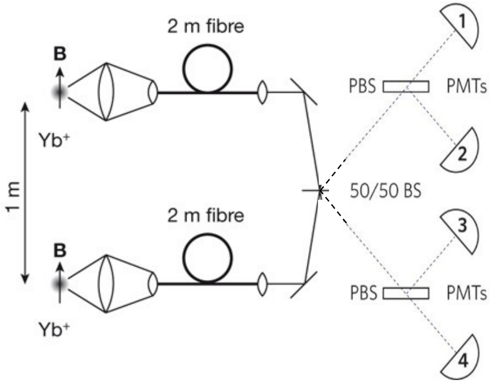
Quantum interfaces for trapped ions



Entanglement of single-atom quantum bits at a distance

D. L. Moehring, P. Maunz, S. Olmschenk, K. C. Younge, D. N. Matsukevich, L.-M. Duan & C. Monroe

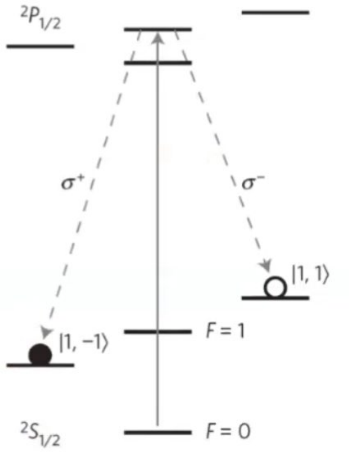
Nature **449**, 68–71 (2007) | Cite this article



1. Excitation pulses @ 550kHz
2. Excitation probability 50%
3. Huge losses in collection/detection
4. Success probability: 3.6e-9

One entangled pair generated every 8.3 minutes

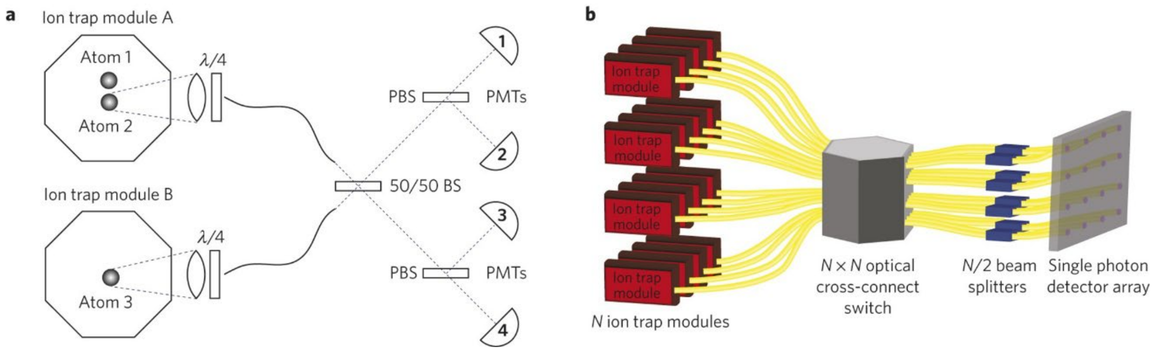
Quantum interfaces for trapped ions



Modular entanglement of atomic qubits using photons and phonons

D. Hucul, I. V. Inlek, G. Vittorini, C. Crocker, S. Debnath, S. M. Clark & C. Monroe

Nature Physics **11**, 37–42 (2015) | Cite this article



1. Fidelity of ion-photon entangled state wrt Bell state 92%
2. Fidelity of ion-ion entangled state: 78(3)%
3. 4.5Hz Entangement generation
4. 1.2s coherence time

Most gains through improved collection and detection efficiency

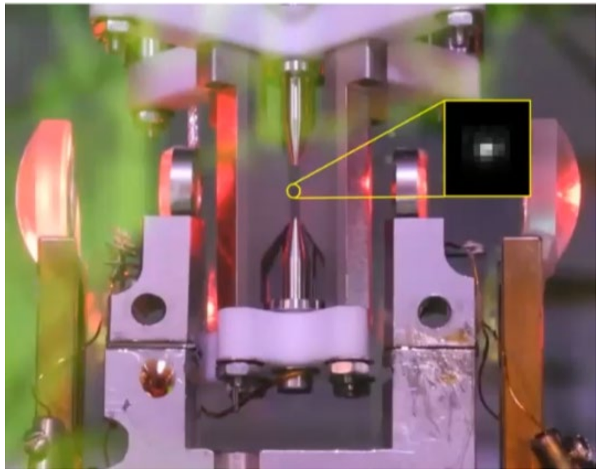
First to demonstrate entanglement generation faster than decoherence

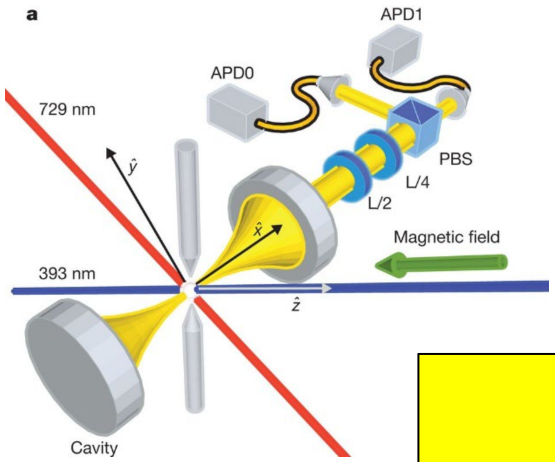
Quantum interfaces for trapped ions

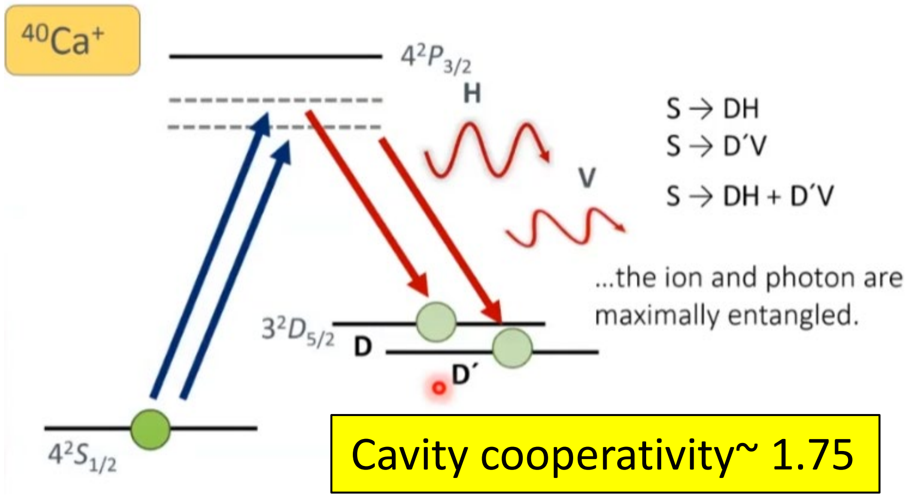
Tunable ion–photon entanglement in an optical cavity

A. Stute, B. Casabone, P. Schindler, T. Monz, P. O. Schmidt, B. Brandstätter, T. E. Northup & R. Blatt

Nature **485**, 482–485 (2012) | Cite this article







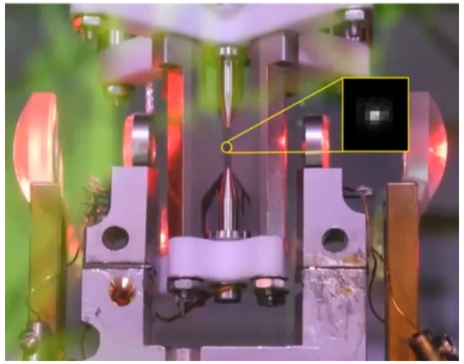
First to use a cavity with ions allowing extremely efficient entanglement generation

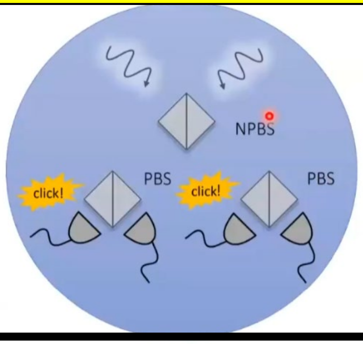
- Cavity allow extremely high entanglement generation probability:
  "primarily limited by the probability of the photon leaving the cavity (16%) and photodiode efficiencies (40%)"
- we control **both amplitude and phase** of entangled state using via two simultaneous cavity-mediated Raman transitions.
- Record ion-photon entanglement fidelity 97.4%
- 40.5 Hz entanglement generation

https://www.youtube.com/watch?v=PIiWZ8RvmkE

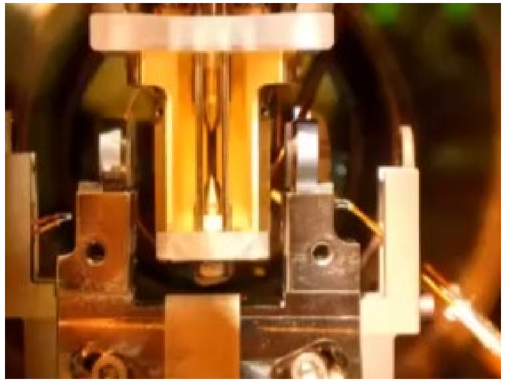
Quantum interfaces for trapped ions

Long distance entanglement generation





230m



PHYSICAL REVIEW LETTERS 130, 050803 (2023)

Editors' Suggestion

Featured in Physics

Entanglement of Trapped-Ion Qubits Separated by 230 Meters

V. Krutyanskiy,[1,2] M. Galli,[2] V. Krcmarsky,[1,2] S. Baier,[2] D. A. Fioretto,[2] Y. Pu,[2] A. Mazloom,[3] P. Sekatski,[4] M. Canteri,[1,2] M. Teller,[2] J. Schupp,[1,2] J. Bate,[2] M. Meraner,[1,2] N. Sangouard,[5] B. P. Lanyon,[1,2,*] and T. E. Northup[2]

[1]*Institut für Quantenoptik und Quanteninformation, Österreichische Akademie der Wissenschaften, Technikerstraße 21a, 6020 Innsbruck, Austria*

[2]*Institut für Experimentalphysik, Universität Innsbruck, Technikerstraße 25, 6020 Innsbruck, Austria*

[3]*Department of Physics, Georgetown University, 37th and O Streets NW, Washington, D.C. 20057, USA*

[4]*Department of Applied Physics, University of Geneva, 1211 Geneva, Switzerland*

[5]*Institut de Physique Théorique, Université Paris-Saclay, CEA, CNRS, 91191 Gif-sur-Yvette, France*

(Received 3 September 2022; accepted 20 December 2022; published 2 February 2023)

target state:

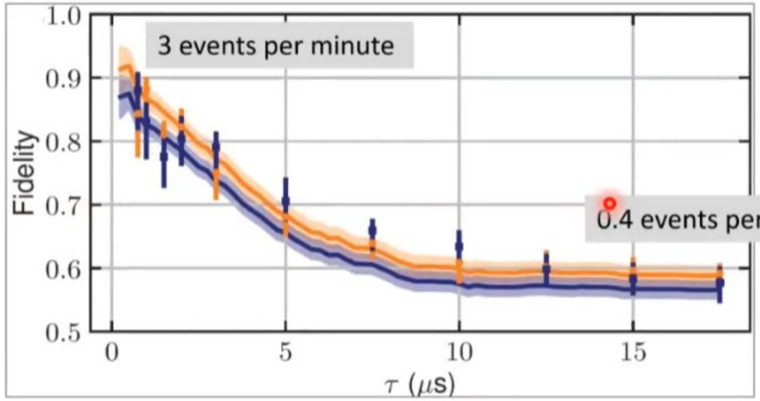
$|\psi^-\rangle = |\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle$

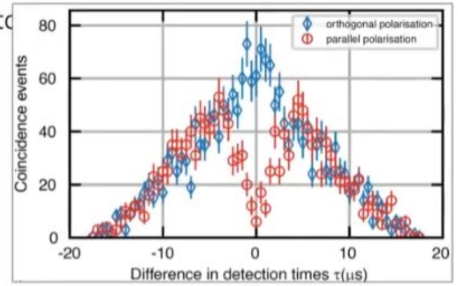
fidelity with respect to target state:

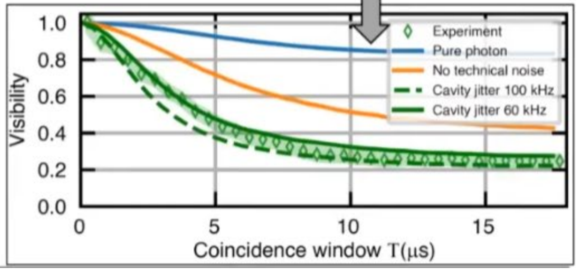
$\langle\psi^-|\rho|\psi^-\rangle = 0.82 + 0.02 - 0.06$

classical bound: 0.5

The fidelity decreases as longer intervals between photons are considered due to photon distinguishability, but it remains nonclassical.







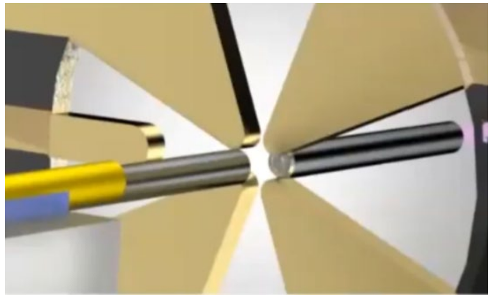
We can understand this trade-off via a master-equation model that reproduces our experimental visibility.

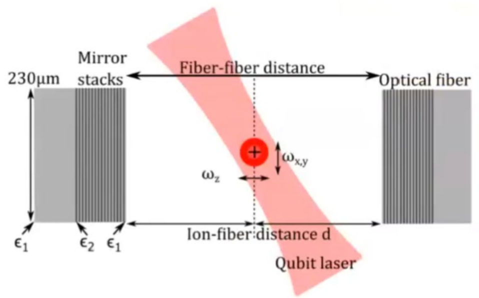
rutyanskiy, M. Galli et al., *Phys. Rev. Lett.* **130**, 050803 (2023)

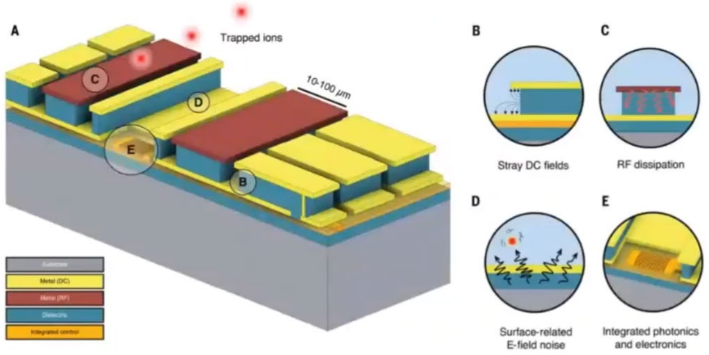
https://www.youtube.com/watch?v=PIiWZ8RvmkE

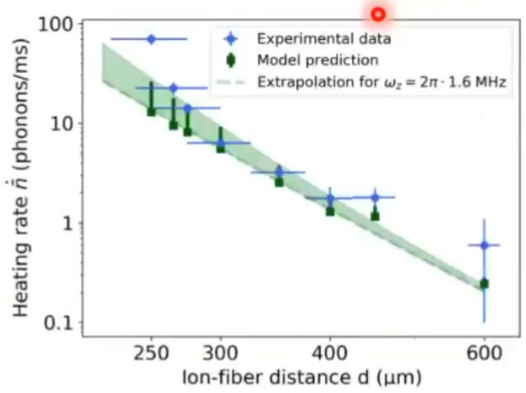
Quantum interfaces for trapped ions - Outlook

Miniaturization – fiber cavities







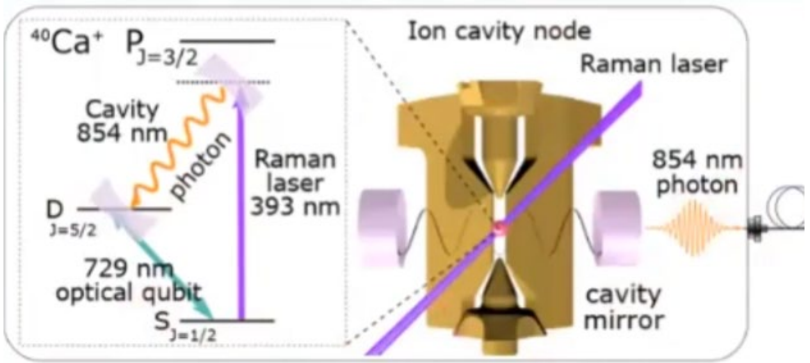


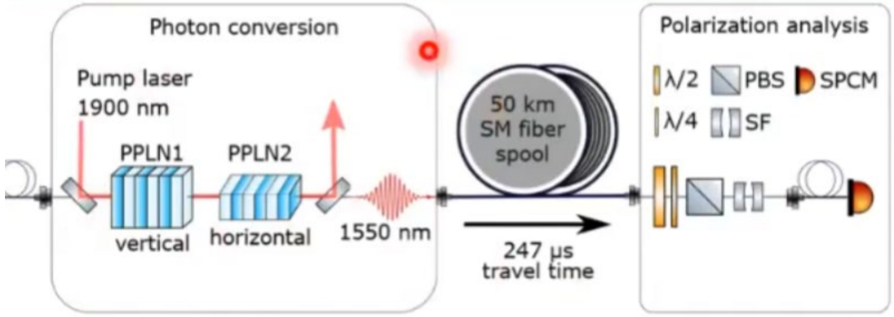
Challenge: heating and surface charges!

M. Teller et al. PRL 126, 230505 (2021)

https://www.youtube.com/watch?v=PIiWZ8RvmkE

Frequency conversion





Transmission >>100m in optical fiber **requires conversion to telecom wavelengths** (1.3um/1.55um) otherwise losses are too high

V. Krutyanskiy et al. NPJ Quantum Inf 5 72 (2019)

Experimental Progress

Interfacing to neutral atoms

References:
Cavity-based quantum networks with single atoms and optical photons, Rev. Mod. Phys. 87, 1379 (2015)
HJ Kimble, The quantum internet. Nature 453, 1023–1030 (2008).
Quantum networks with neutral atom processing nodes, npj Quantum Information **9**, 90 (2023)
Quantum matter built from nanoscopic lattices of atoms and photons, Rev. Mod. Phys. 90, 031002 (2018)

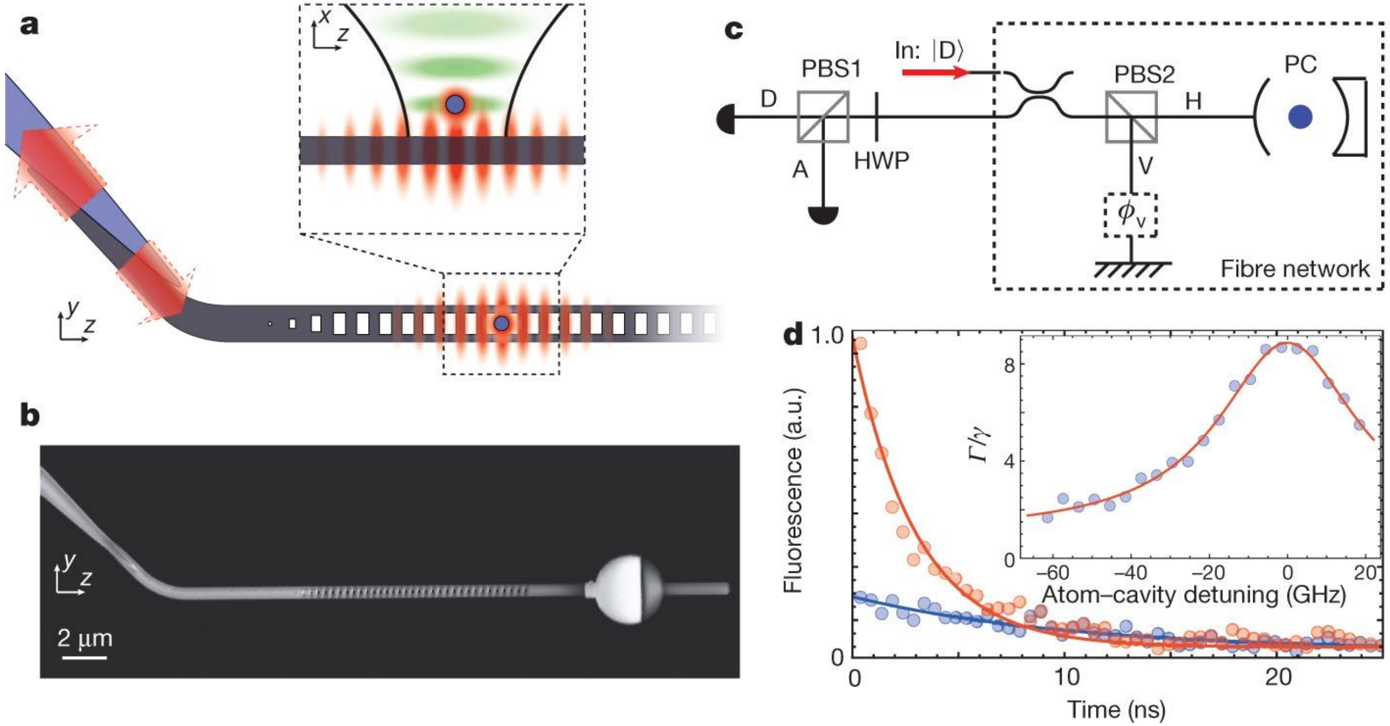
Quantum interfaces for neutral atoms

Nanophotonic quantum phase switch with a single atom

T. G. Tiecke, J. D. Thompson, N. P. de Leon, L. R. Liu, V. Vuletić & M. D. Lukin

Nature **508**, 241–244 (2014) | Cite this article

First atomic state - photonic cavity controlled phase gate interface



- Prepare atom in superposition $|u\rangle+|c\rangle$
- Reflect one single photon
- Atomic superposition is shifted by nearly π
- Verified via Ramsey interference

Cooperativity C=7.7

- A single atom can flip the phase of a photon by (1.1)π

Quantum interfaces for neutral atoms

Entanglement transport and a nanophotonic interface for atoms in optical tweezers

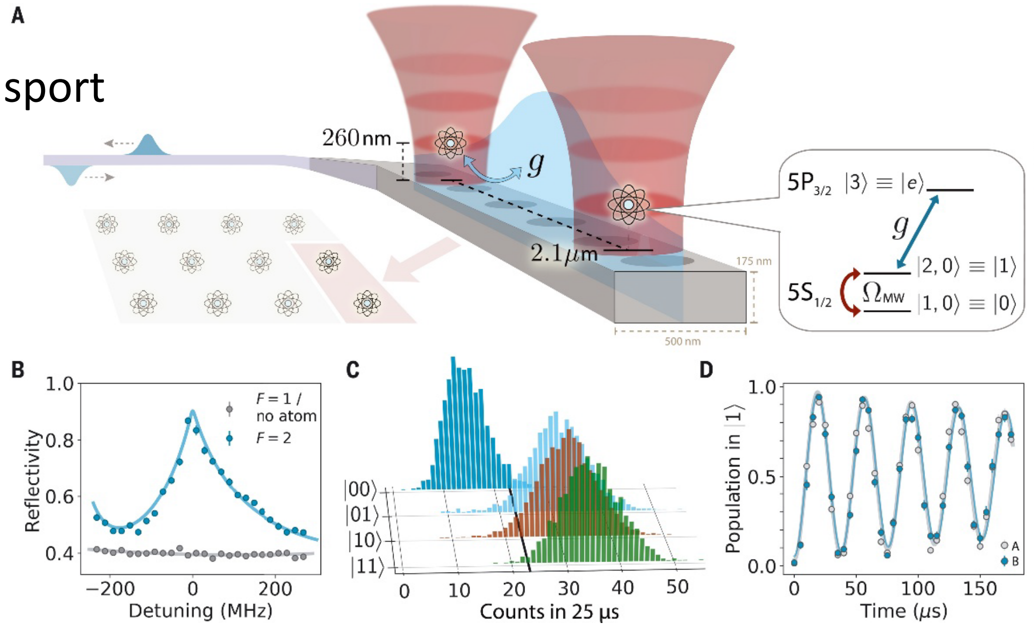
TAMARA ĐORĐEVIĆ, POLNOP SAMUTPRAPHOOT, PALOMA L. OCOLA, HANNES BERNIEN, BRANDON GRINKEMEYER, IVANA DIMITROVA, VLADAN VULETIĆ, AND MIKHAIL D. LUKIN Authors Info & Affiliations

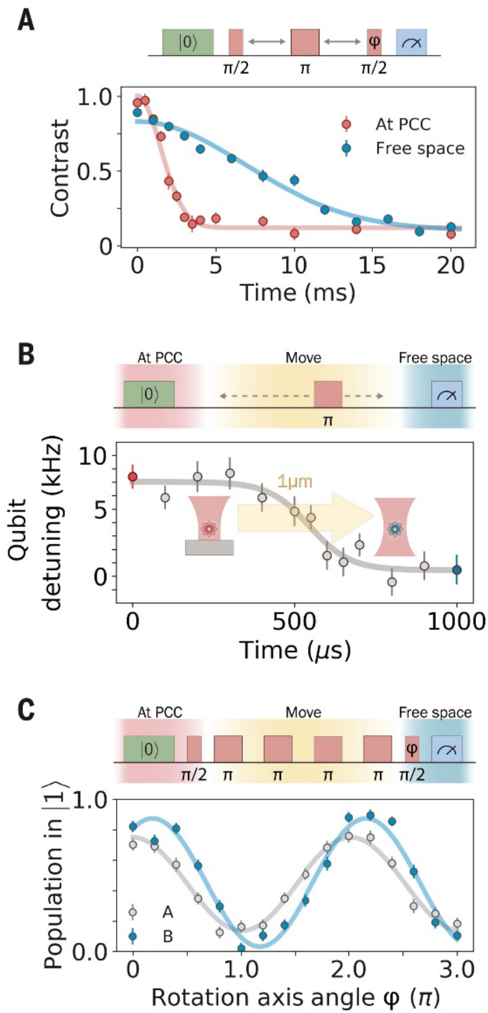
SCIENCE • 13 Aug 2021 • Vol 373, Issue 6562 • pp. 1511-1514 • DOI: 10.1126/science.abi9917

First demonstration of entanglement preservation after transporting atoms away from cavity into free space

Cooperativity C=27

- 15% loss probability for each atom during transport
- Fidelity 72% near cavity
- Fidelity 65% after 1um transport
- Much longer lifetime away from cavity





Quantum interfaces for neutral atoms

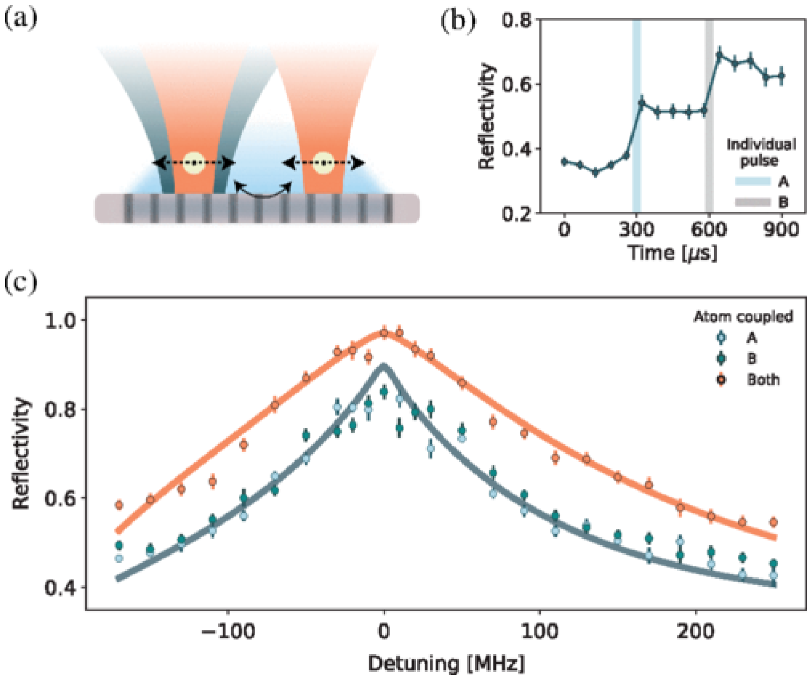
Strong Coupling of Two Individually Controlled Atoms via a Nanophotonic Cavity

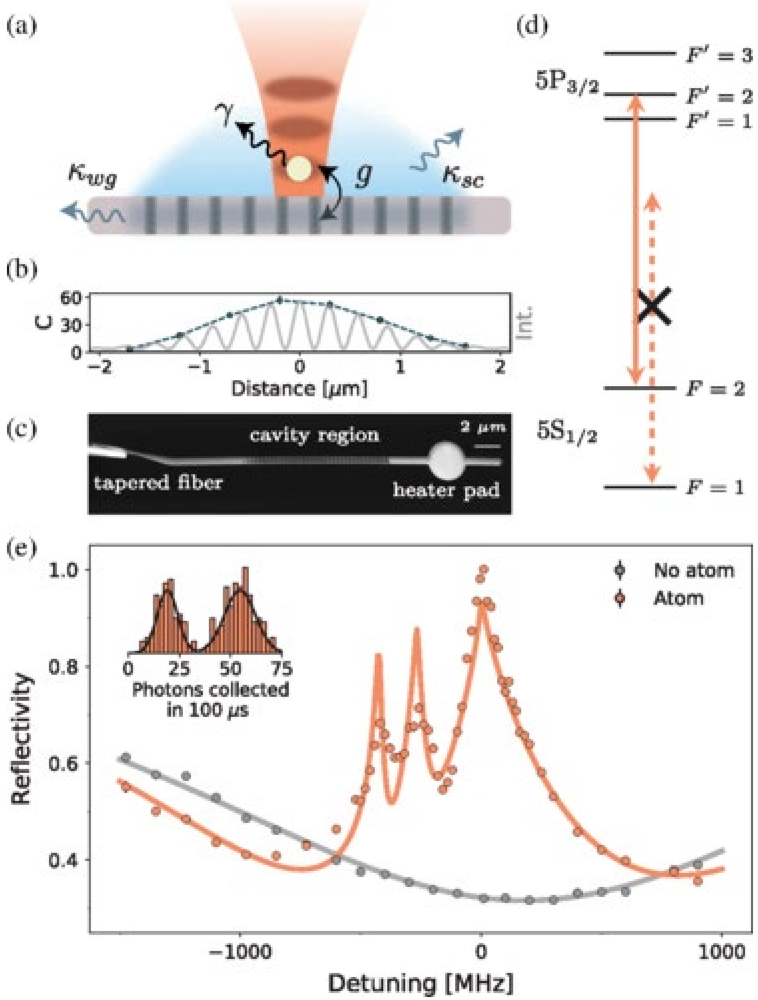
Polnop Samutpraphoot[1,*], Tamara Đorđević[1,*], Paloma L. Ocola[1,*], Hannes Bernien[2], Crystal Senko[3,4], Vladan Vuletić[5], and Mikhail D. Lukin[1,†]

Show more

Phys. Rev. Lett. **124**, 063602 – **Published 11 February, 2020**

Long range cavity mediated interactions between 2 atoms in PCC





Cooperativity C=71

Single photon Rabi frequency 1.24GHz!

Quantum interfaces for neutral atoms

Atom–atom interactions around the band edge of a photonic crystal waveguide

Jonathan D. Hood, Akihisa Goban, Ana Asenjo-Garcia, +3 , and H. J. Kimble Authors Info & Affiliations

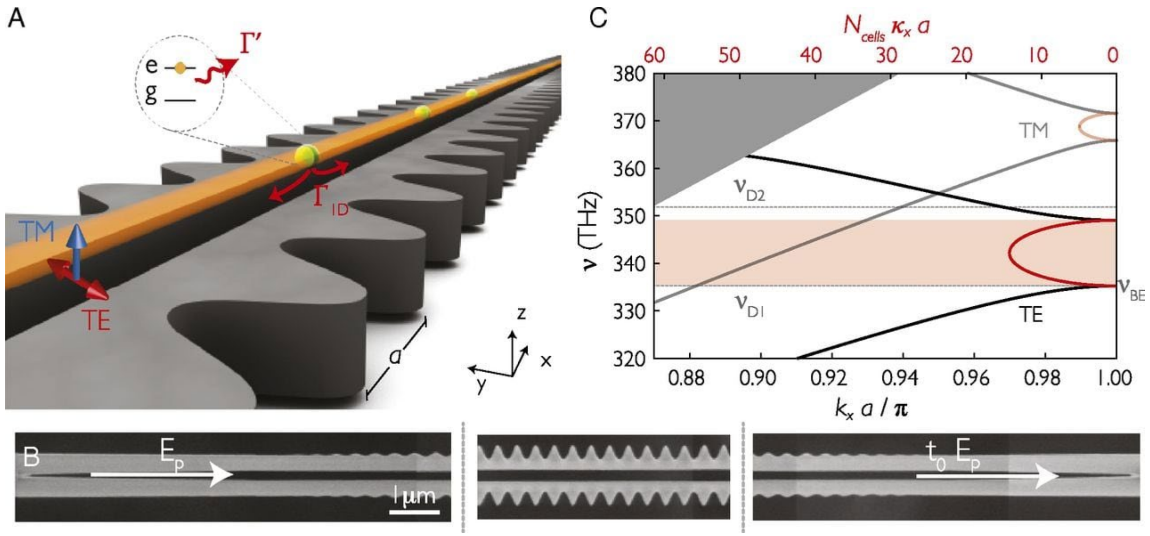
Contributed by H. Jeffrey Kimble, June 17, 2016 (sent for review March 7, 2016; reviewed by Eugene Polzik and Dan Stamper-Kurn)

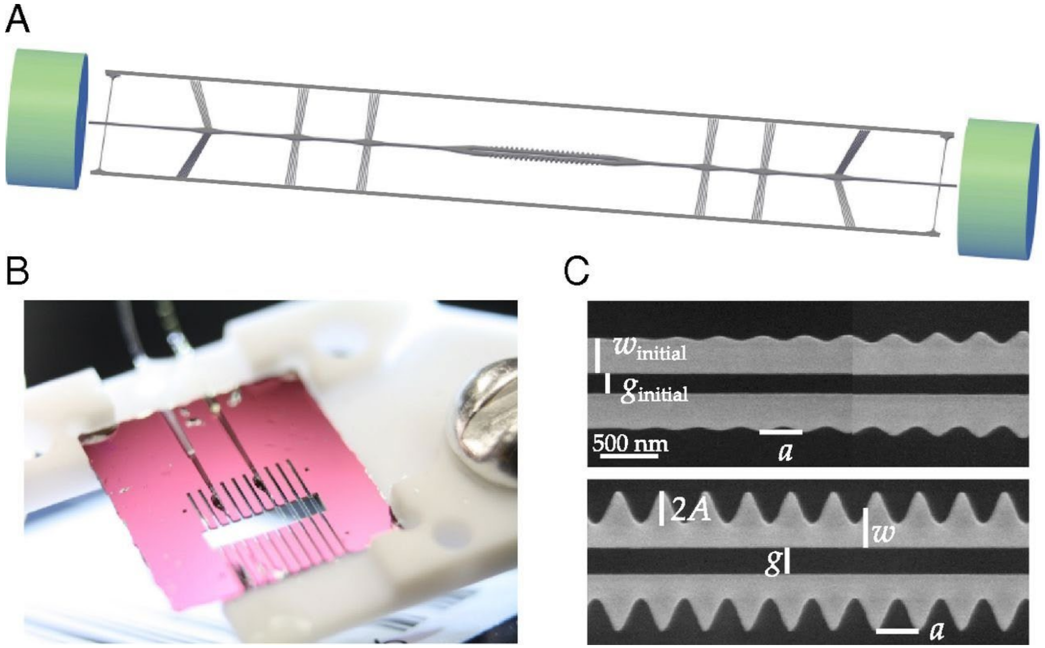
August 31, 2016 | 113 (38) 10507-10512 | https://doi.org/10.1073/pnas.1603788113

Long range strong coupling in a waveguide (NOT cavity!!)

Famous “alligator waveguide”

$\beta = 0.3$



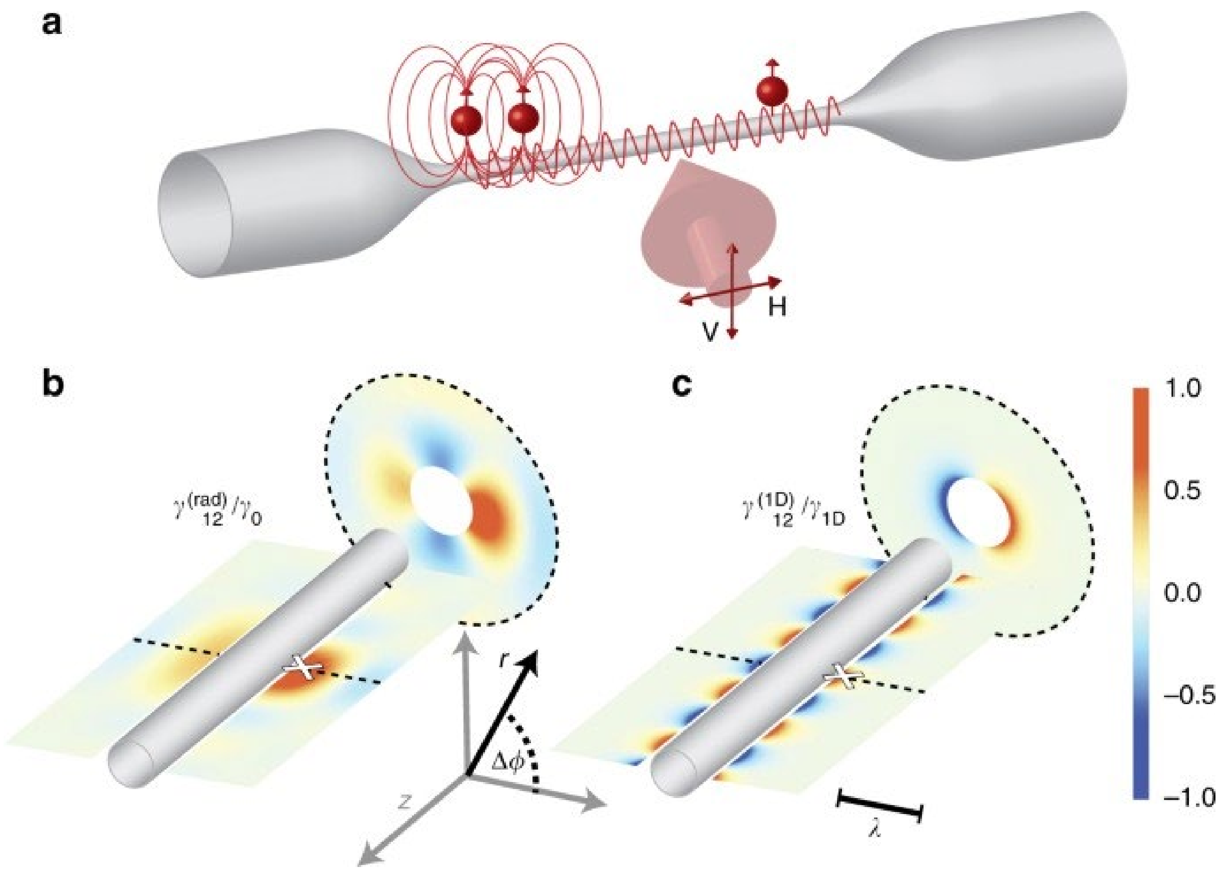


Quantum interfaces for neutral atoms

Super-radiance reveals infinite-range dipole interactions through a nanofiber

P. Solano, P. Barberis-Blostein, F. K. Fatemi, L. A. Orozco & S. L. Rolston

Nature Communications **8**, Article number: 1857 (2017) | Cite this article



See also:

Ultra-low-loss nanofiber Fabry–Perot cavities optimized for cavity quantum electrodynamics, Optics Letters Vol. 45, Issue 17, pp. 4875-4878 (2020)

High-finesse nanofiber Fabry–Pérot resonator in a portable storage container, Rev. Sci. Instrum. 95, 073103 (2024)

Quantum interfaces for neutral atoms

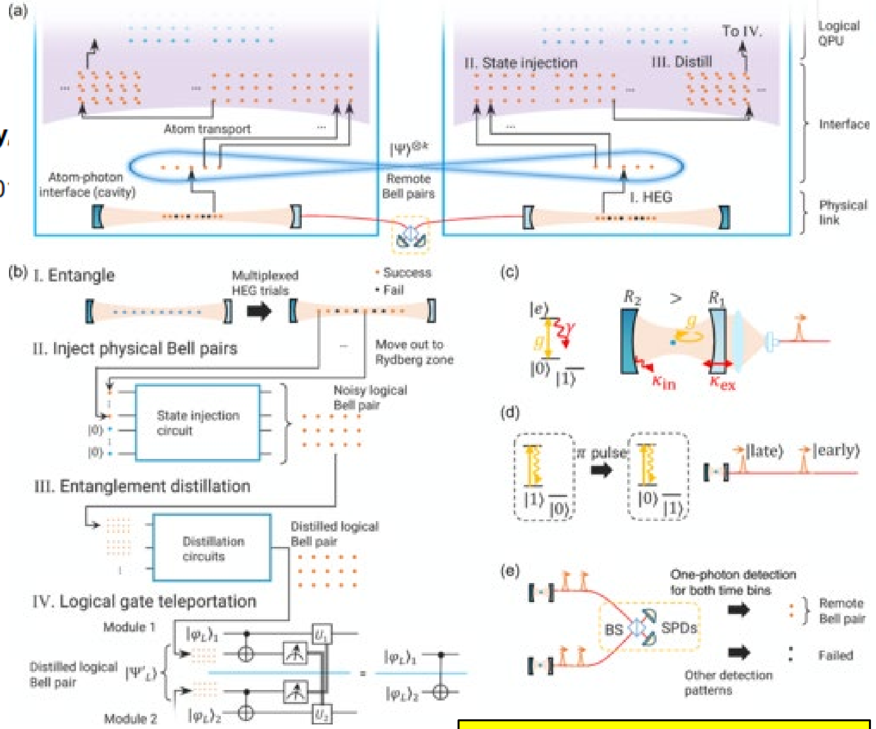
Scalable Networking of Neutral-Atom Qubits: Nanofiber-Based Approach for Multiprocessor Fault-Tolerant Quantum Computers

Shinichi Sunami[1,2,*], Shiro Tamiya[1], Ryotaro Inoue[1], Hayata Yamasaki[1,3,†], and Akihisa Goban[1,‡]

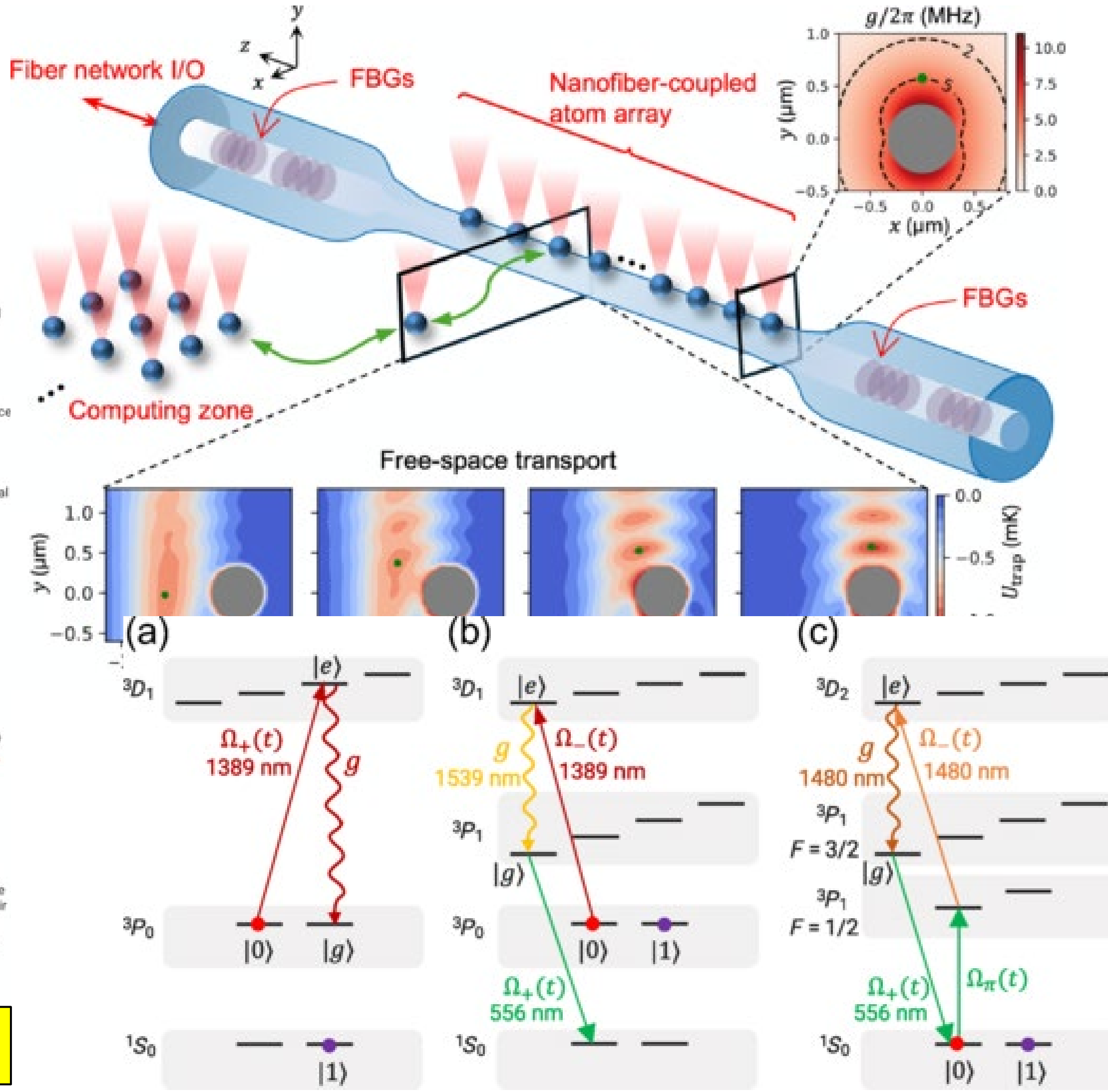
Show more

PRX Quantum **6**, 010101 – **Published 4 February**

DOI: https://doi.org/10.1103/PRXQuantum.6.010



Cooperativity ~100



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Ultra-low-loss nanofiber Fabry–Perot cavities optimized for cavity quantum electrodynamics, Optics Letters Vol. 45, Issue 17, pp. 4875-4878 (2020)

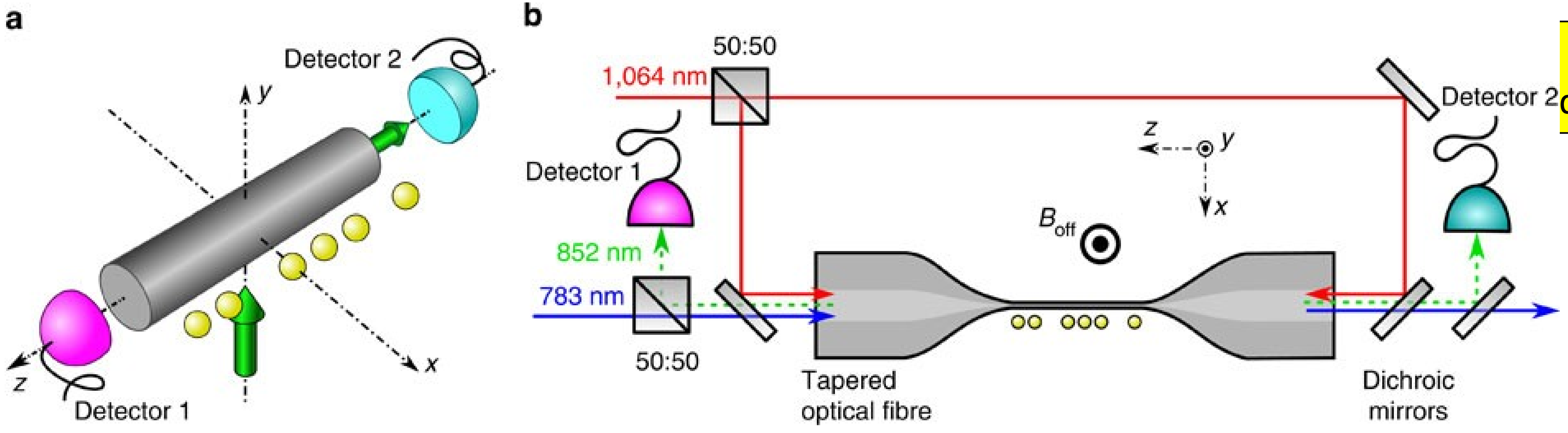
High-finesse nanofiber Fabry–Pérot resonator in a portable storage container, Rev. Sci. Instrum. 95, 073103 (2024)

Quantum interfaces for neutral atoms

Quantum state-controlled directional spontaneous emission of photons into a nanophotonic waveguide

R. Mitsch, C. Sayrin, B. Albrecht, P. Schneeweiss & A. Rauschenbeutel

Nature Communications **5**, Article number: 5713 (2014) | Cite this article



strong coupling in a
de (NOT cavity!!)

Advances In Atomic, Molecular, and Optical Physics **66**, 439, (2017)

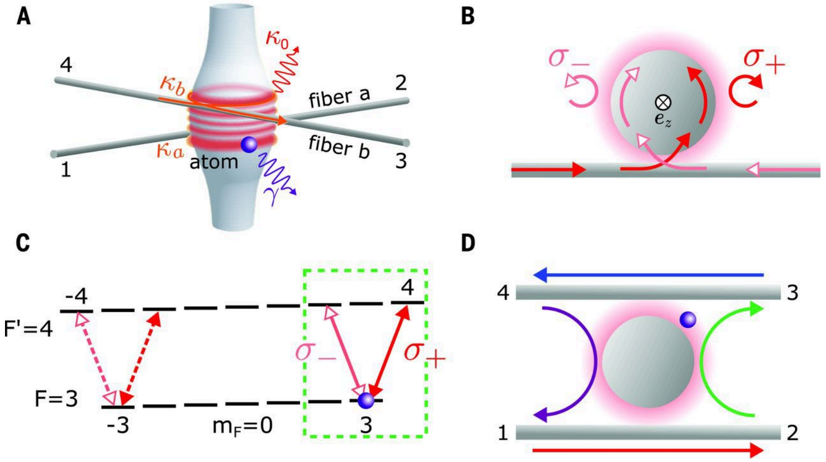
Quantum interfaces for neutral atoms

Quantum optical circulator controlled by a single chi-rally coupled atom

MICHAEL SCHEUCHER, ADÈLE HILICO, ELISA WILL, JÜRGEN VOLZ, AND ARNO RAUSCHENBEUTEL Authors Info & Affiliations

SCIENCE • 8 Dec 2016 • Vol 354, Issue 6319 • pp. 1577-1580 • DOI: 10.1126/science.aaj2118

Long range strong coupling in a waveguide (NOT cavity!!)



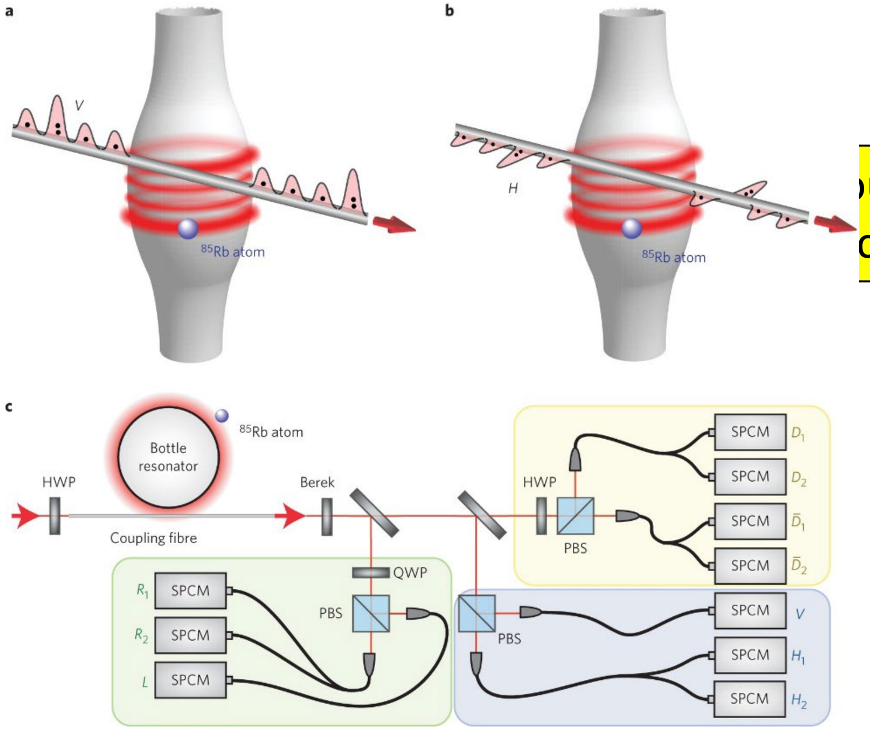
Advances In Atomic, Molecular, and Optical Physics **66**, 439, (2017)

Quantum interfaces for neutral atoms

Nonlinear π phase shift for single fibre-guided photons interacting with a single resonator-enhanced atom

Jürgen Volz, Michael Scheucher, Christian Junge & Arno Rauschenbeutel

Nature Photonics **8**, 965–970 (2014) | Cite this article



upling in a
cavity!!)

Advances In Atomic, Molecular, and Optical Physics **66**, 439, (2017)

Quantum interfaces for neutral atoms

Error-detected quantum operations with neutral atoms mediated by an optical cavity

BRANDON GRINKEMEYER, ELMER GUARDADO-SANCHEZ, IVANA DIMITROVA, DANILO SHCHEPANOVICH, G. EIRINI MANDOPOULOU, JOHANNES BORREGAARD, VLADAN VULETIĆ, AND MIKHAIL D. LUKIN Authors Info & Affiliations

SCIENCE • 20 Mar 2025 • Vol 387, Issue 6740 • pp. 1301-1305 • DOI: 10.1126/science.adr7075

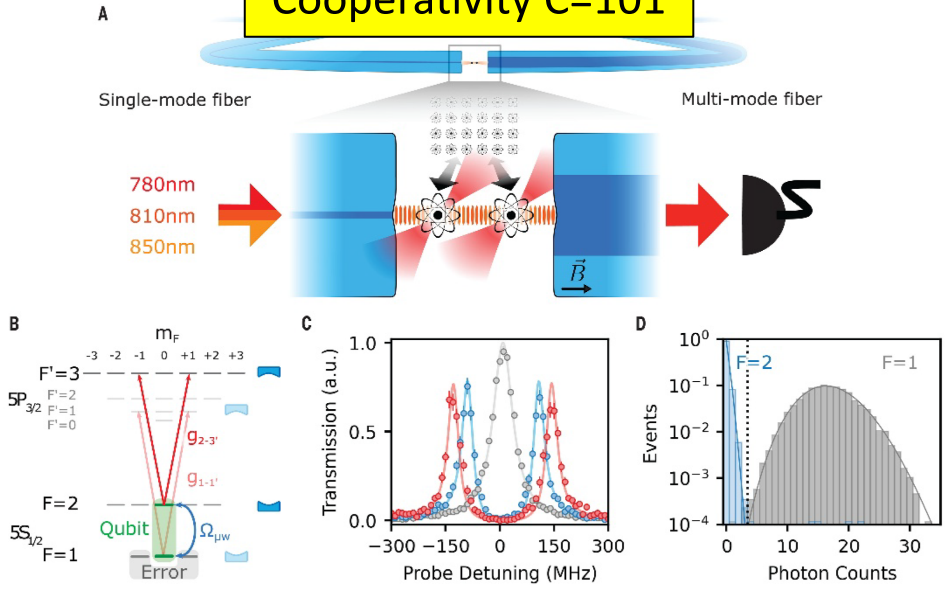
Demonstrates high fidelity operations between 2 atoms in a cavity with built in error detection

Entanglement generation
- Fidelity = 91%
- 32% deterministic success (no heralding!)

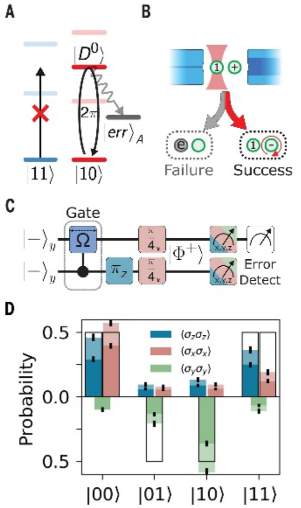
Deterministic controlled phase gates
- Deterministic fidelity 52%
- Error detected fidelity 76%

High fidelity readout
- 99.96% readout fidelity in 10us
- Just 0.034% loss

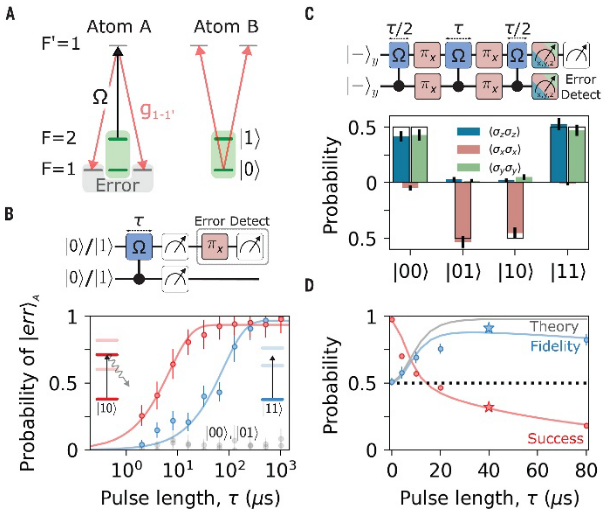
Cooperativity C=101



Deterministic phase gate:



Entanglement:



Quantum interfaces for neutral atoms

Heralded Entanglement Between Widely Separated Atoms

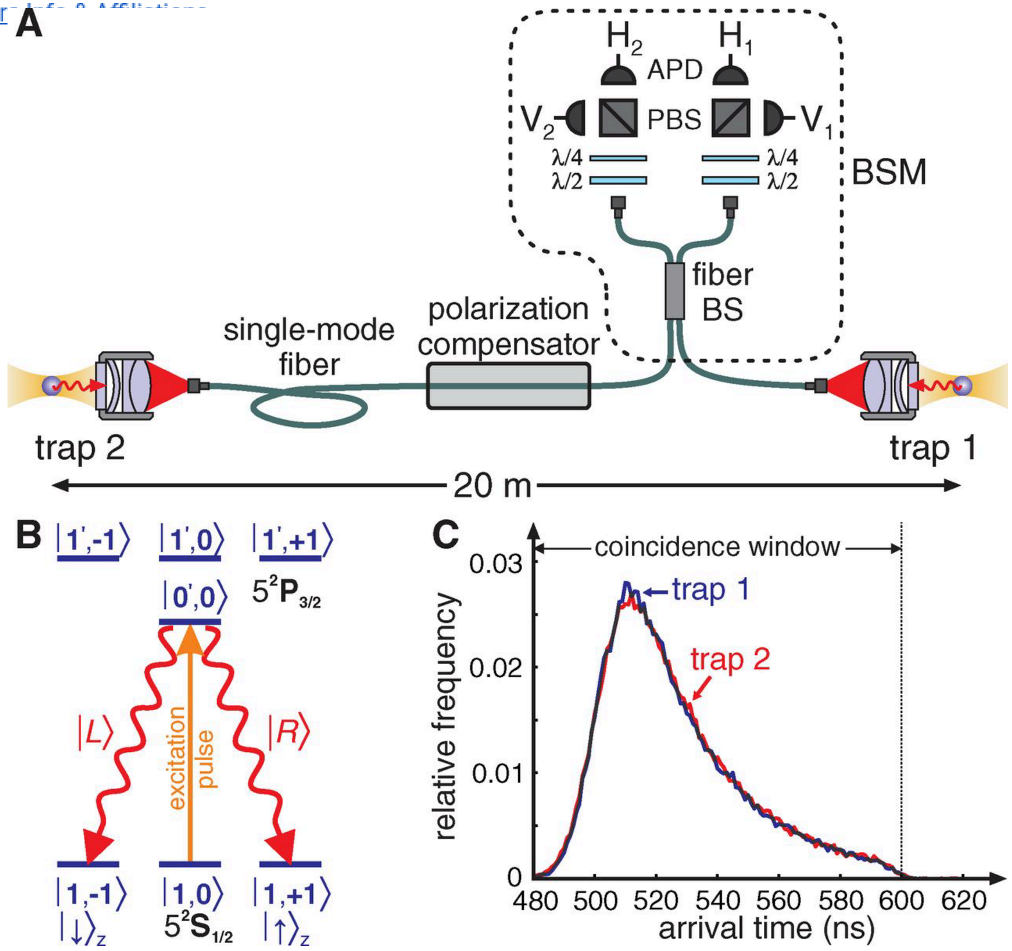
JULIAN HOFMANN, MICHAEL KRUG, NORBERT ORTEGEL, LEA GÉRARD, MARKUS WEBER, WENJAMIN ROSENFELD, AND HARALD WEINFURTER Author

SCIENCE • 6 Jul 2012 • Vol 337, Issue 6090 • pp. 72-75 • DOI: 10.1126/science.1221856

First high-fidelity, heralded entanglement between two individually trapped remote atoms

Performance:

- Detection efficiency 1e-3 (no cavity, low solid angle)
- Operate at 50kHz excitation rate
- 4.4e-7 Heralded success probability (2 photons of different polarizations)
- 1 entanglement generated every 100 seconds



Quantum interfaces for neutral atoms

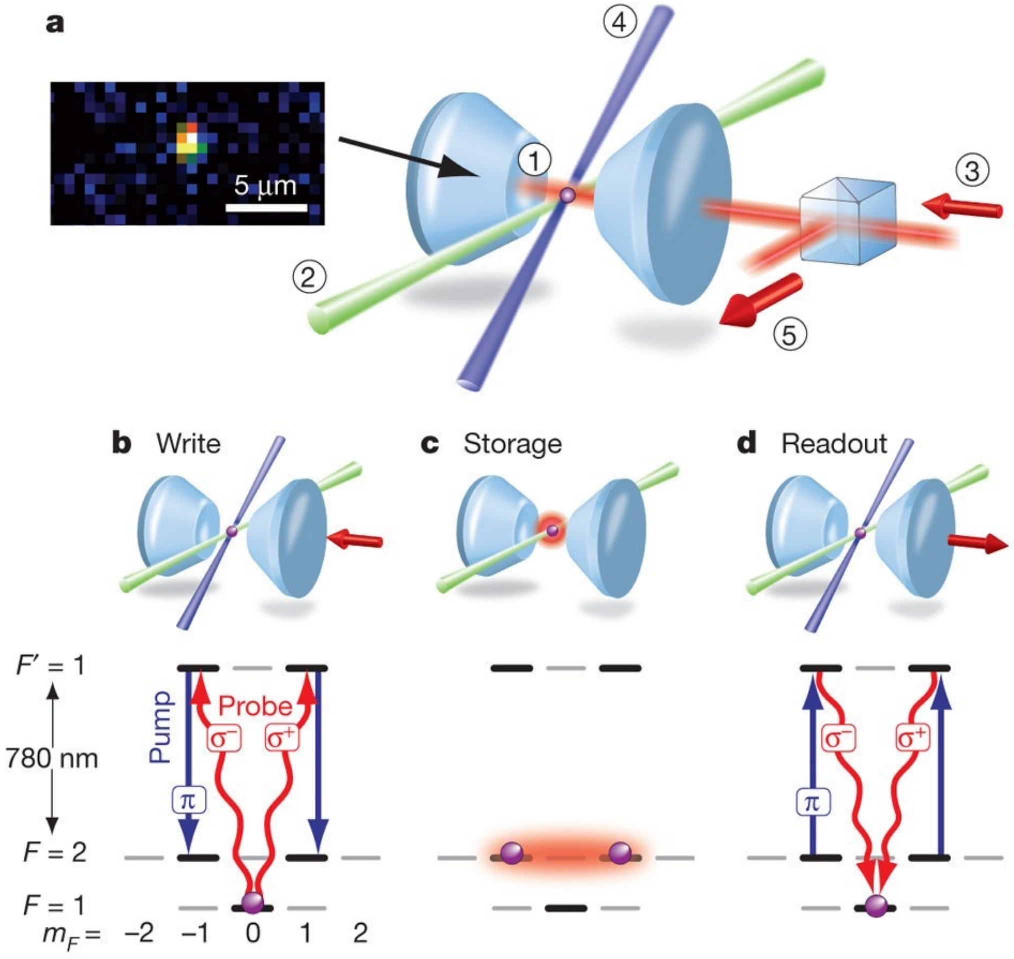
A single-atom quantum memory

Holger P. Specht, Christian Nölleke, Andreas Reiserer, Manuel Uphoff, Eden Figueroa, Stephan Ritter & Gerhard Rempe

Nature **473**, 190–193 (2011) | Cite this article

First single atom quantum memory

Deterministic storage, not heralded



Performance:

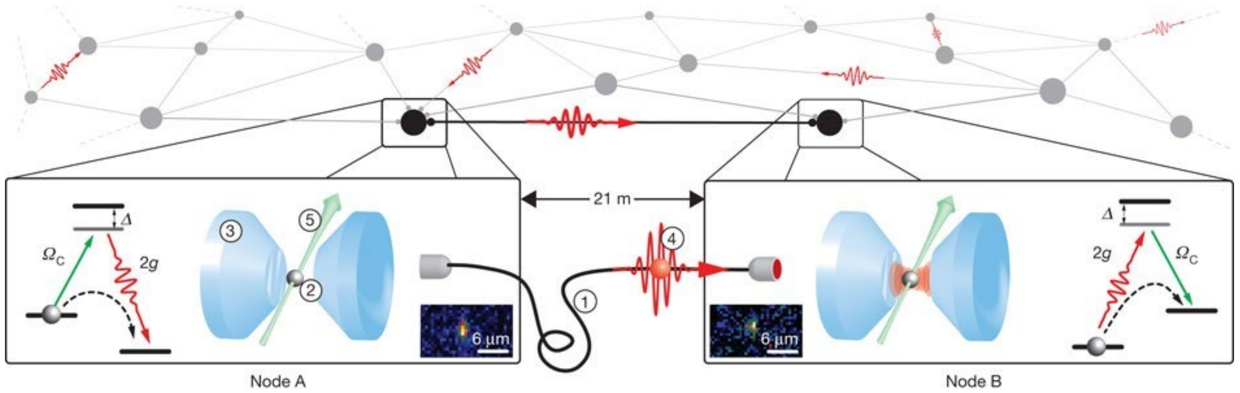
- 92.7% fidelity
- 184us storage time
- Uses control beam to store photon to atomic state
- 56% retrieval efficiency
- Storage+retrieval efficiency 9.3%

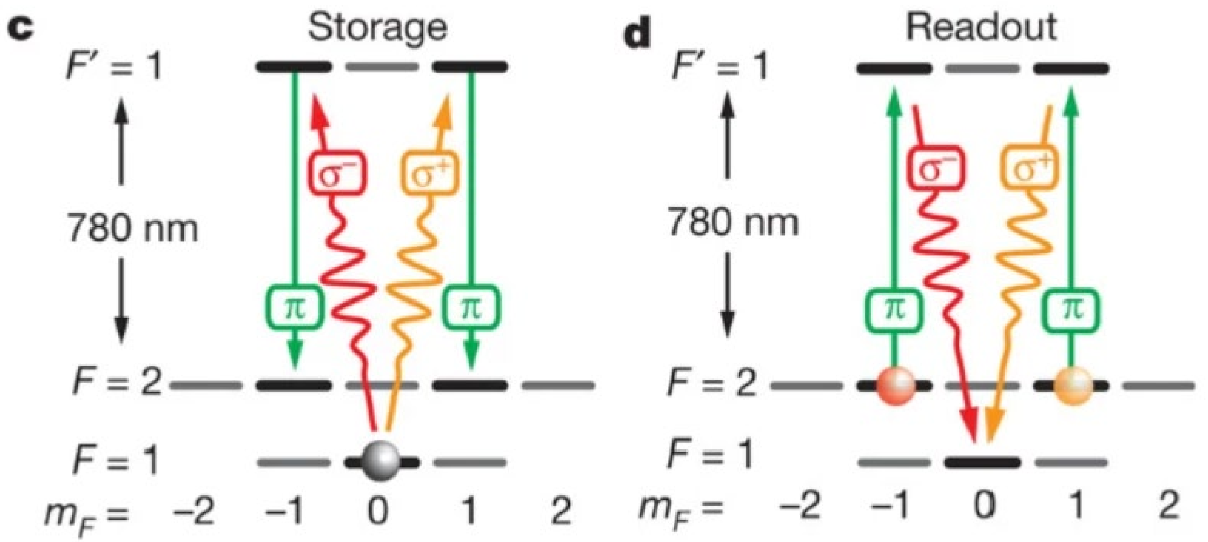
Quantum interfaces for neutral atoms

An elementary quantum network of single atoms in optical cavities

Stephan Ritter, Christian Nölleke, Carolin Hahn, Andreas Reiserer, Andreas Neuzner, Manuel Uphoff, Martin Mücke, Eden Figueroa, Joerg Bochmann & Gerhard Rempe

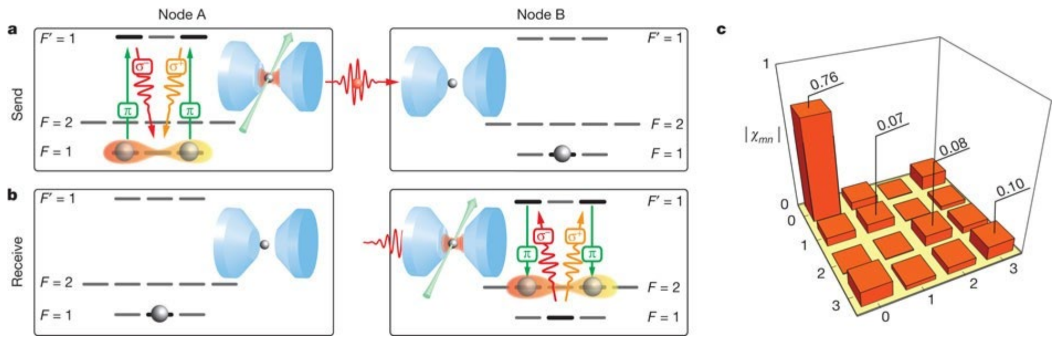
Nature **484**, 195–200 (2012) | Cite this article





First quantum network

- Able to send, receive and store quantum information between two remote nodes
- 60% photon creation efficiency
- 17% photon storage efficiency (STIRAP)
- State transfer fidelity 84%



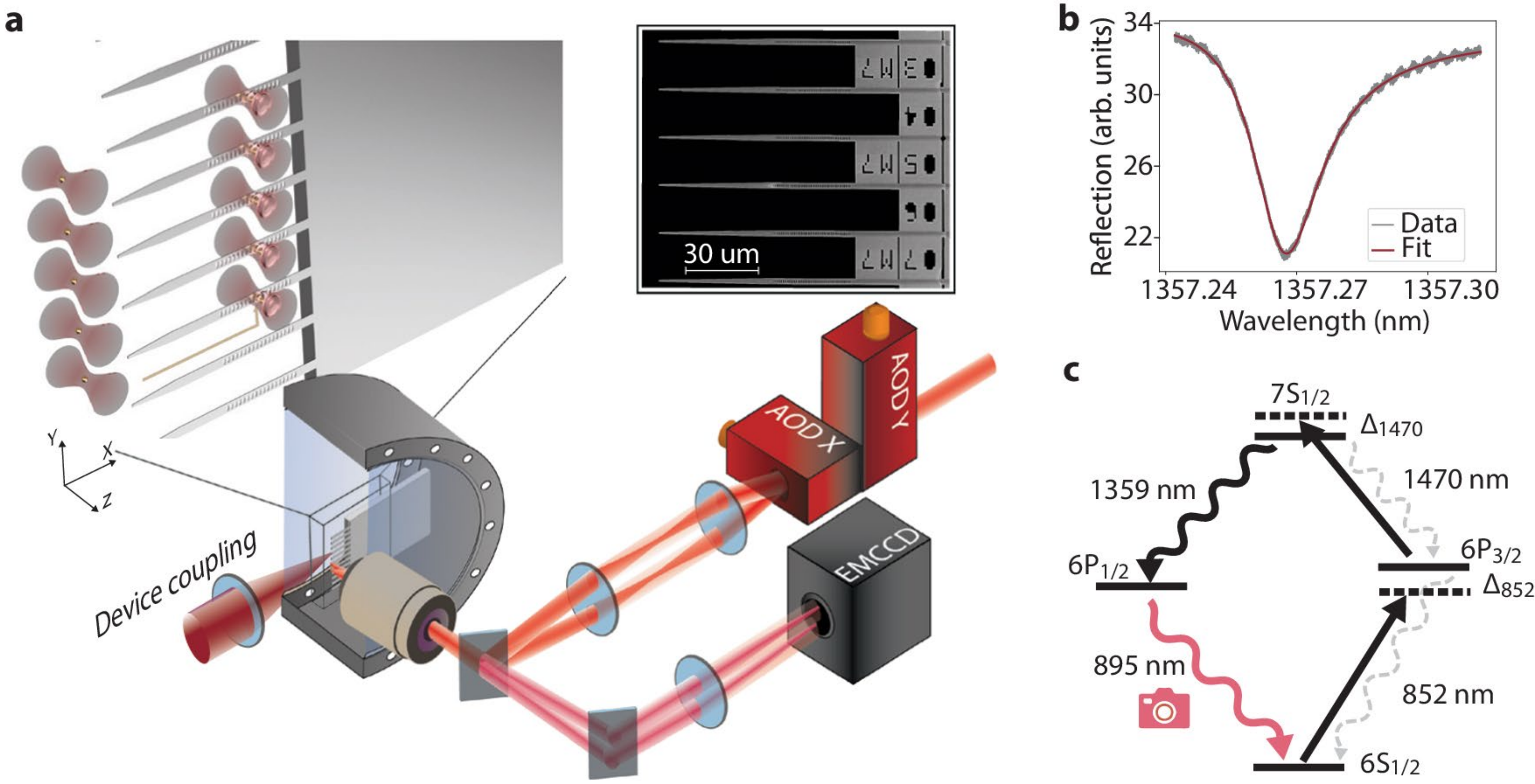
Cooperativity = 1.4

Quantum interfaces for neutral atoms

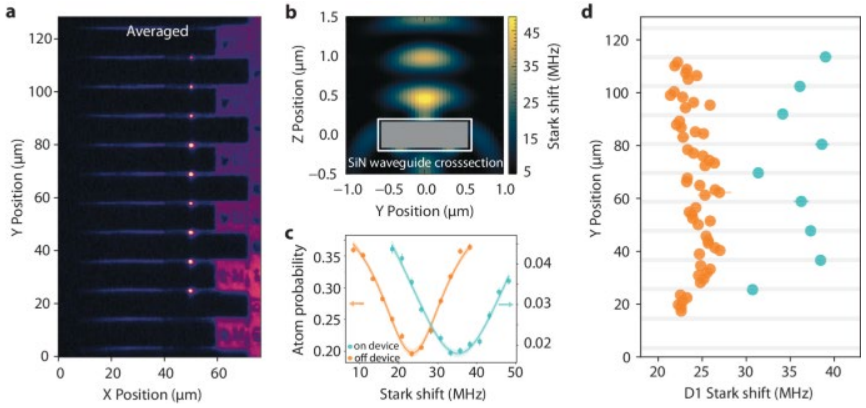
An integrated atom array-nanophotonic chip platform with background-free imaging

Shankar G. Menon, Noah Glachman, Matteo Pompili, Alan Dibos & Hannes Bernien

Nature Communications 15, Article number: 6156 (2024) | Cite this article



Cavity arrays (proof-of-principle)



- Demonstrated scalable approach able to make hundreds of cavities
- 29% single atom loading probability

Estimated cooperativity C~50

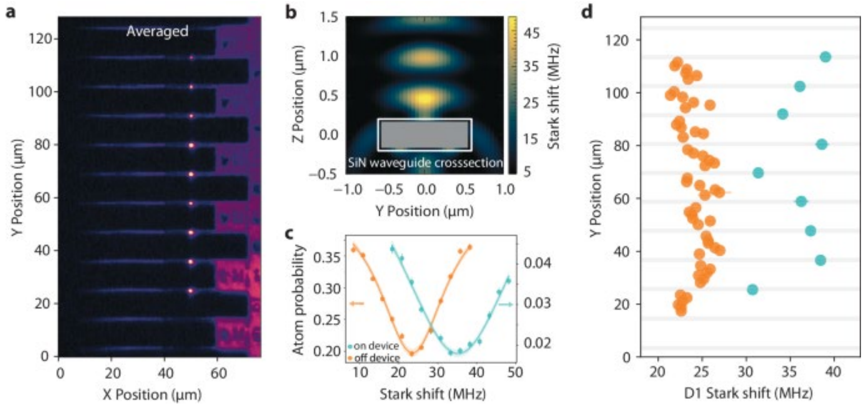
Quantum interfaces for neutral atoms

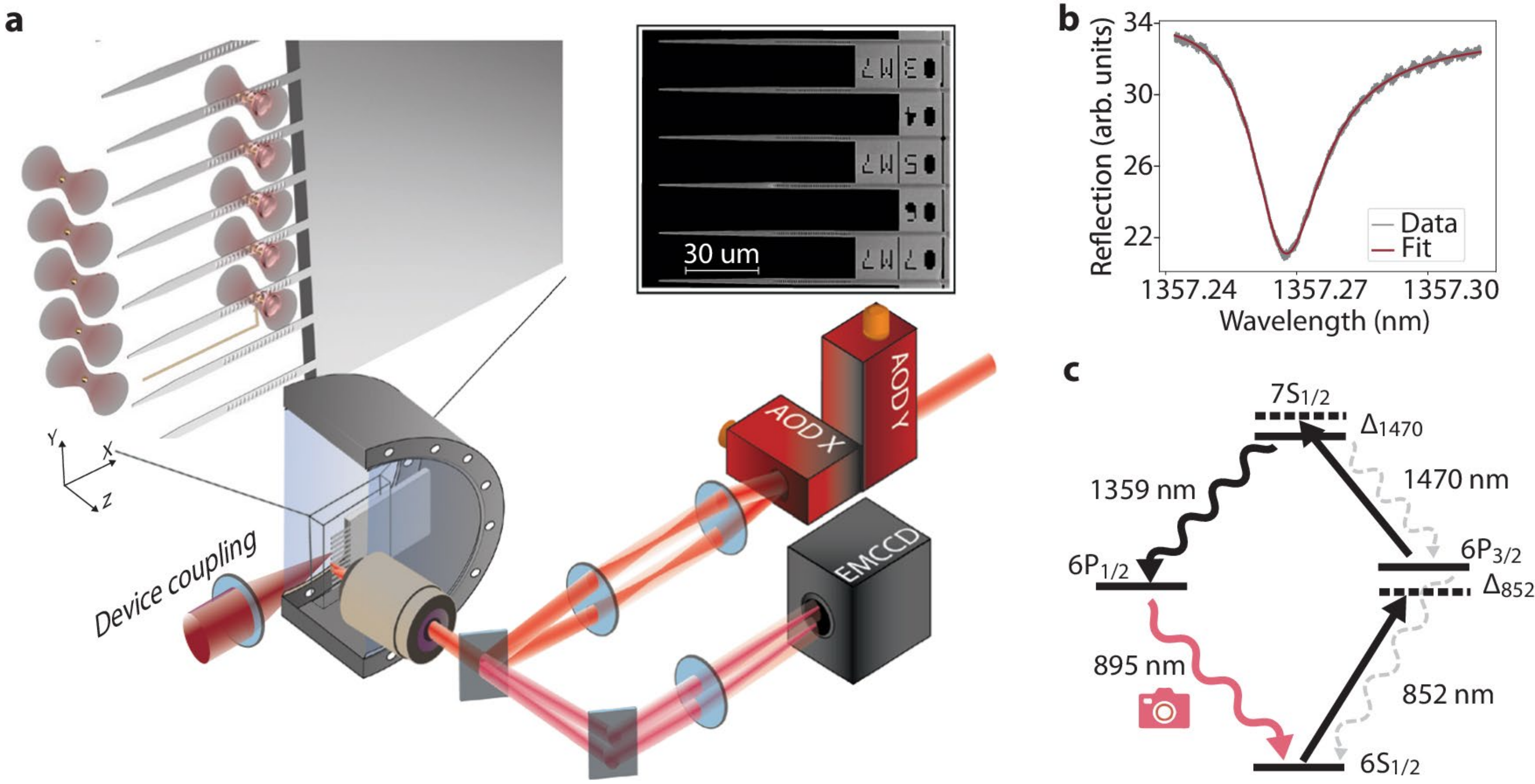
An integrated atom array-nanophotonic chip platform with background-free imaging

Shankar G. Menon, Noah Glachman, Matteo Pompili, Alan Dibos & Hannes Bernien

Nature Communications **15**, Article number: 6156 (2024) | Cite this article

Cavity arrays (proof-of-principle)c





- Demonstrated scalable approach able to make hundreds of cavities
- 29% single atom loading probability

Estimated cooperativity C~50

Quantum interfaces for neutral atoms

A quantum gate between a flying optical photon and a single trapped atom

Andreas Reiserer, Norbert Kalb, Gerhard Rempe & Stephan Ritter

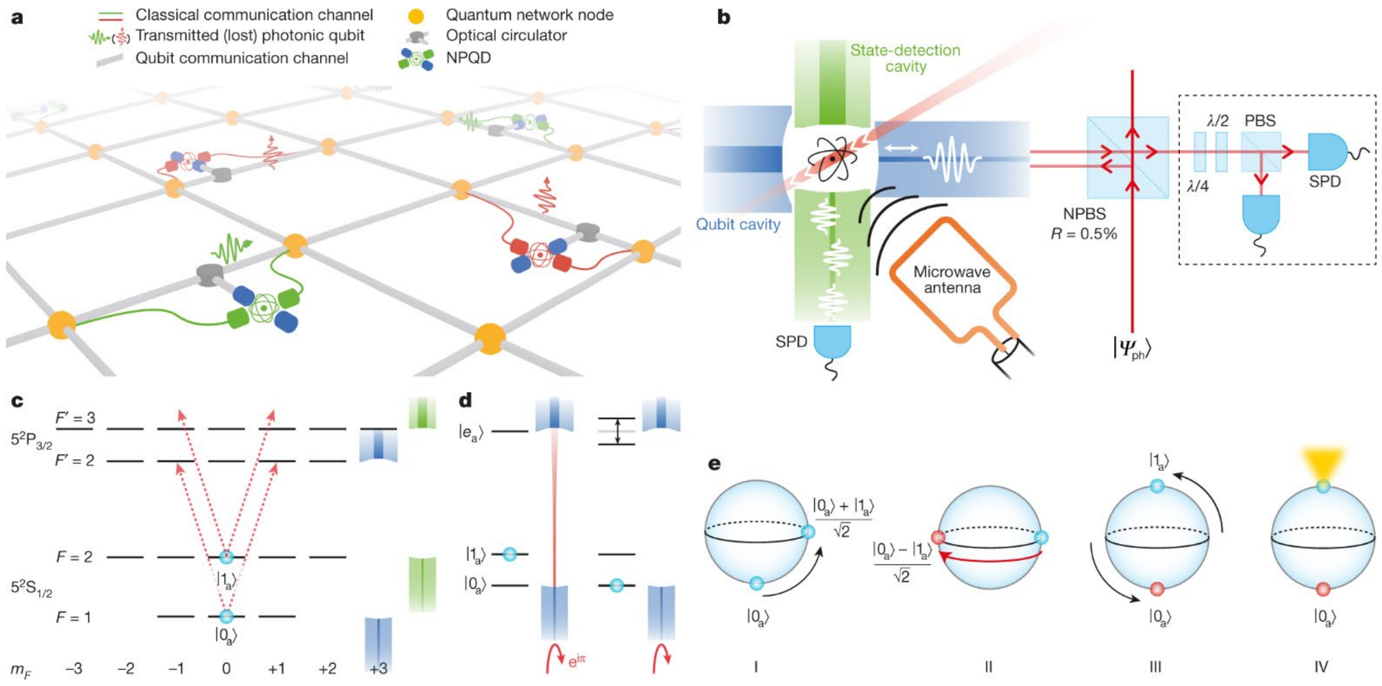
Nature **508**, 237–240 (2014) | Cite this article

Quantum interfaces for neutral atoms

Nondestructive detection of photonic qubits

Dominik Niemietz, Pau Farrera, Stefan Langenfeld & Gerhard Rempe

Nature **591**, 570–574 (2021) | Cite this article

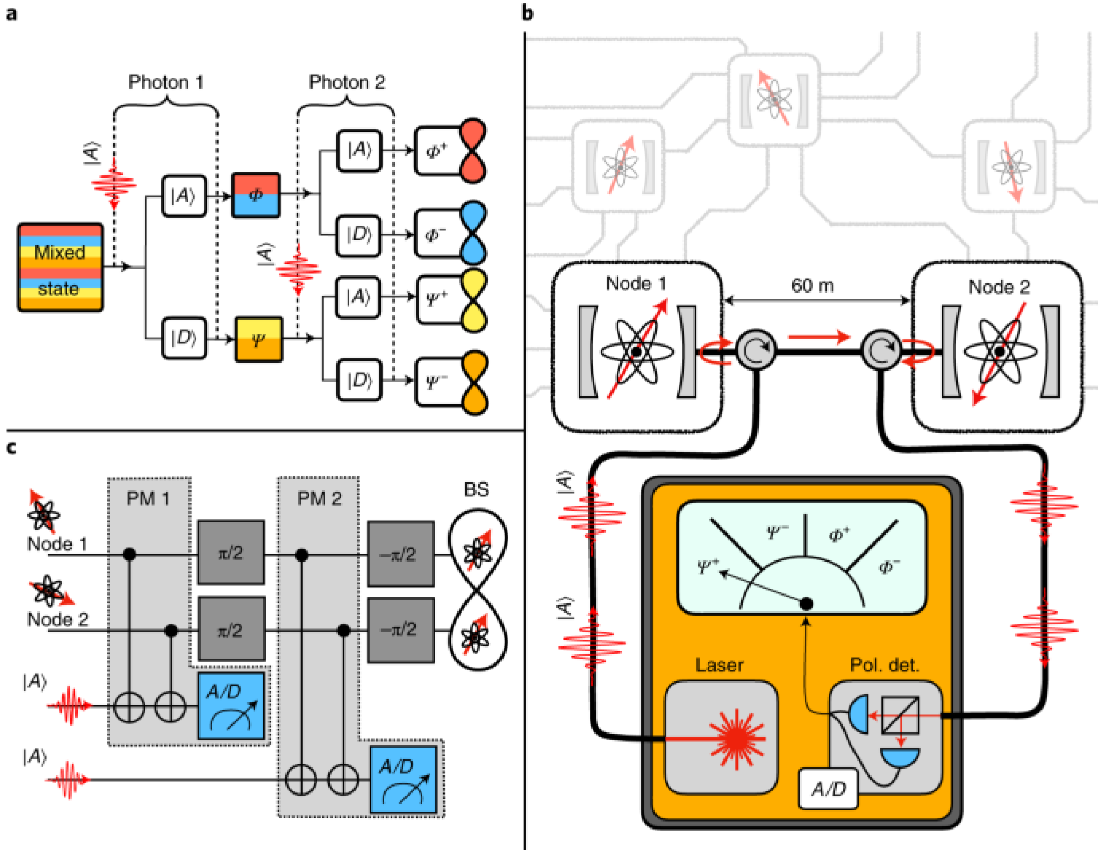


Quantum interfaces for neutral atoms

A nondestructive Bell-state measurement on two distant atomic qubits

Stephan Welte, Philip Thomas, Lukas Hartung, Severin Daiss, Stefan Langenfeld, Olivier Morin, Gerhard Rempe & Emanuele Distante

Nature Photonics **15**, 504–509 (2021) | Cite this article



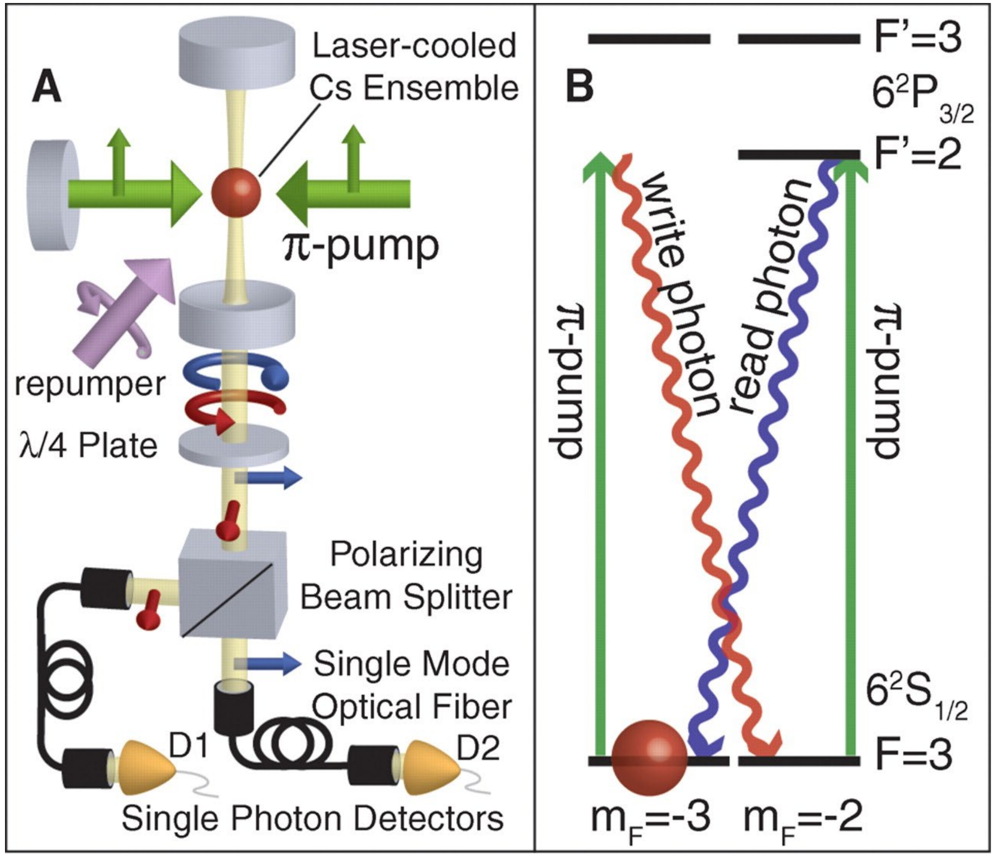
Quantum interfaces for neutral atoms

A High-Brightness Source of Narrowband, Identical-Photon Pairs

JAMES K. THOMPSON, JONATHAN SIMON, HUANQIAN LOH, AND VLADAN VULETIĆ Authors Info & Affiliations

SCIENCE • 7 Jul 2006 • Vol 313, Issue 5783 • pp. 74-77 • DOI: 10.1126/science.1127676

Cooperativity = 1.4



Experimental Progress

Interfacing to NV centers and quantum dots

References:

P Lodahl, S Mahmoodian, S Stobbe, Interfacing single photons and single quantum dots with photonic nanostructures. *Rev Mod Phys* **87**, 347 (2015).

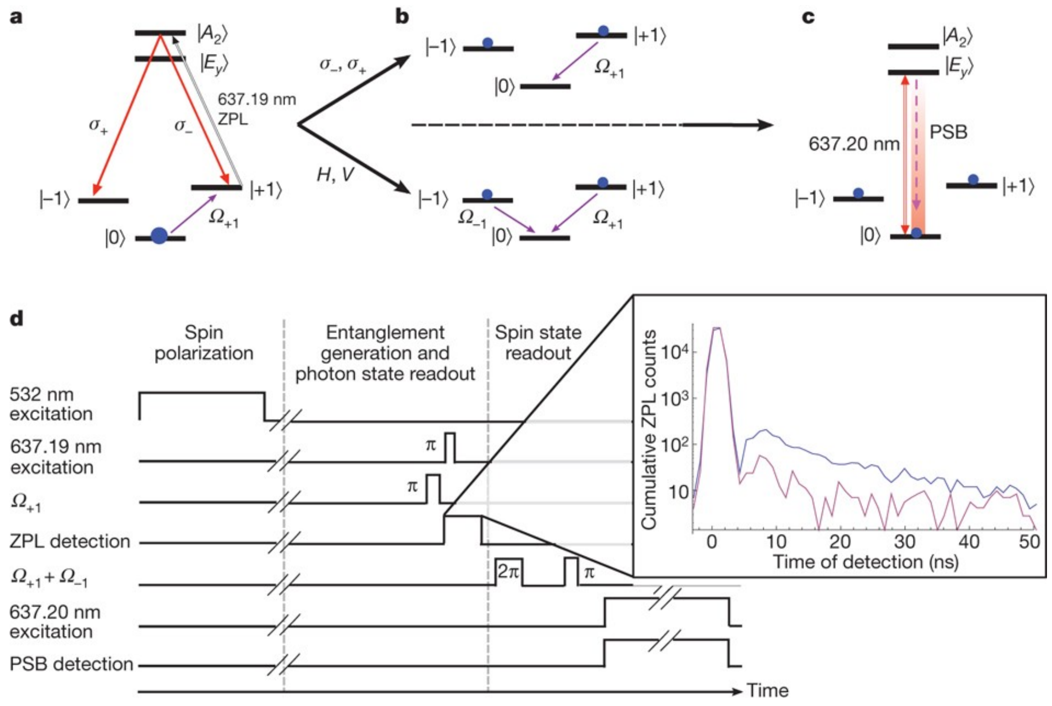
Quantum interfaces for NV centers

Quantum entanglement between an optical photon and a solid-state spin qubit

E. Togan, Y. Chu, A. S. Trifonov, L. Jiang, J. Maze, L. Childress, M. V. G. Dutt, A. S. Sørensen, P. R. Hemmer, A. S. Zibrov & M. D. Lukin

Nature 466, 730–734 (2010) | Cite this article

First entanglement between a NV center and a photon



1. Prepare the NV in |0> then flip to |+1> with a microwave π pulse
2. Excite to $|A_2>$ with optical π pulse
3. If it decays to |-1> a σ- photon is emitted, if it decays to |+1> a σ+ photon is emitted.

$$\frac{1}{\sqrt{2}}\left(|\sigma^+\rangle|-1\rangle + |\sigma^-\rangle|+1\rangle\right)$$

4. Collect photon and send to a PBS optionally with or without waveplate rotation to select measurement basis (H/V vs σ+/σ- measurement basis).
5. Apply microwave pulse to map the measured state to |0>
6. Confirm NV is in |0> by measuring fluorescence on cycling $|0>-|E_y>$

- Per event success probability 1e-6
- 100kHz repetition rate
- Entanglement every 10 seconds

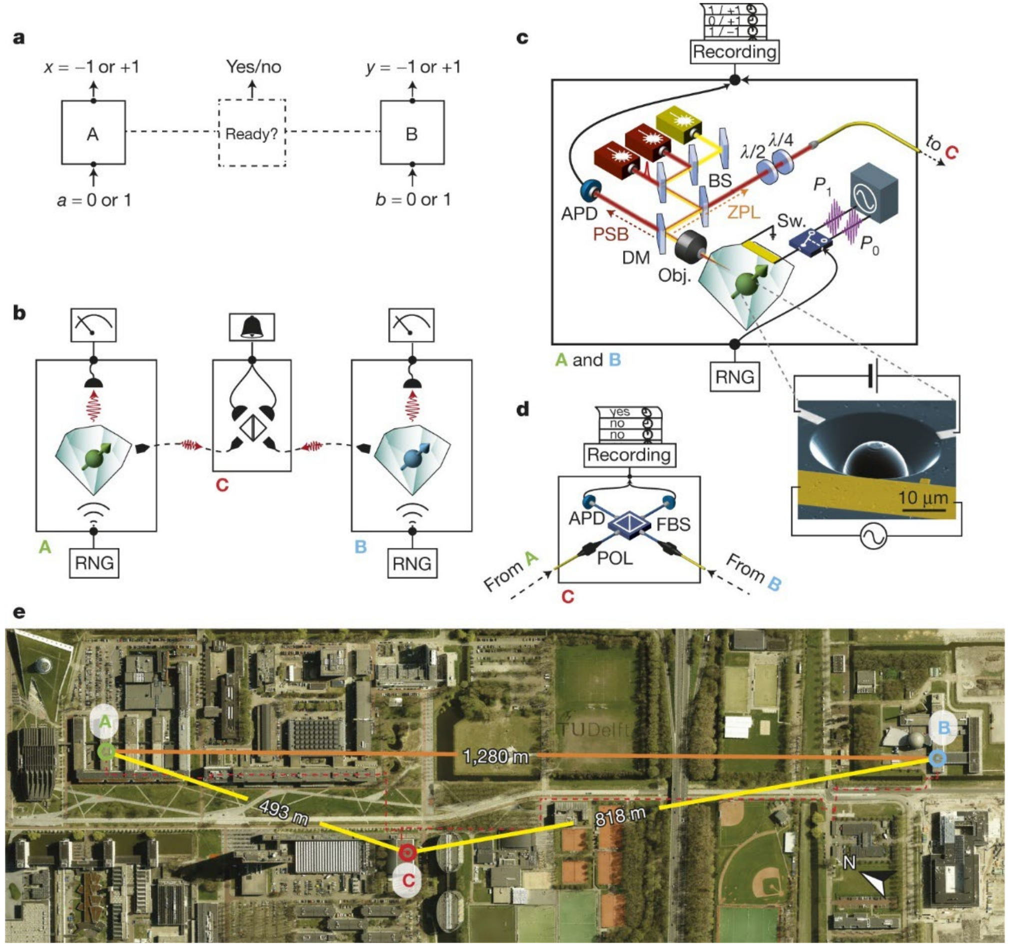
Quantum interfaces for NV centers

Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres

B. Hensen, H. Bernien, A. E. Dréau, A. Reiserer, N. Kalb, M. S. Blok, J. Ruitenberg, R. F. L. Vermeulen, R. N. Schouten, C. Abellán, W. Amaya, V. Pruneri, M. W. Mitchell, M. Markham, D. J. Twitchen, D. Elkouss, S. Wehner, T. H. Taminiau & R. Hanson

Nature 526, 682–686 (2015) | Cite this article

First fully loophole-free Bell test



1. Start with two identical remote setups NV centers in diamond in 4K cryostats.
2. Pump each in a known spin $|m_s=0>$ state.
3. Use a microwave $\pi/2$ pulse to put each into a superposition state $|m_s=0>+|m_s=-1>$.
4. Apply two time separated resonant optical pulses first one for $|m_s=0>$ and later $|m_s=-1>$, if in $|m_s=0>$ emits photon earlier, if in $|m_s=-1>$ fluorescent photon emitted later.
5. The fluorescence decay photon is collected by a microscope and sent to a beamsplitter at C 500m/800m away from the two setups.
6. Use a random number to choose one of two microwave pulses. These pulses rotate the spin so that reading out in the usual $|m_s=0>$ or $m_s=-1>$ basis effectively measures the spin either along 0 or tilted by ±45°.
7. Apply a resonant optical π pulse and read the resulting fluorescence photon.
8. At the beamsplitter if photon is early and one is late the two NV centres are in an entangled state (**heralded**).
9. Post select only those measurements in which entanglement was heralded.

- Per event success probability 6e-9
- Ran experiment for 220 hours with 245 valid Bell tests

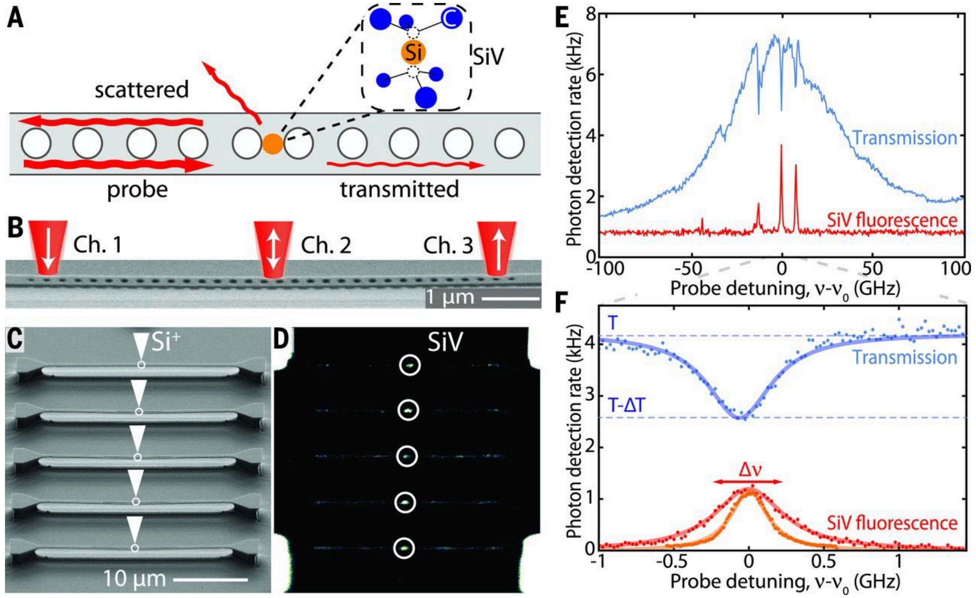
Quantum interfaces for NV centers

An integrated diamond nanophotonics platform for quantum-optical networks

A. SIPAHIGIL, R. E. EVANS, D. D. SUKACHEV, M. J. BUREK, J. BORREGAARD, M. K. BHASKAR, C. T. NGUYEN, J. L. PACHECO, H. A. ATIKIAN, [...] , AND M. D. LUKIN +6 authors

Authors Info & Affiliations

SCIENCE • 13 Oct 2016 • Vol 354, Issue 6314 • pp. 847-850 • DOI: 10.1126/science.aah6875



Strong coupling - Cooperativity C~1

Spectrally tunable single photon source (Raman)

Entanglement between NV centers fiber coupled

Single photon switching

This single platform had 2000 NV center/cavity nodes and demonstrated:

- deterministic emitter placement
- cavity-enhanced emission
- single-photon switching
- tunable single-photon source
- waveguide-mediated two-emitter entanglement

Raman photons ~15kHz
Entanglement ~0.5Hz (entanglement lifetime T2~2.5ns)

Challenges:

- 4K operation
- Coherence time <50ns
- SiV lifetime ~1.8ns
- SiV centers demonstrated ~100MHz variation (vs GHz/10 GHz for NV centers), ~0 for atoms/ions

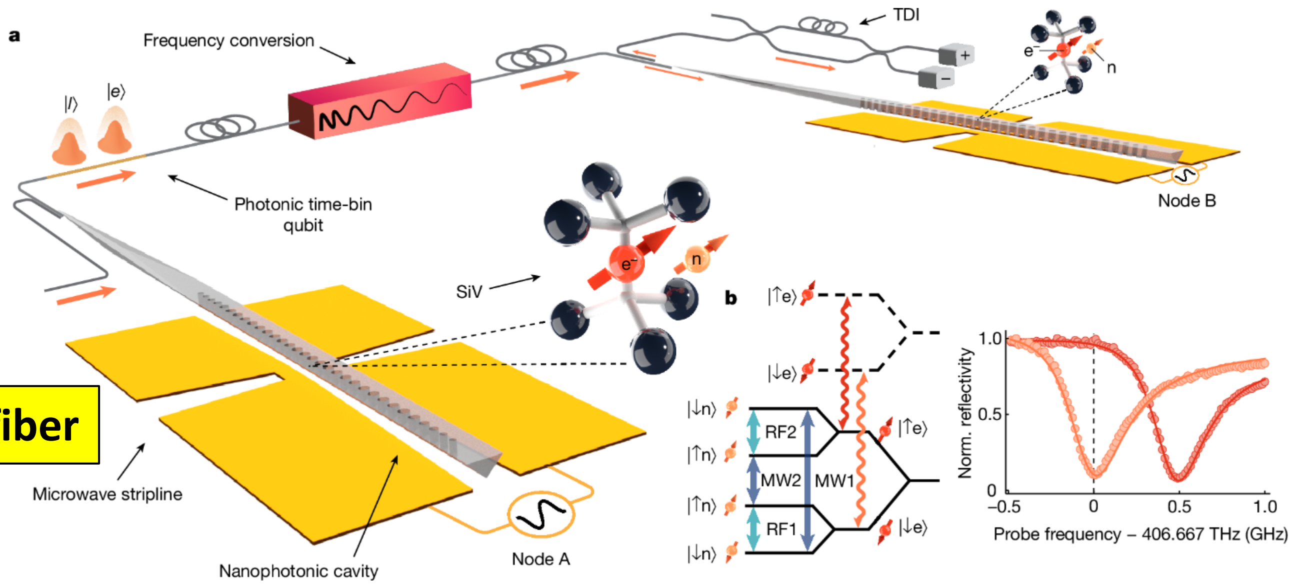
See also: High quality-factor optical nanocavities in bulk single-crystal diamond, Nature Comms **5, 5718 (2014)

Quantum interfaces for NV centers

Entanglement of nanophotonic quantum memory nodes in a telecom network

C. M. Knaut, A. Suleymanzade, Y.-C. Wei, D. R. Assumpcao, P.-J. Stas, Y. Q. Huan, B. Machielse, E. N. Knall, M. Sutula, G. Baranes, N. Sinclair, C. De-Eknamkul, D. S. Levonian, M. K. Bhaskar, H. Park, M. Lončar & M. D. Lukin

Nature **629**, 573–578 (2024) | Cite this article



Entanglement distribution @ 1350nm over **40km deployed fiber**

Deterministic! Cooperativity=12

Each node is a **multi-qubit register**:

- 1 electron spin (communication qubit, fast, cavity-coupled)
- 1 nuclear spin (long-lived quantum memory, >2 s coherence)

Entanglement of electron spins at ~1Hz, fidelity 86%
Transfer to nuclear spins which have a lifetime up to 2s
Nuclear spin entanglement rate 1 per 170s (6mHz)

See also:

Experimental demonstration of memory-enhanced quantum communication

M. K. Bhaskar, R. Riedinger, B. Machielse, D. S. Levonian, C. T. Nguyen, E. N. Knall, H. Park, D. Englund, M. Lončar, D. D. Sukachev & M. D. Lukin

Nature **580**, 60–64 (2020) | Cite this article

Demonstrated memory assisted

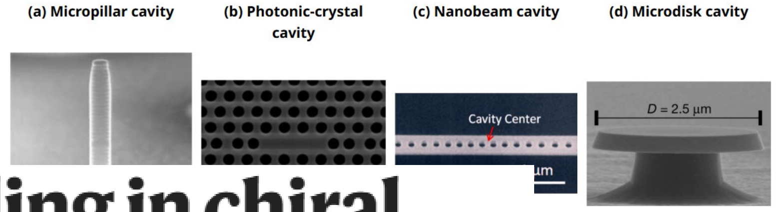
Deterministic! Cooperativity=105

Quantum interfaces for quantum dots

Controlling the dynamics of spontaneous emission from quantum dots by photonic crystals

Peter Lodahl, A. Floris van Driel, Ivan S. Nikolaev, Arie Irman, Karin Overgaag, Willem L. Vos

Nature **430**, 654–657 (2004)



Srinivasan and Painter, 2007

Near-Unity Coupling ... a Photonic Crystal W...

M. Arcari[1], I. Söllner[1,*], A. Javadi[1], S. L... Lee[2], J. D. Song[2] *et al.*

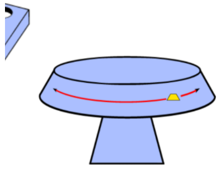
Show more

Phys. Rev. Lett. **113**, 093603 – Published 2...

Deterministic photon–emitter coupling in chiral photonic circuits

Immo Söllner, Sahand Mahmoodian, Sofie Lindskov Hansen, Leonardo Midolo, Alisa Javadi, Gabija Kiršanskė, Tommaso Pregnolato, Haitham El-Ella, Eun Hye Lee, Jin Dong Song, Søren Stobbe & Peter Lodahl

Nature Nanotechnology **10**, 775–778 (2015) | Cite this article



$g/2\pi = 3$ GHz

$\kappa/2\pi = 1$ GHz

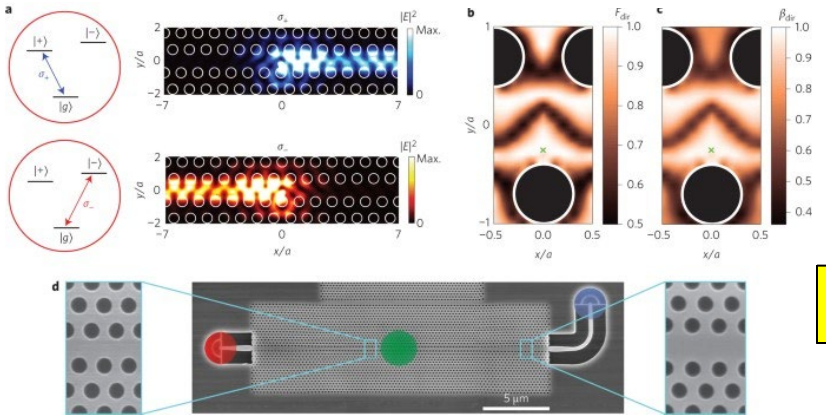
$\gamma/2\pi \sim 0.6$ GHz

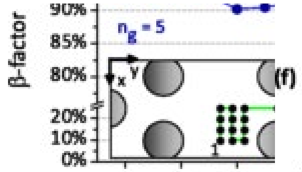
...gle quantum dots

...tonic nanostructures

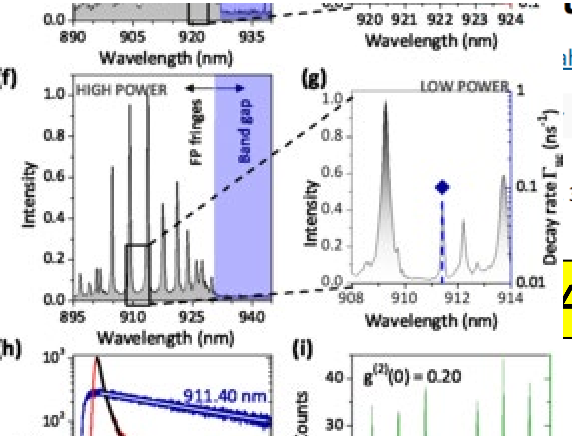
...hand Mahmoodian, and Søren Stobbe

...347 – Published 11 May, 2015





Entanglement



40km deployed fiber

Experimental Progress

Interfacing to supercontucting qubits

References:

Optomechanics for quantum technologies, Nature Physics **18**, 15 (2022)

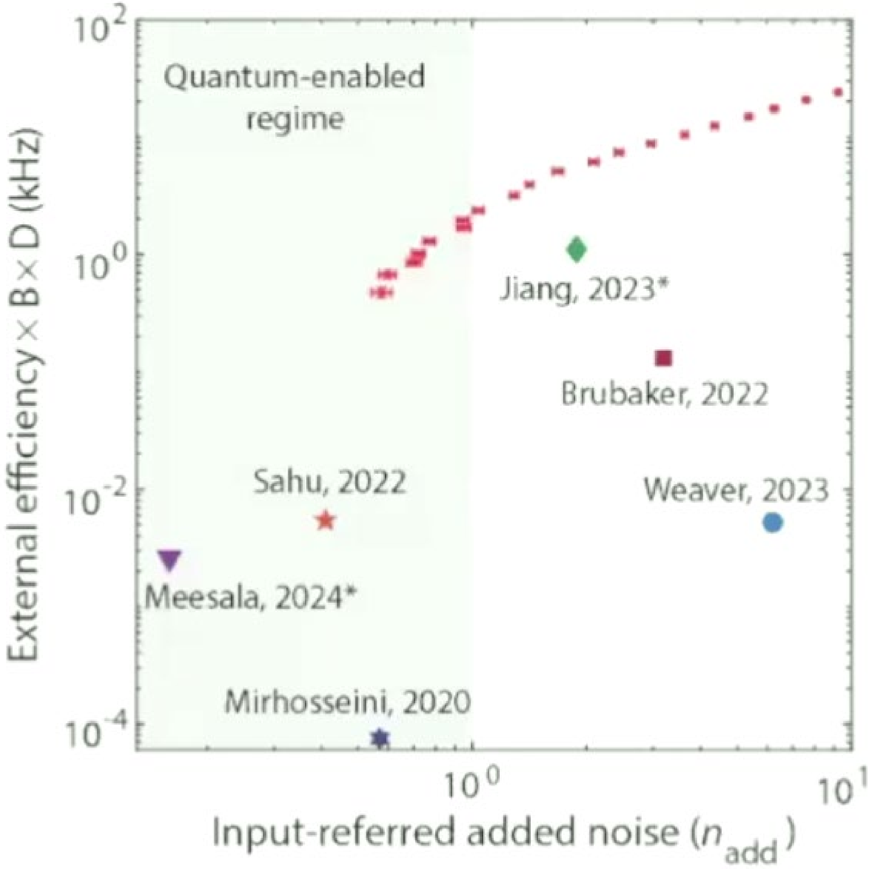
[Interfacing superconducting qubits with light | Johannes Fink (IST Austria)](https://www.youtube.com/watch?v=JHZA2UMqe8s)

https://www.youtube.com/watch?v=JHZA2UMqe8s

Progress in quantum interfaces for superconducting qubits

Microwave-optical conversion rate:

Microwave–Optical photon conversion rate
Efficiency*Bandwidth*Duty Cycle



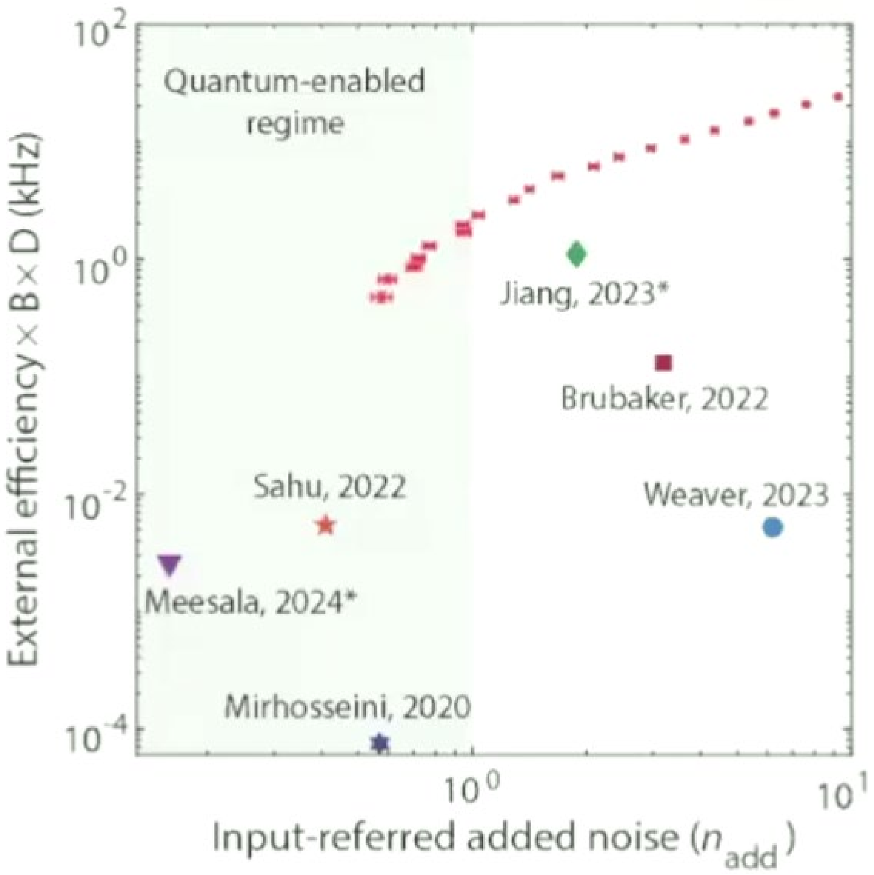
Should be able to operate close to the "clock speed of the system you want to interface to: MHz for superconducting qubits, kHz for ions

https://www.youtube.com/watch?v=JHZA2UMqe8s

Progress in quantum interfaces for superconducting qubits

Microwave-optical conversion rate:

Microwave–Optical photon conversion rate
Efficiency*Bandwidth*Duty Cycle

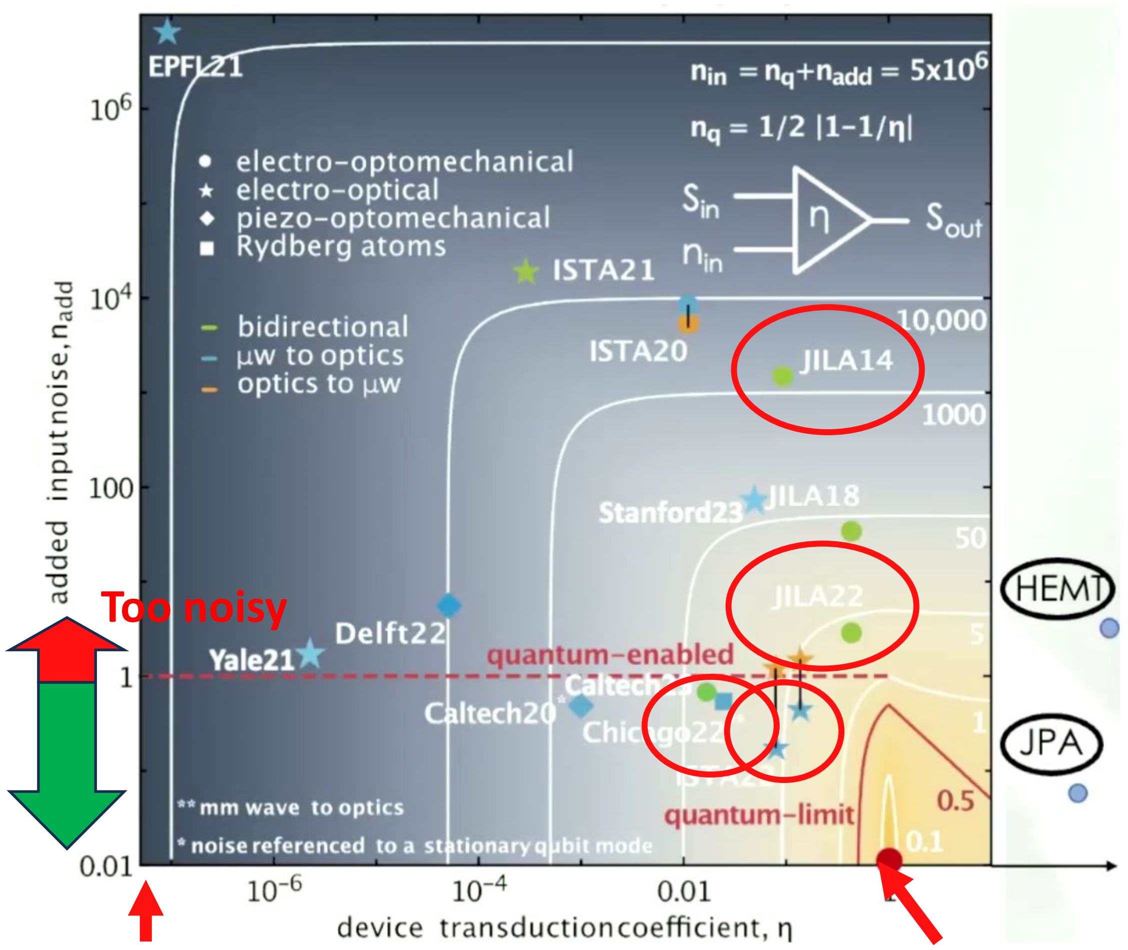


Should be able to operate close to the "clock speed of the system you want to interface to: MHz for superconducting qubits, kHz for ions

If added noise is <1photon it becomes possible to

- Distribute entanglement and teleport
- Herald on successful events

Total input noise:



Want no added noise

Want 1:1 output:input photons, no gain, no loss

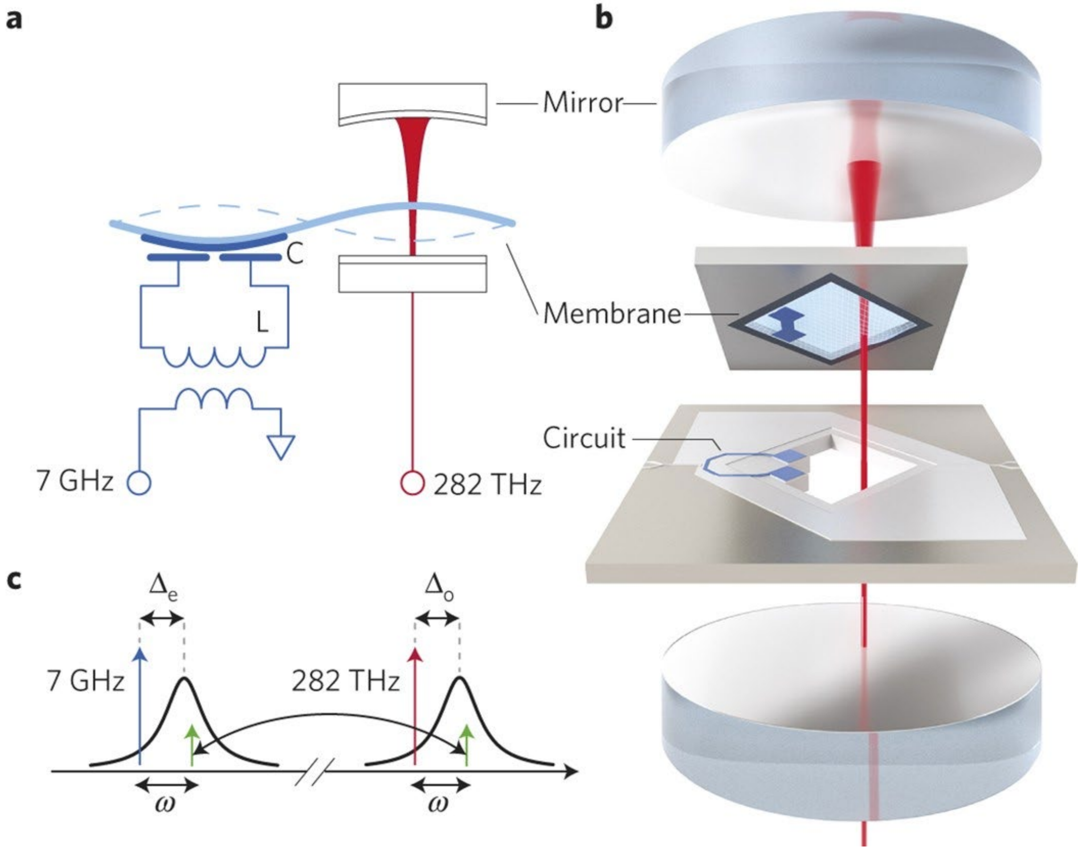
https://www.youtube.com/watch?v=JHZA2UMqe8s

Opto-mechanical microwave-optical transducers

Bidirectional and efficient conversion between microwave and optical light

R. W. Andrews, R. W. Peterson, T. P. Purdy, K. Cicak, R. W. Simmonds, C. A. Regal & K. W. Lehnert

Nature Physics **10**, 321–326 (2014) | Cite this article



Proof-of-principle demonstration
(~1e3 photons of added noise)

First bidirectional electro-optomechanical microwave-optical transducer

Opto-mechanical microwave-optical transducers

PHYSICAL REVIEW X **12,** 021062 (2022)

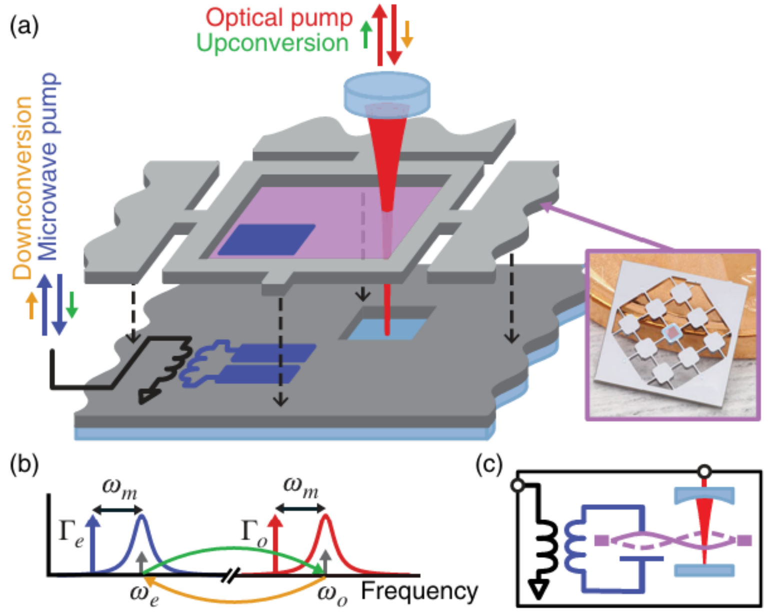
Optomechanical Ground-State Cooling in a Continuous and Efficient Electro-Optic Transducer

B. M. Brubaker,[1,2,*] J. M. Kindem,[1,2,*] M. D. Urmey,[1,2,*,†] S. Mittal,[1,2] R. D. Delaney,[1,2]
P. S. Burns,[1,2] M. R. Vissers,[3] K. W. Lehnert,[1,2,3] and C. A. Regal[1,2]
[1]*JILA, National Institute of Standards and Technology and the University of Colorado, Boulder, Colorado 80309, USA*
[2]*Department of Physics, University of Colorado, Boulder, Colorado 80309, USA*
[3]*National Institute of Standards and Technology, Boulder, Colorado 80305, USA*

First continuous, high-efficiency, nearly ground-state, low-noise electro-optomechanical transducer

Key improvements:

- Operating temperature 4K -> 40mK
- Mechanical mode 1e5 phonon population -> 0.34 phonons
- Added noise: 1e3 photons -> 3.2photons
- Efficiency 8.6% -> 47% by reducing optical cavity losses and better modematching
- Heating of suprconductors from optical scatter largely eliminated
- Much higher Q mechanical membrane Q=1.3e7
- Use of NbTiN superconductor low loss to optical light (very loss optical loss)



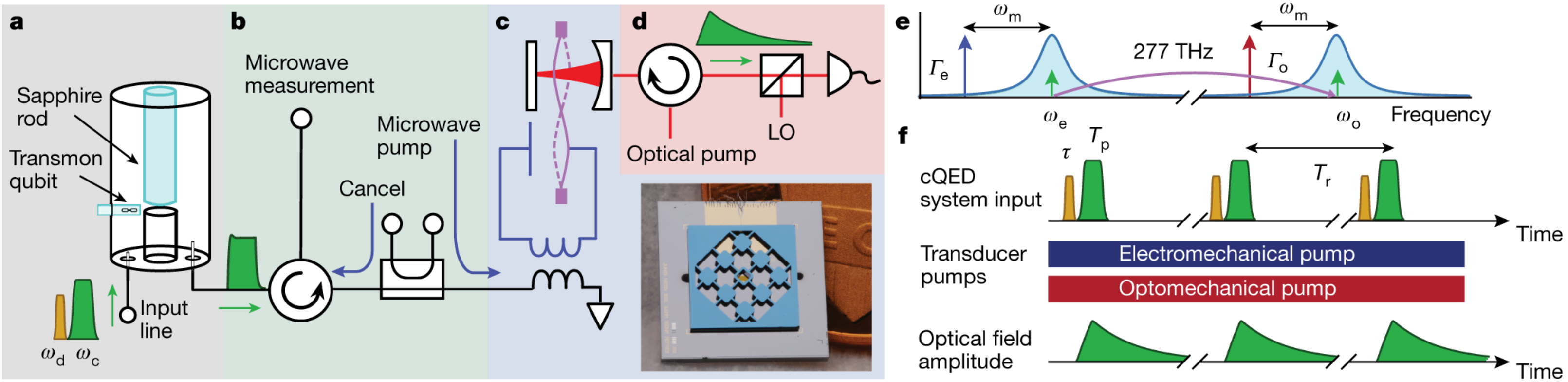
Optomechanical Ground-State Cooling in a Continuous and Efficient Electro-Optic Transducer, PRX 12, 021062 (2022)

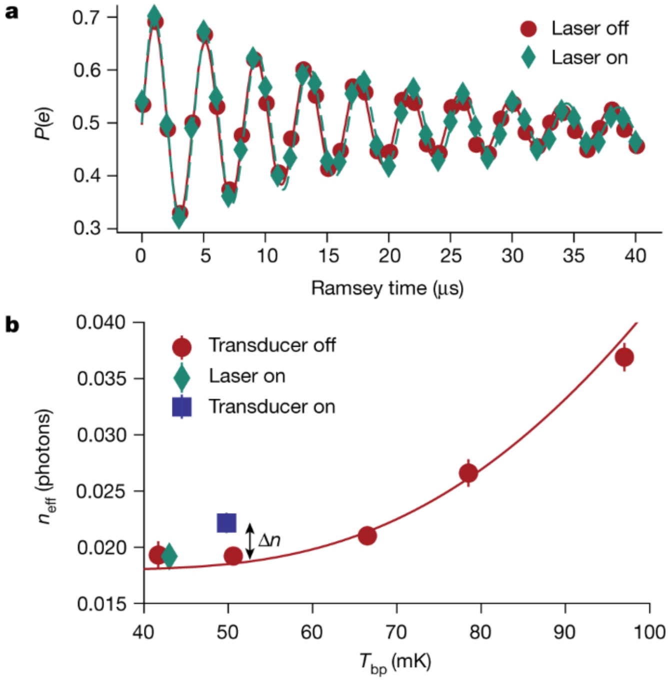
Opto-mechanical microwave-optical transducers

Superconducting-qubit readout via low-backaction electro-optic transduction

R. D. Delaney, M. D. Urmey, S. Mittal, B. M. Brubaker, J. M. Kindem, P. S. Burns, C. A. Regal & K. W. Lehnert

Nature **606**, 489–493 (2022) | Cite this article





First demonstration of a superconducting qubit readout via optical upconversion.

Demonstrated low backaction on superconducting qubit (Δn~3e-3 photons)

Piezo-optomechanical microwave-optical transducers

Superconducting qubit to optical photon transduction

Mohammad Mirhosseini, Alp Sipahigil, Mahmoud Kalaee & Oskar Painter

Nature **588**, 599–603 (2020) | Cite this article

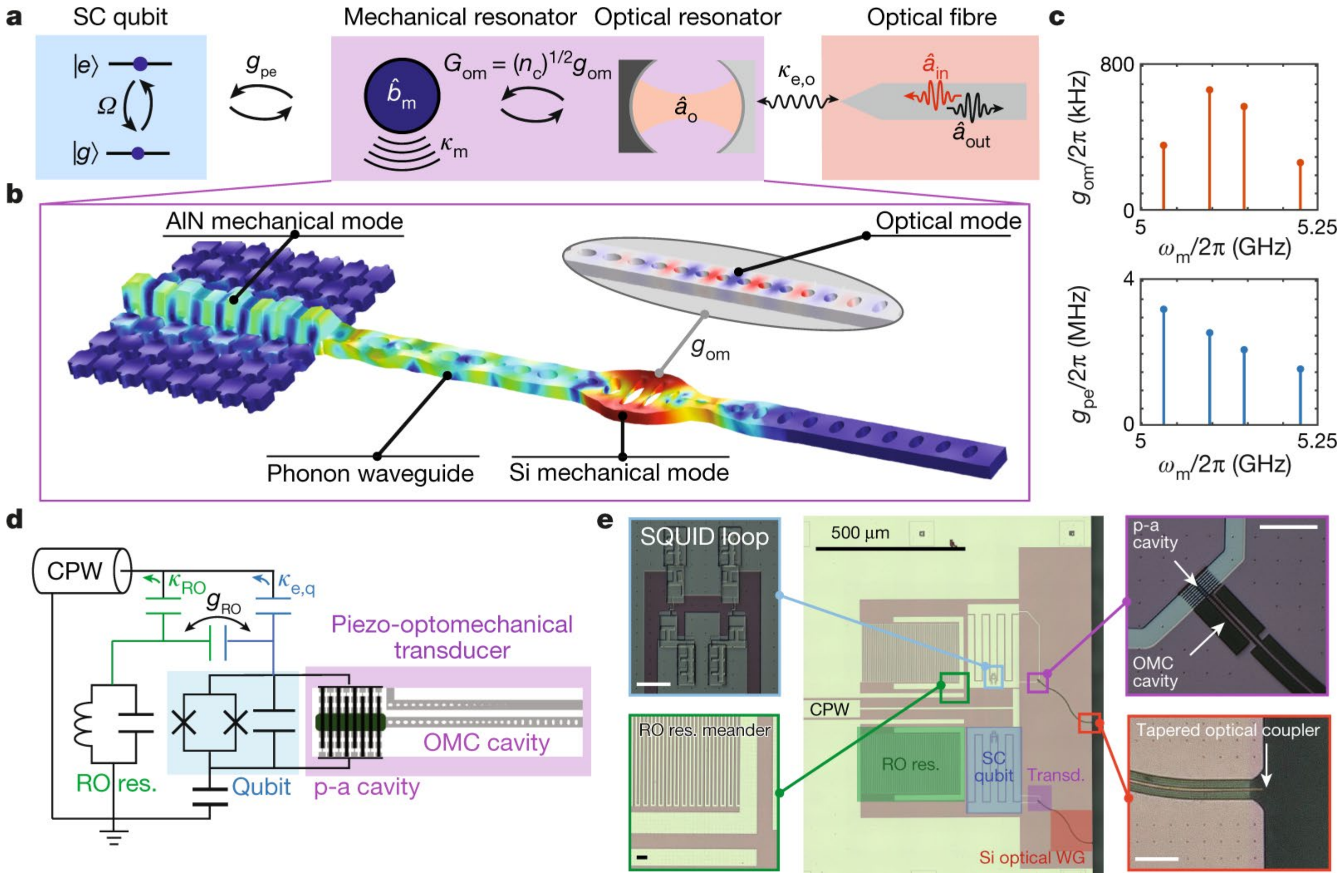
First demonstration of a quantum enabled superconducting microwave to optical transducer (noise added < 1photon)
Added noise just 0.57photons!

Transduction **efficiency just 1e-5 so very lossy**!

First demonstration of low noise transducer using piezo-optomechanical resonator and directly to telecom band.

Made possible by:

- Operating temperature 15mK
- Tiny optical cavity mode volume with few photons 2uW peak power giving very low heating
- Very low readout duty cycle (1e-5) to reduce heating from laser
- Fast qubit-mechanical phonon swap to minimize loss



Cooperativity ~1
(not higher to reduce heating)

Rydberg atom microwave-optical transducers

Quantum-enabled millimetre wave to optical transduction using neutral atoms

Aishwarya Kumar, Aziza Suleymanzade, Mark Stone, Lavanya Taneja, Alexander Anferov, David I. Schuster & Jonathan Simon

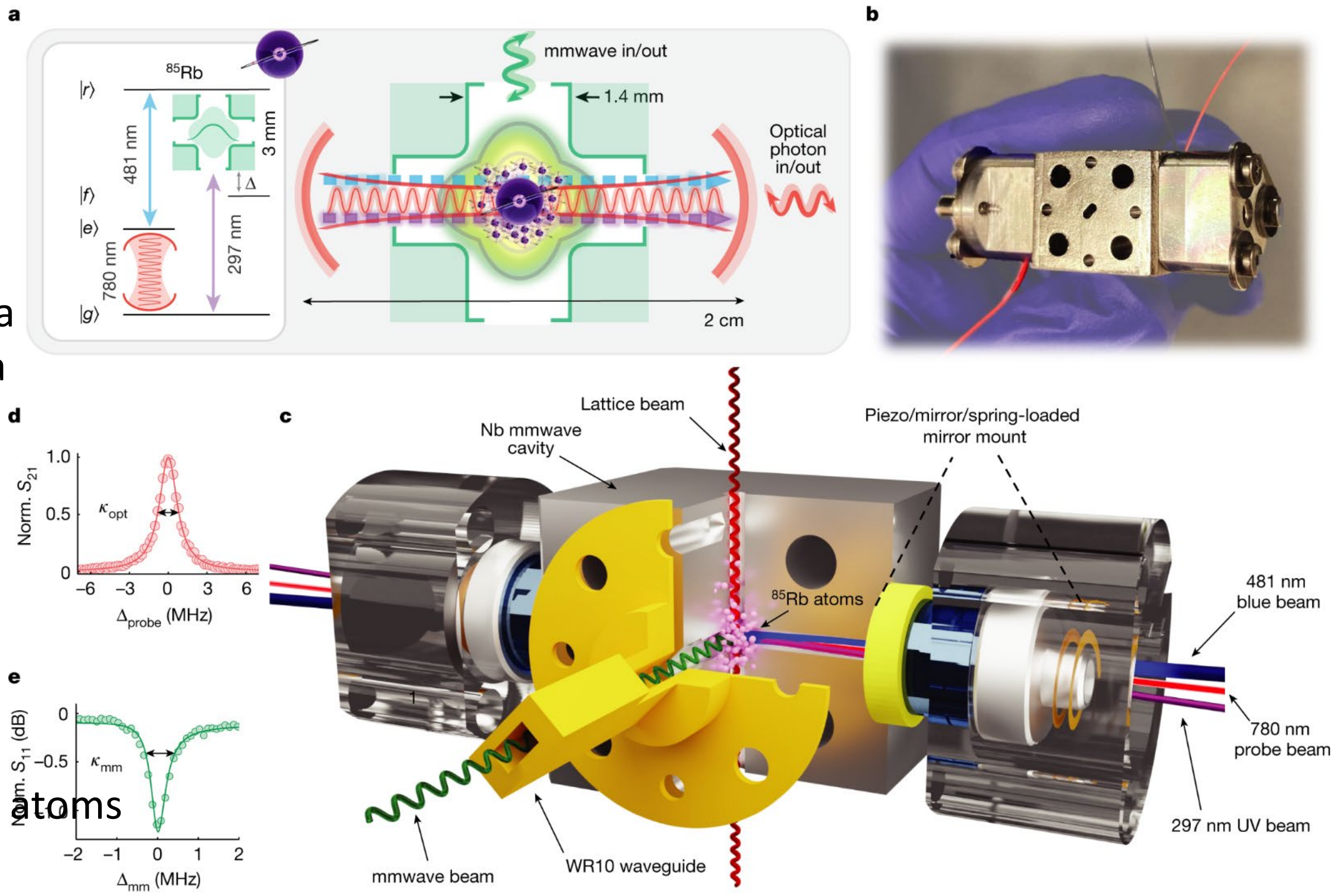
Nature **615**, 614–619 (2023) | Cite this article

Cold Rydberg atoms simultaneously coupled to both a mm-wave (100GHz) superconducting resonator and a high-finesse optical cavity

Added noise 0.6photons
360kHz bandwidth
58% internal conversion efficiency (2.5% conversion)

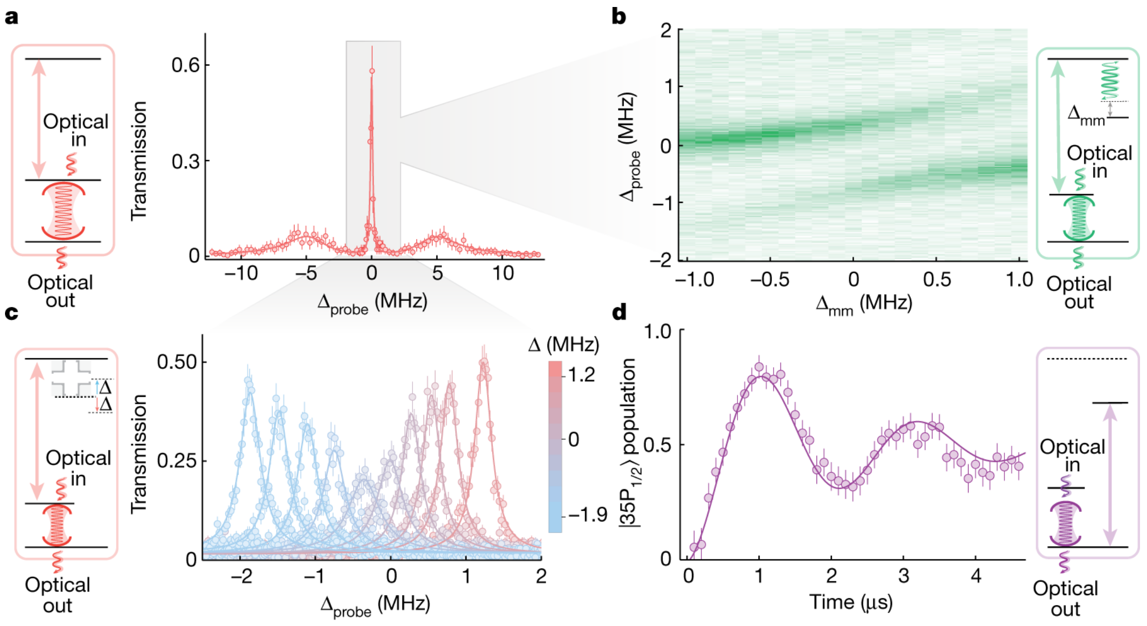
Possible because:

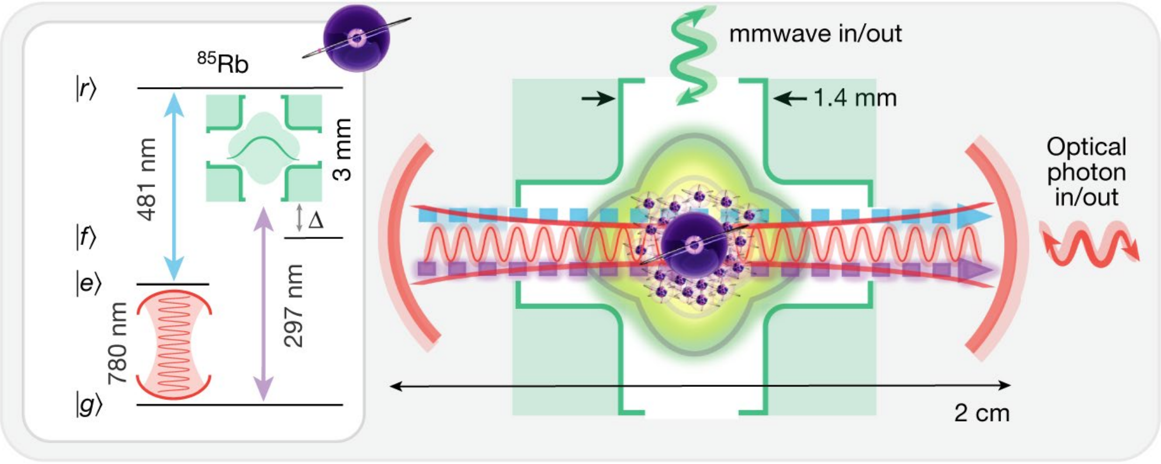
- Optical cavity creates superradiant emission from atoms
- N~500-600 atoms in cavity couple collectively
- Optical cooperativity ~3
- Mm-wave cooperativity 3-6
- 481nm cavity cooperativity 20-40
- Impedance matching gives near ideal efficiency



Quantum enabled noise with dramatically improved conversion efficiency!

Rydberg atom microwave-optical transducers





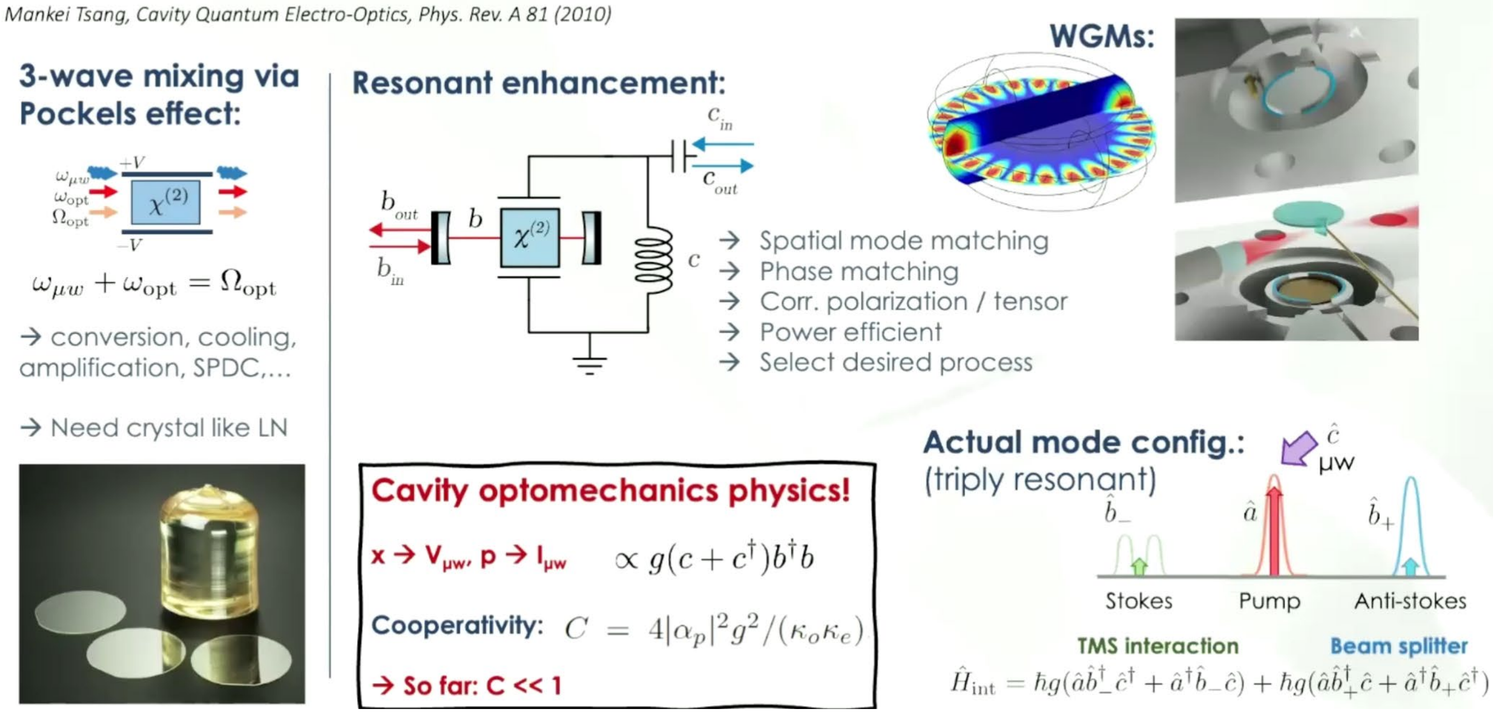
1. 780 nm probe couples |g⟩ → |e⟩, injecting optical cavity photons into the atomic ensemble.
2. 481 nm control beam couples |e⟩ → |r⟩, creating Rydberg-EIT and forming a dark polariton (|e⟩ ↔ |r⟩ hybrid).
3. EIT dark polariton suppresses loss via 5P decay and provides a long-lived coherence linking optical photons to Rydberg excitations.
4. mm-wave cavity couples |r⟩ → |f⟩ via the 36S ↔ 35P transition, hybridizing the dark polariton with the 100 GHz resonator field.
5. 297 nm UV beam couples |f⟩ → |g⟩, closing the four-wave-mixing loop and returning the atom to the ground state.
6. Cycle converts a mm-wave photon into an optical cavity photon (and is fully reversible).
7. Impedance matching via EIT (tuning $\Omega_{(481)}$) ensures optimal mixing of optical and Rydberg components → high-efficiency transduction.

Quantum interfaces for superconducting qubits

Entangling microwaves with light

R. SAHU, L. QIU, W. HEASE, G. ARNOLD, Y. MINOGUCHI, P. RABL, AND J. M. FINK

SCIENCE • 18 May 2023 • Vol 380, Issue 6646 • pp. 718-721 • DOI: 10.1126/science.adg3812



Use a pulsed (1W) pump (for 100ns) to achieve efficient (99%) conversion of microwave photon to an optical photon 15% effective conversion efficiency due to in/out coupling losses

Only 0.4Hz entanglement at these performances due to heating

Cooperativity ~ 1 (close to strong interaction between optical and microwave photons)

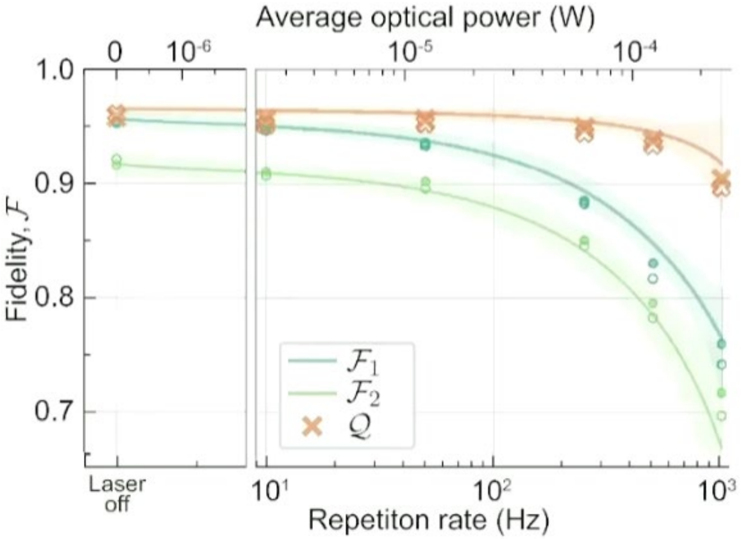
https://www.youtube.com/watch?v=JHZA2UMqe8s

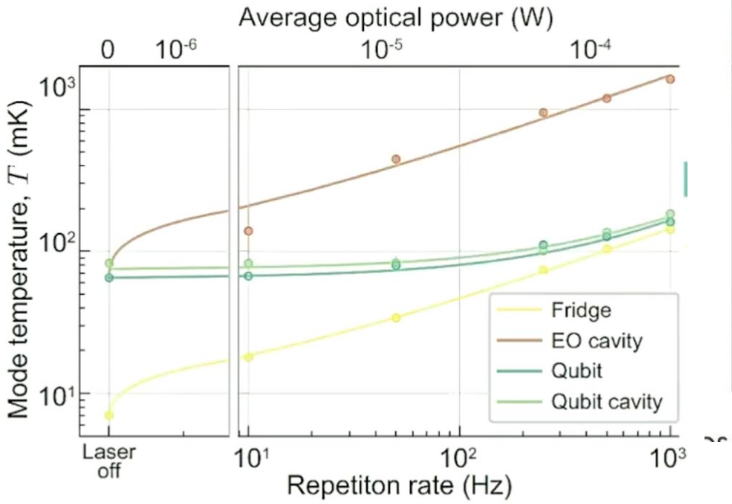
Quantum interfaces for superconducting qubits

All-optical superconducting qubit readout

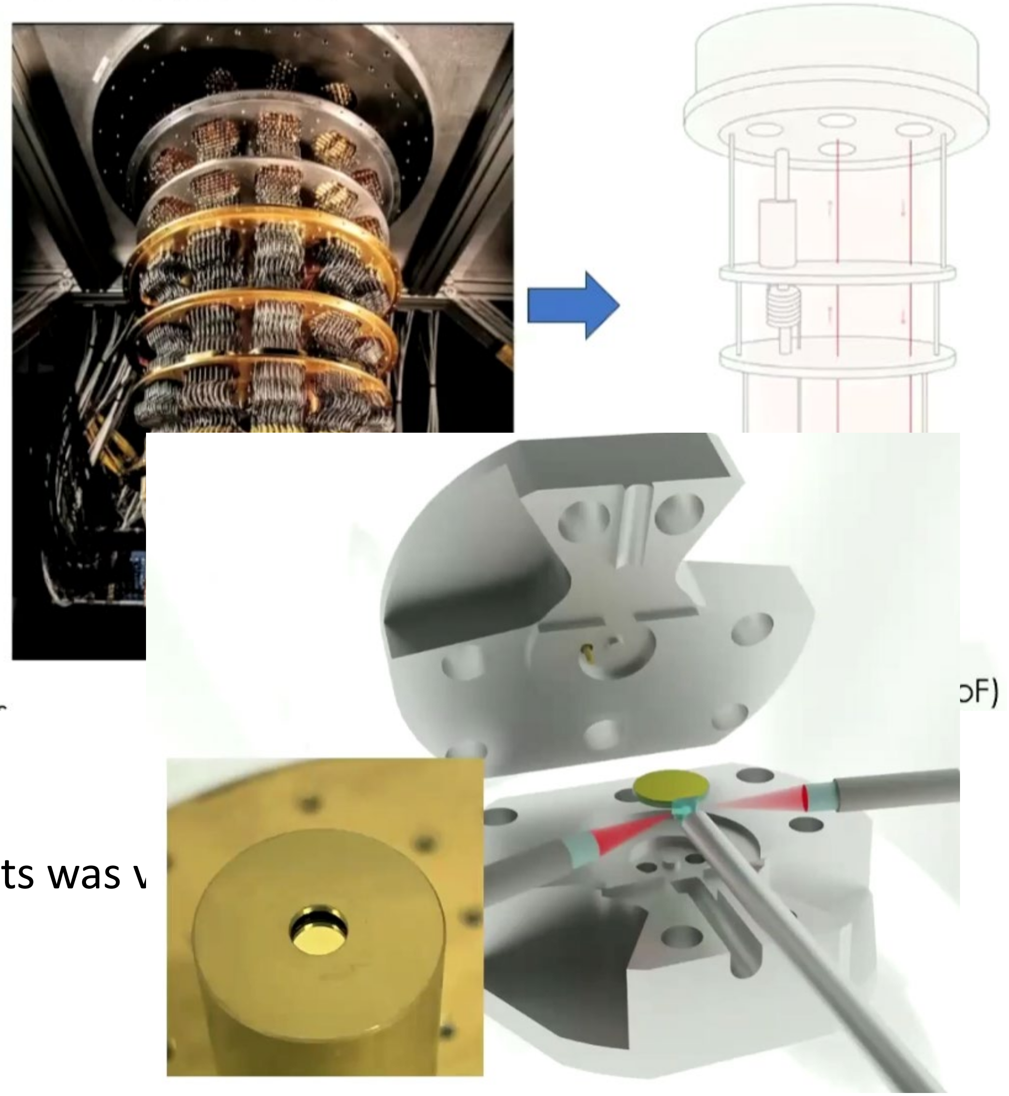
Georg Arnold, Thomas Werner, Rishabh Sahu, Lucky N. Kapoor, Liu Qiu & Johannes M. Fink

Nature Physics **21**, 393–400 (2025) | Cite this article





How about **removing the clutter** in the fridge with **cryogenic microwave photonics?**



Demonstrated that the loss in coherence time of superconducting qubits was v

-> Challenge is how to reduce the pump power to reduce heating…

https://www.youtube.com/watch?v=JHZA2UMqe8s

Experimental Progress Summary

Entanglement Generation Rate

- Ions: 40 Hz (cavity-enhanced)
- NV centers: ~1 Hz, telecom ready
- Neutral atoms: 32% deterministic survival, but low photon collection in free-space
- Superconductors: <1 Hz due to heating constraints

Requirement for network-scale quantum computers: kHz+Fidelity

- Ions: 97% ion–photon, 78% ion–ion
- NV centers: 86% nuclear-spin, loophole-free Bell performed
- Neutral atoms: >90% atom–atom (with error detection)
- Requirement for fault tolerance: >99.9%
- Superconducting qubits: still low fidelity

Noise / Loss

- Best microwave–optical transducers: 0.57 photons added (significant challenge!)
- Best optical memories: 92% retrieval
- Requirement: near unit efficiency and << sub-photon noise

No platform integrates high fidelity, high rate, telecom compatibility, and scalability simultaneously…yet

Summary

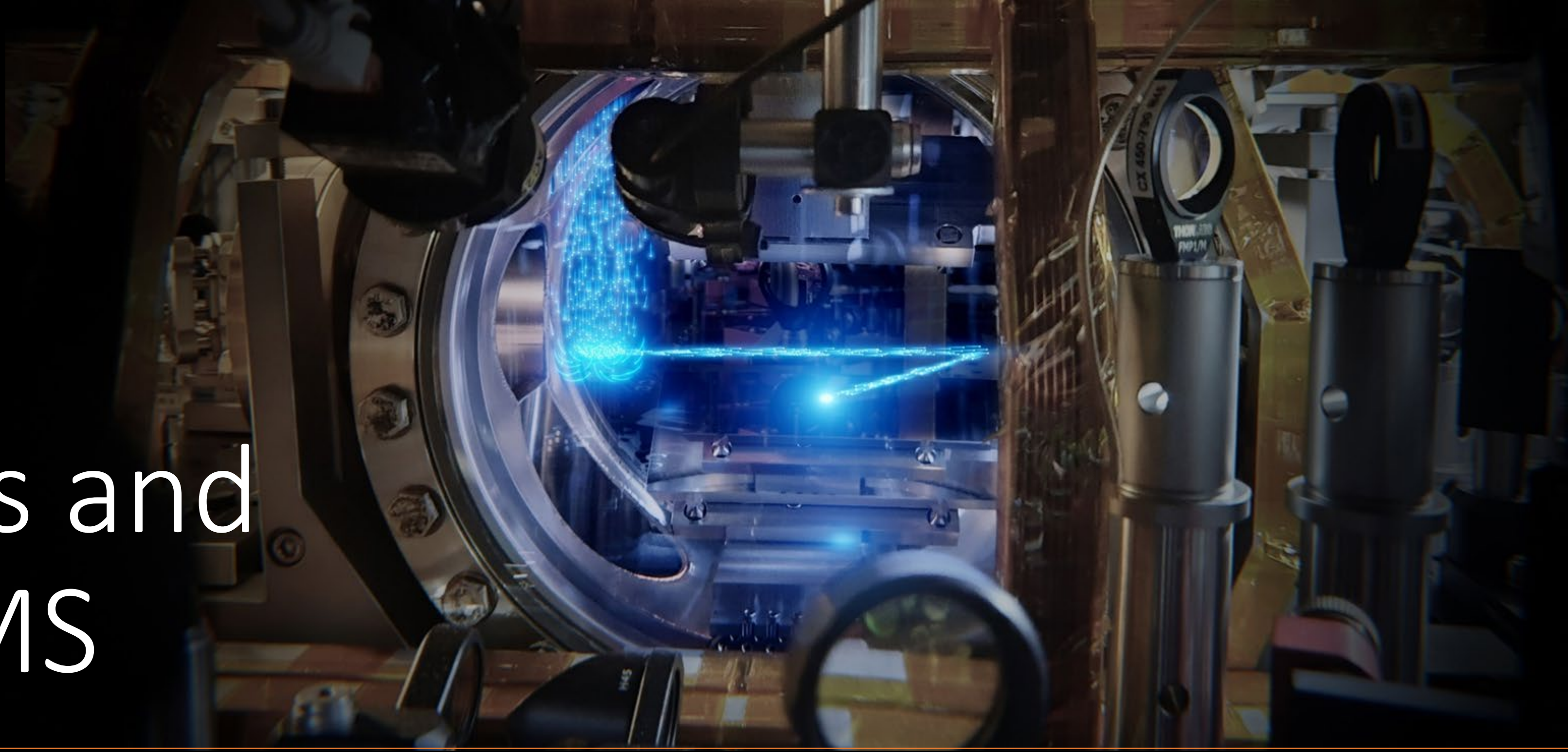
- Introduction

Foundations and Theory

- Free space
- Waveguides
- Cavities

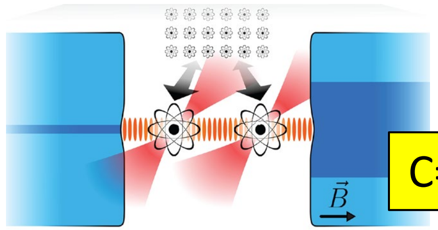
Experimental Progress

Current efforts and outlook at IAMS



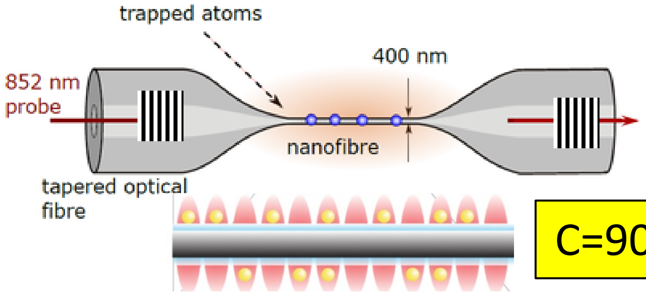
Hybrid atom-photonic architectures

Free space fiber cavities



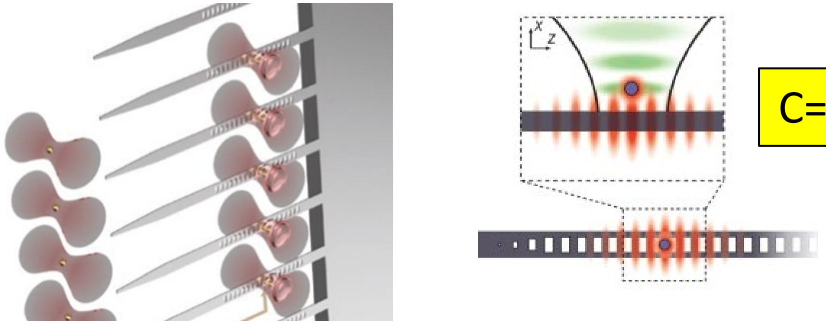
C=105

Nanofiber cavities



C=90

Photonic chip cavity arrays



C=27

- ✓ No dielectric interfaces, can operate tweezer arrays directly within cavity modes.
- ✓ No special fabrication, few sources of cavity mirrors.

- ! Diffraction limits scalability.
- ! Individual crafting of each cavity.

- ✓ Simple/low cost fabrication (except for gratings).
- ✓ Can trap thousands of atoms with mW power (evanescent config).

- ! Dielectric interfaces/Casimir forces can disturb.

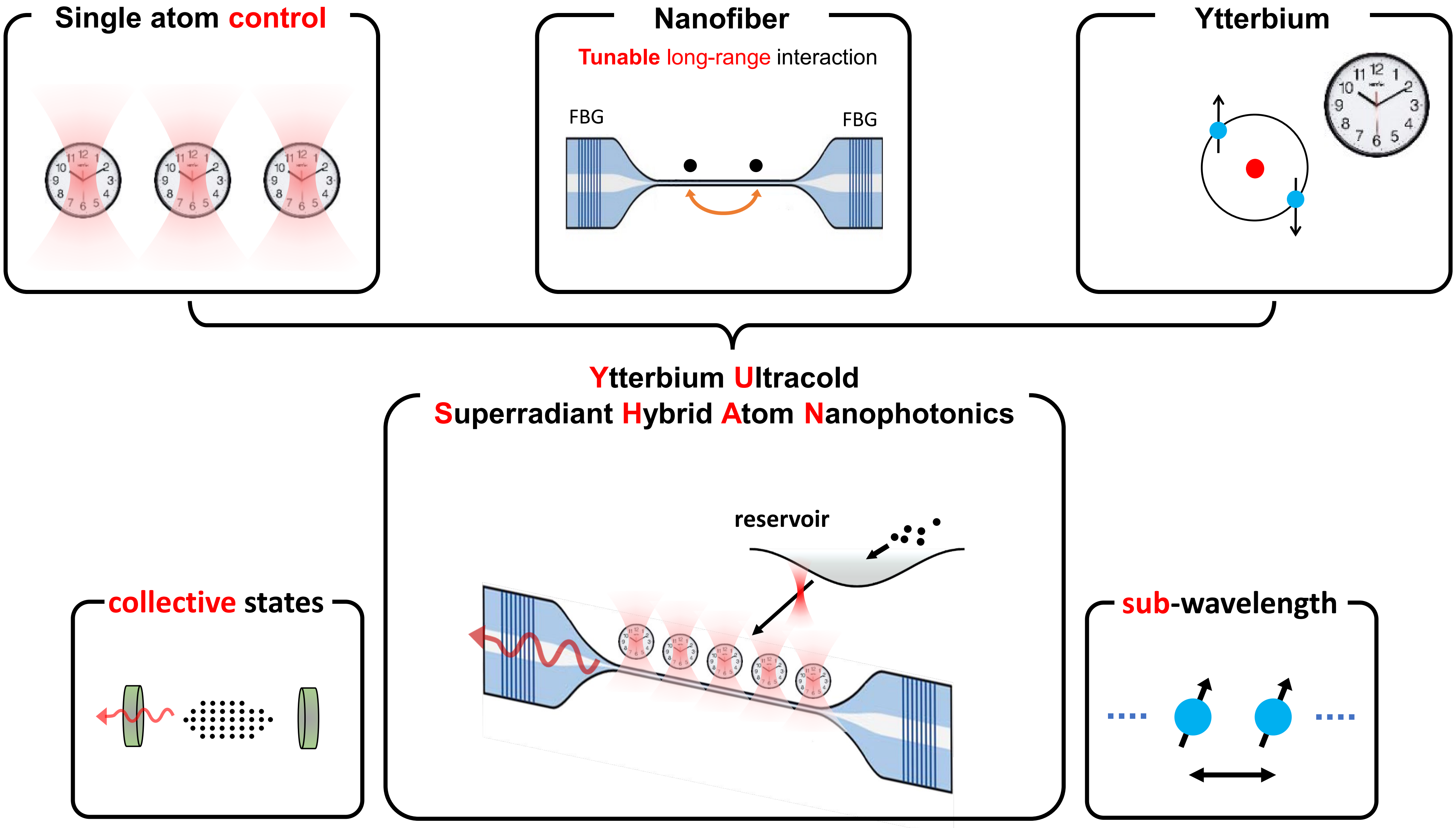
Our baseline objective

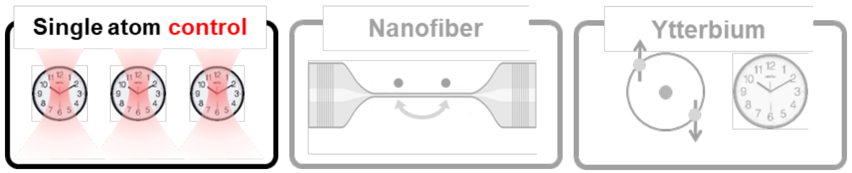
- ✓ Can interface complex integrated circuits.
- ✓ Can implement complex functionality.
- ✓ Excellent scaling potential, can fabricate many cavities/interfaces.

- ! Difficult/risky/expensive fabrication!
- ! Dielectric interfaces/Casimir forces can disturb.
- ! Work needed to get cooperativity up.

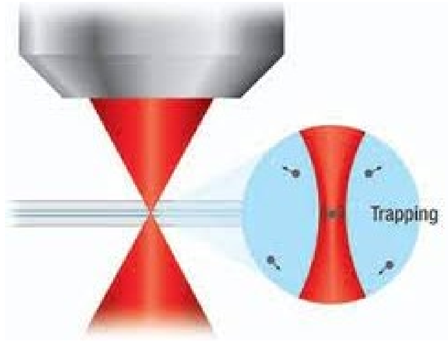
Long term objective

Our new platform at IAMS





Tweezer arrays of neutral atoms



Optical tweezers - Trap and control single atoms in the focus of light beams.

Allows:

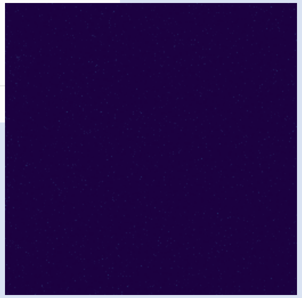
- Easily create and rearrange large arrays (using SLM)
- Defect free arrays or added defects
- Control the spacing (along the nanofiber)
- Individual control/tuning of interactions across nanofiber

Tweezer arrays have delivered explosive progress in neutral atom quantum computing

FORBES > INNOVATION > CLOUD

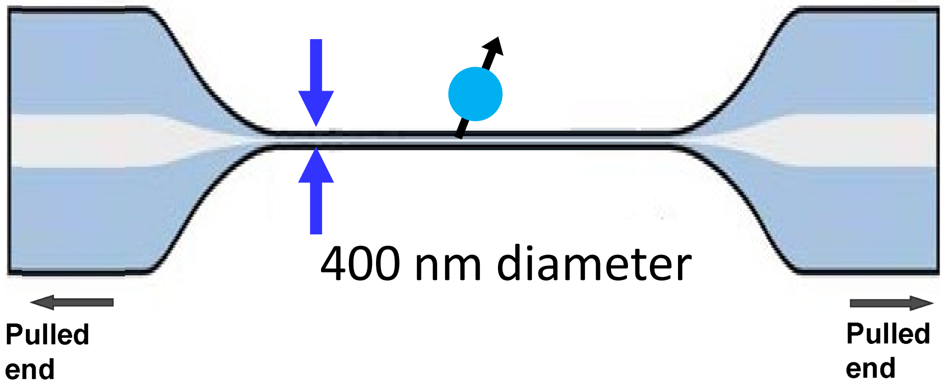
Atom Computing Announces Record-Breaking 1,225-Qubit Quantum Computer

Paul Smith-Goodson Contributor
Moor Insights and Strategy Contributor Group



Oct 2023: First >1000 qubit Quantum Processor

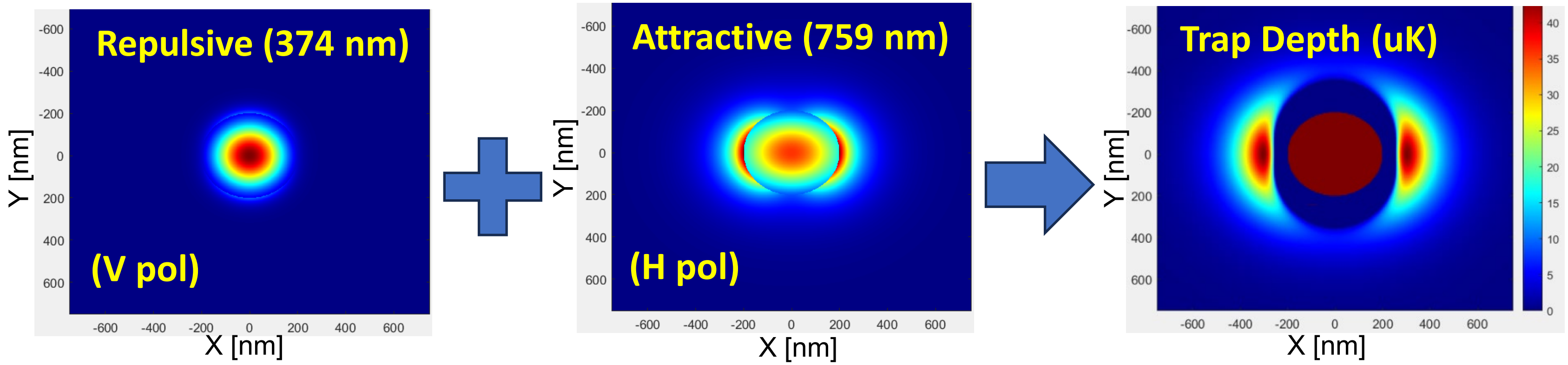
Tapered optical nanofibers



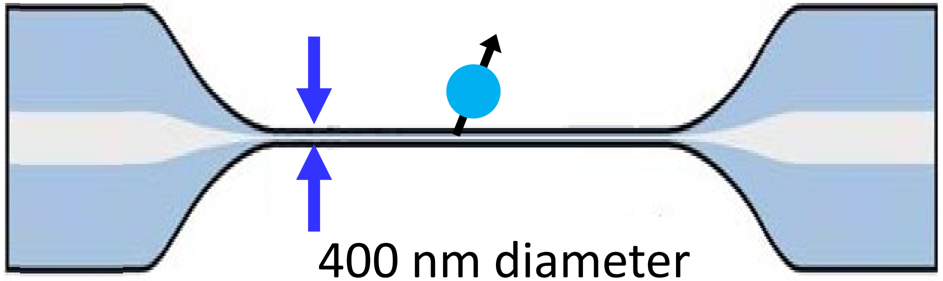
Allows:

- Simple, low cost, easy initial nanophotonic platform
- Can trap thousands of atoms with mW power
- Opportunity: New platform

Trapping potential formed by double magic wavelength trap at 759nm (red) and 375nm (or 553.5nm) for blue



Tapered optical nanofibers



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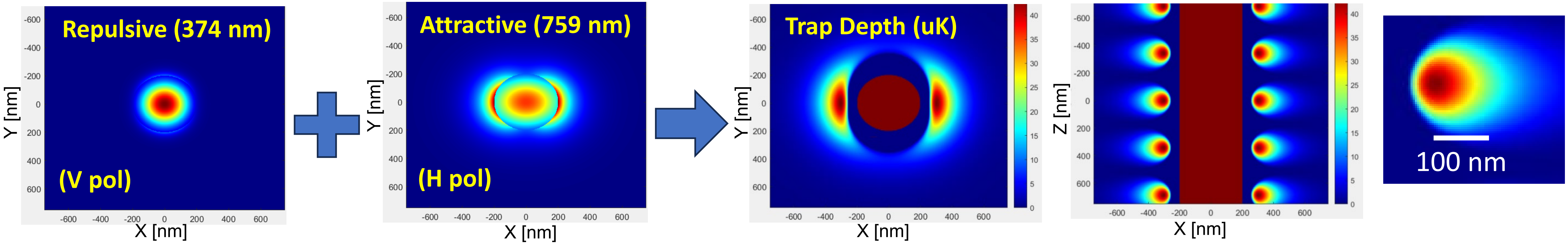
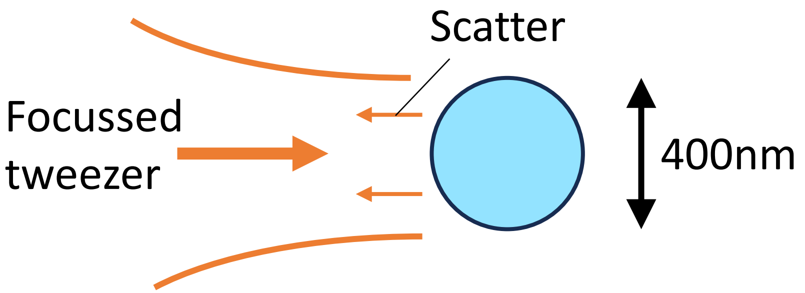
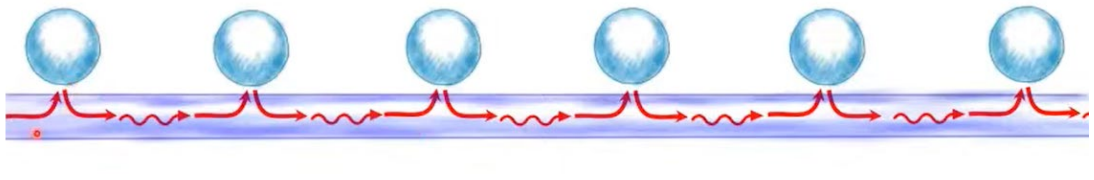


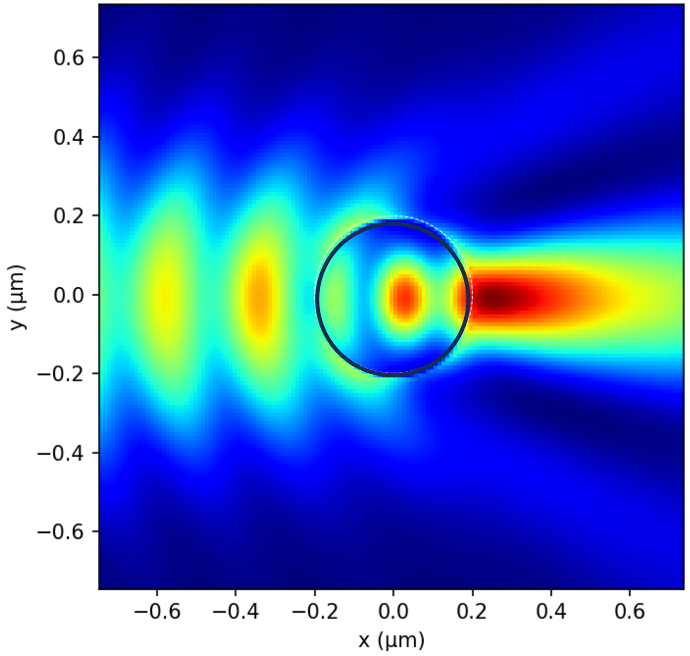
Tapered optical nanofibers

Tweezers from the side allow:

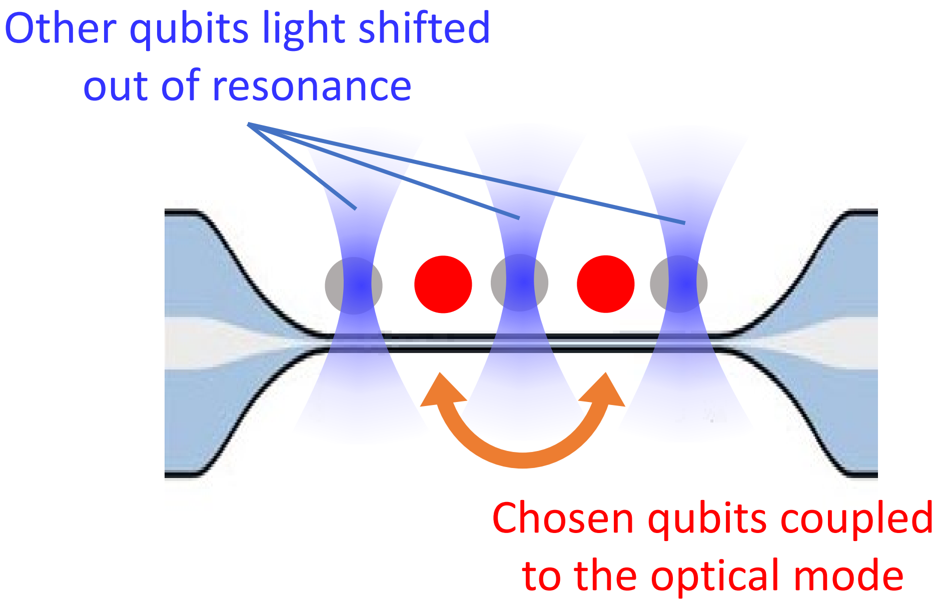
- Defect free arrays
- Sub-wavelength arrays for telecom band transitions
- Arbitrary geometries
- Move atoms into or out of communication with nanofiber
- Individual control over atomic state and coupling to propagating mode





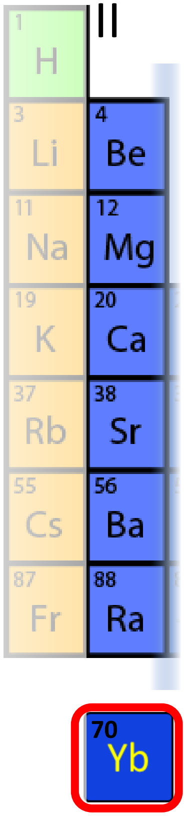


Tunable long-range interactions



Any-to-any long-range **tunable** interactions

Ytterbium - Features



Telecom wavelengths perfect for a quantum communication

- 309 kHz, 1388 nm
- 170 kHz, 1539 nm

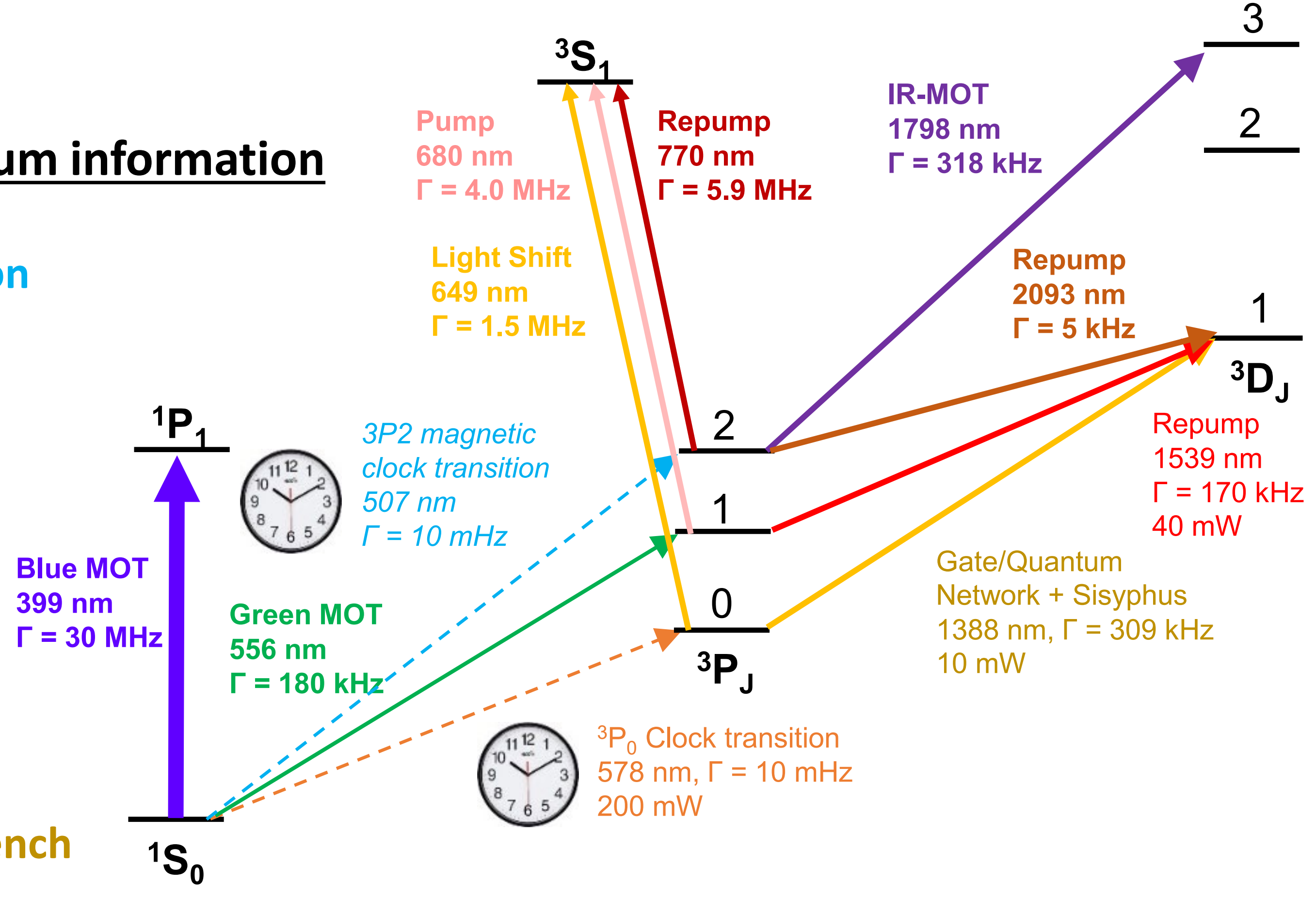
Metastable states for processing quantum information

- 10 mHz, 578 nm clock transition
- 10 mHz, 507 nm magnetic clock transition

Rich laser cooling capabilities

- 30 MHz, 399 nm for high flux
- Narrow 180 kHz, 556 nm for low temperature
- 318 kHz, 1798nm metastable MOT
  -> light doesn't affect other atoms!
- Recoil limited using 309 kHz 1388nm quench

  Quenched clock line cooling
  Sisyphus clock line cooling



Ytterbium - Features

Yb-171

- I=1/2, $^1S_0$ ground state is split $m_F=-1/2$, $m_F=+1/2$

Yb-171: A nuclear spin qubit
Environmentally insensitive
Excellent coherence
Ideal Quantum Memory

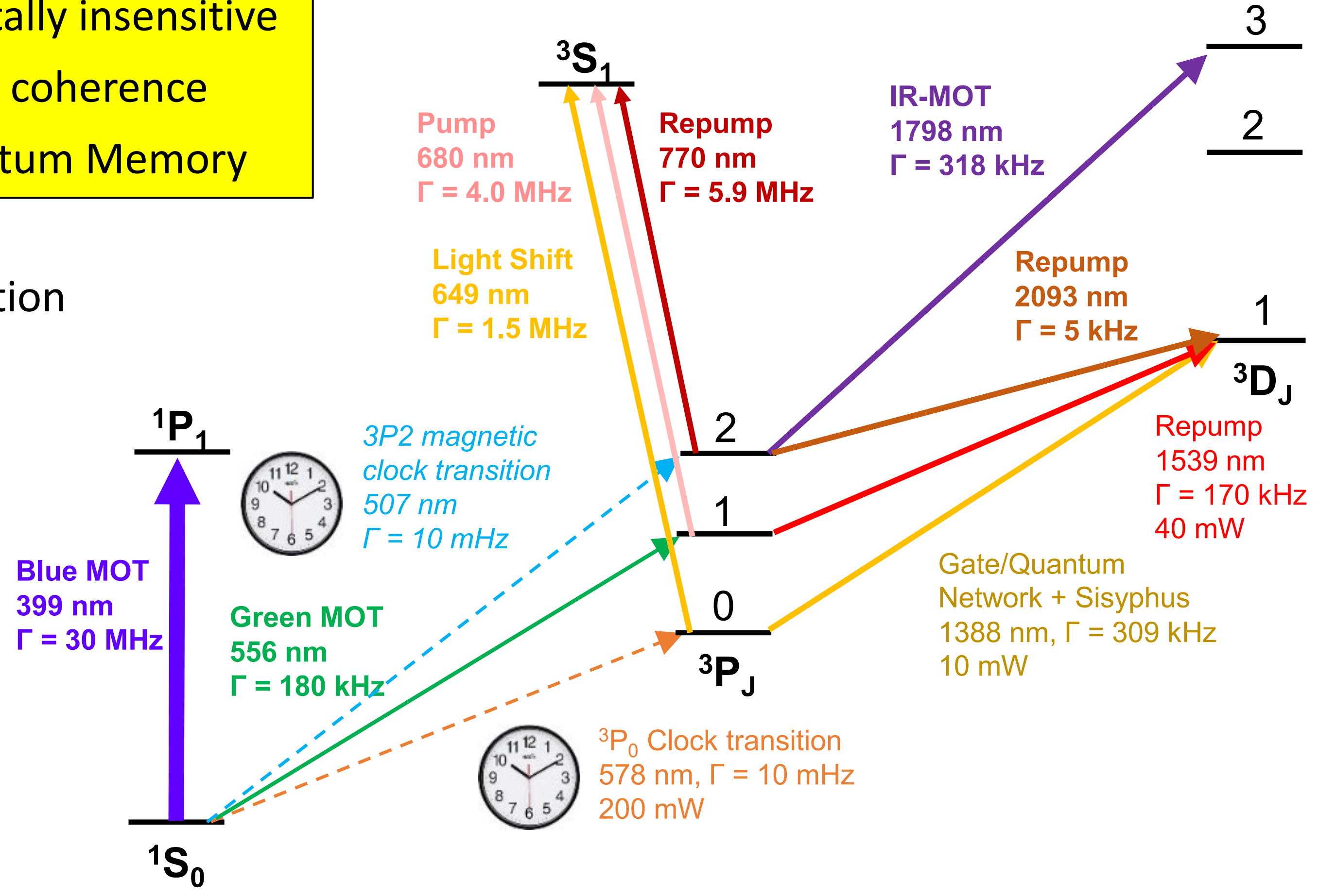
- 14% abundant
- Easier to cool and easier to store quantum information compared to Sr-87 (I=9/2 -> 10 ground states)

Yb-174

- Ideal for BEC, atom lasers
- S-wave scattering length 105 a_0
- 32% abundance (vs 0.6% for Sr-84)

7 stable isotopes (5 bosons I=0, 2 fermions)

- Proposed for searches for physics beyond the Standard Model*



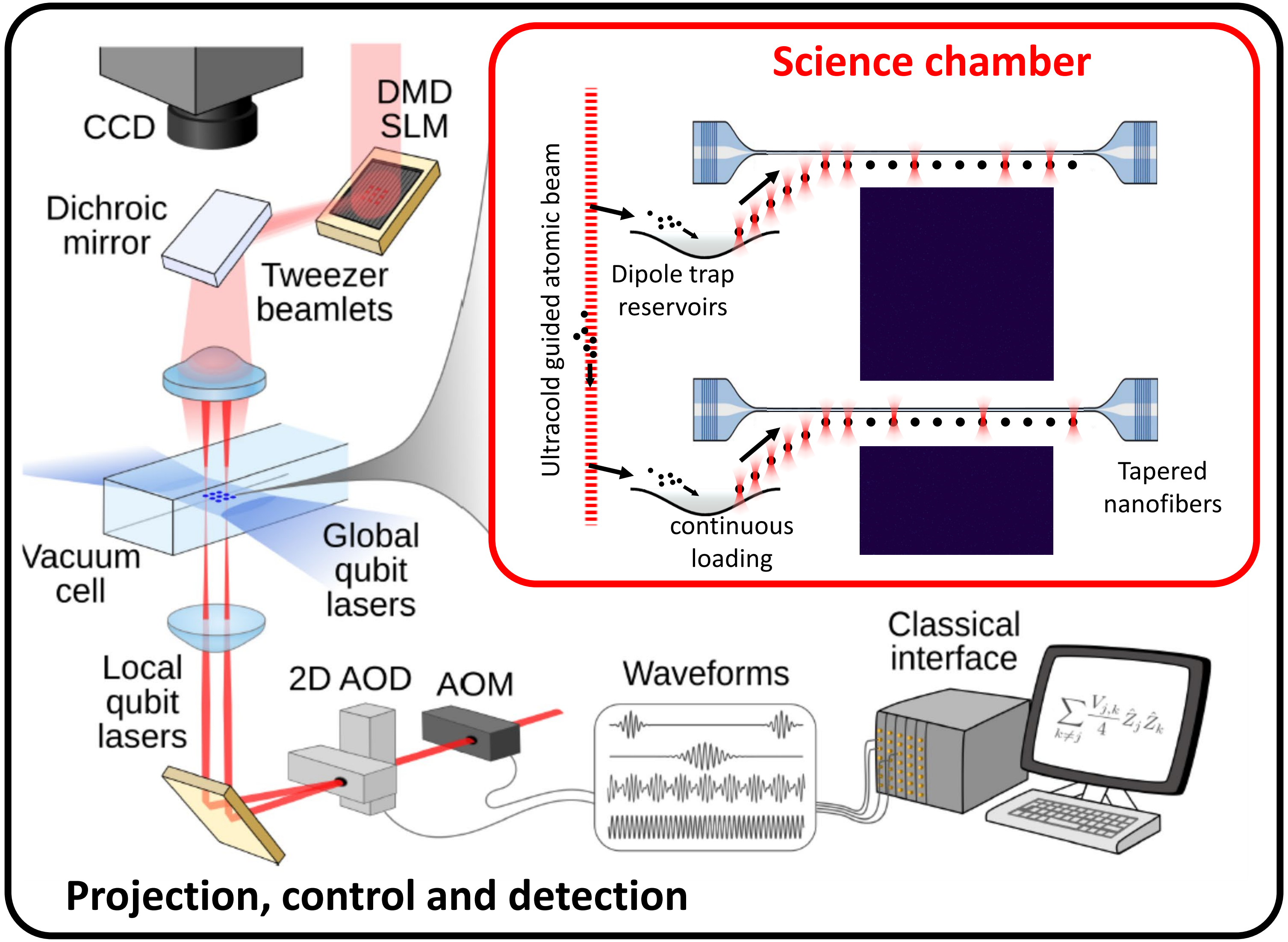
*PRL **125**, 123002 (2020).

Conceptual Design

Ytterbium Ultracold
Superradiant Hybrid Atom Nanophotonics

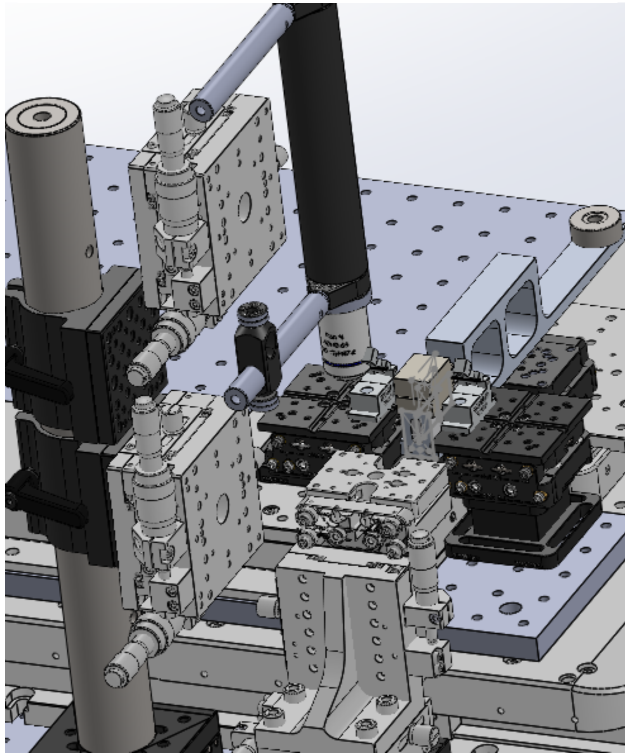
Features:

- Tweezer traps @ 460nm 1S0-3P0 magic (initially AOD, later SLM+AOD)
- Evanescent trap arrays 375nm+759nm
- 2x 0.5NA objectives (Mitutoyo)
- EMCCD iXon Ultra imaging with background free imaging



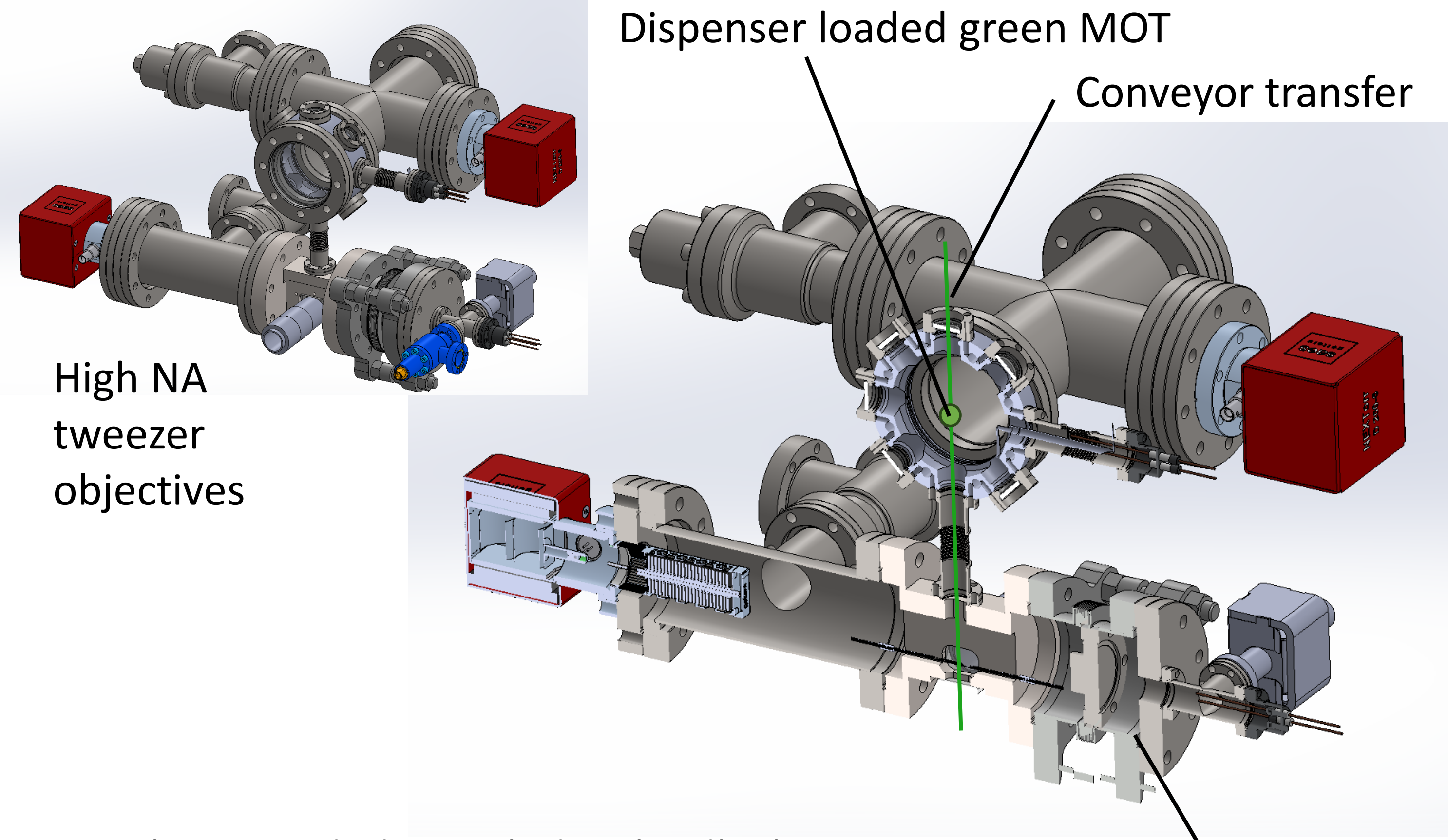
Physics Platform

Nanofiber fabrication



Electric furnace instead of hydrogen torch to eliminate OH loss at 1388nm and to attempt to reduce bend loss.

Nanofiber fabrication



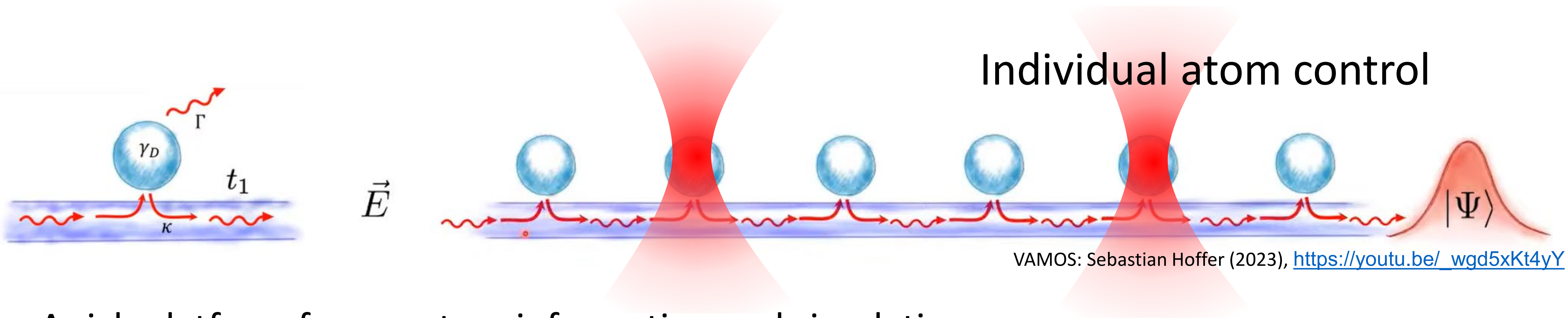
Can be upgraded to include a loadlock to swap out chips and nanofiber targets.

An platform for quantum Information processing

Strongly coupled quantum information processors

linked by single photons

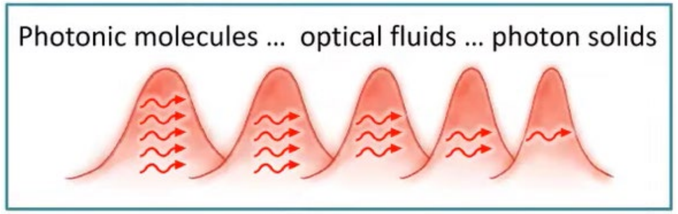
Individual atom control



VAMOS: Sebastian Hoffer (2023), https://youtu.be/_wgd5xKt4yY

A rich platform for quantum information and simulation:

- Photon-atom interactions
- Atom-photon-atom interactions
- Superradiance
- Chiral quantum optics
- …

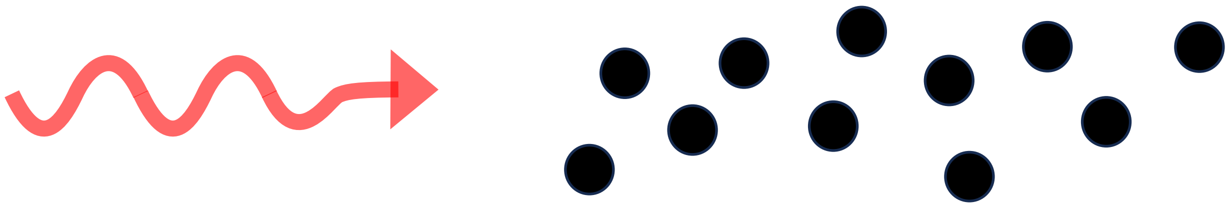


Many local theory groups to partner with

Quantum memory: Limits to storing a single photon?

Lots of great local work

Single photon carrying quantum information



$$\text{Error} \geq \frac{5.8}{OD} \qquad \propto \frac{1}{N}$$

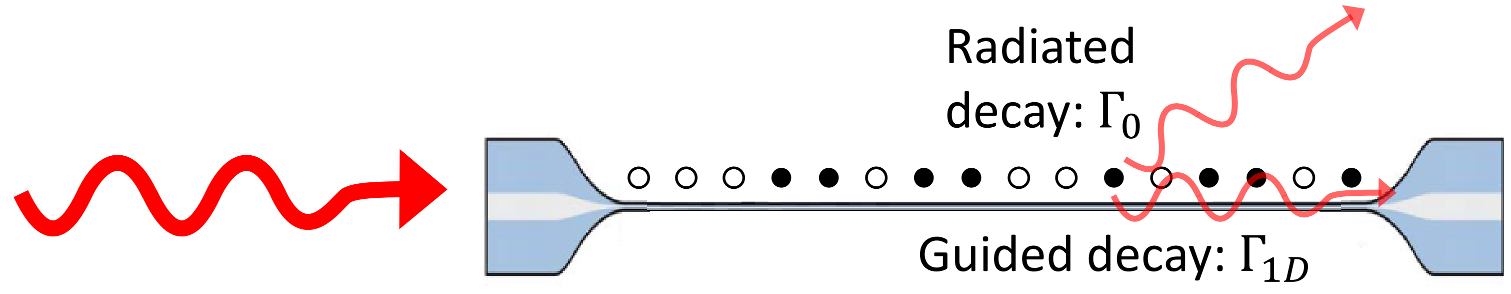
A. V. Gorshkov *et al.*, PRL **98**, 123601 (2007)

OD = Optical Density/Orders of magnitude of absorption

N = Number of atoms

Limits of storing a single photon?

Single photon carrying quantum information



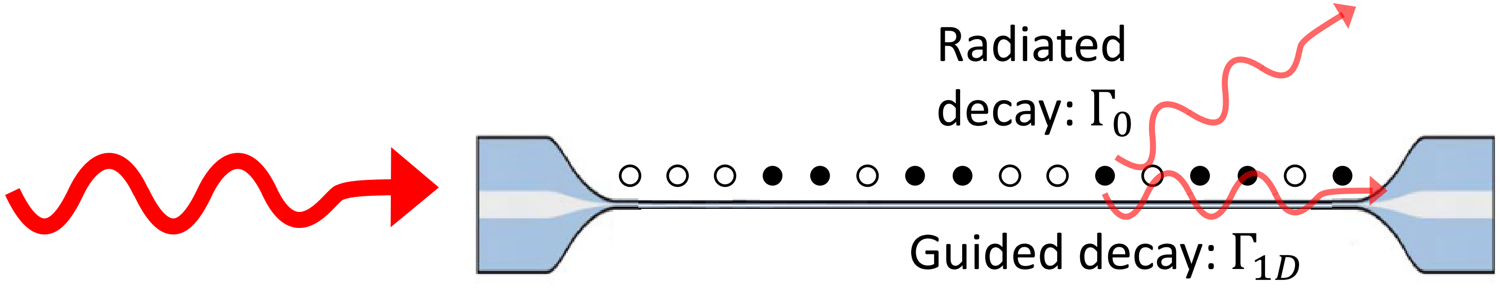
$$\text{Error} \geq \frac{5.8}{OD} = \frac{5.8}{2N}\frac{\Gamma_0}{\Gamma_{1D}} \propto \frac{1}{N}$$

$$C_1 = \frac{\Gamma_{1D}}{\Gamma_{Tot} - \Gamma_{1D}} \sim 0.05$$

$$\frac{\Gamma_{1D}}{\Gamma_0} \sim 0.05$$

Limits of storing a single photon?

Single photon carrying quantum information



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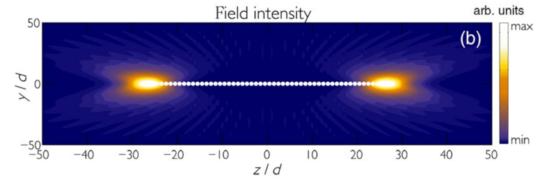
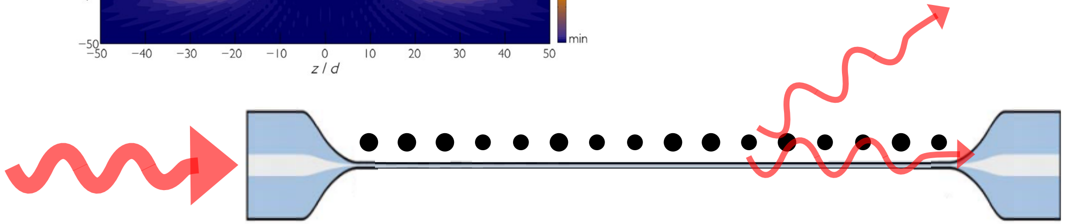
$$C_1 = \frac{\Gamma_{1D}}{\Gamma_{Tot}-\Gamma_{1D}} \sim 0.05$$

$$\frac{\Gamma_{1D}}{\Gamma_0} \sim 0.05$$

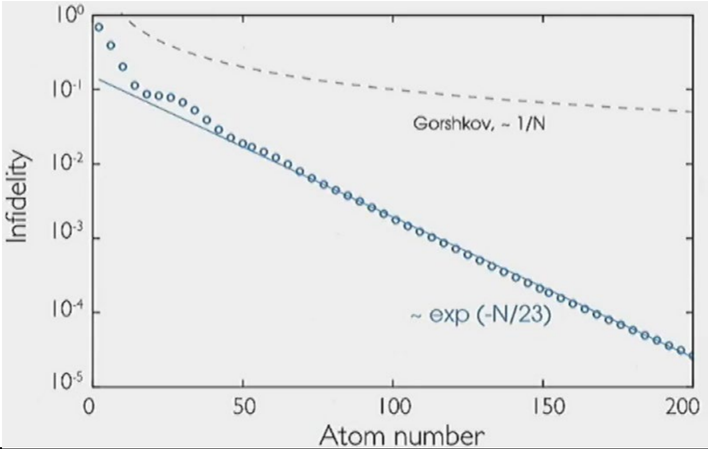
In a defect free, sub-wavelength nanofiber coupled atomic cloud:





$$\text{Error} \propto \frac{1}{e^{N/23}}$$

Exponentially **better fidelity per atom**



A. Asenjo-Garcia *et al.*, PRX 7, 031024 (2017).

Long term dream…marry photonic chips with neutral atoms

Phase 1:

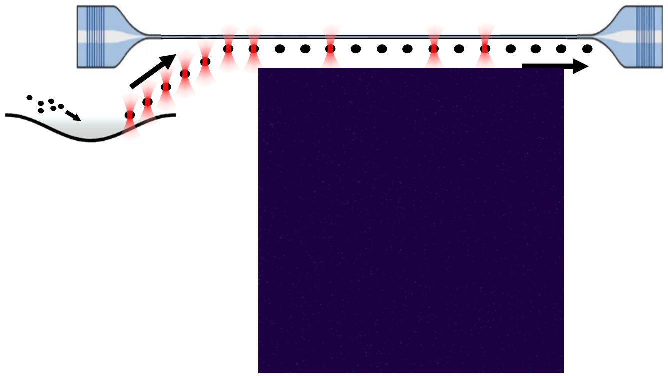
Nanofibers are a great starting place:

- High performance
- Simple behaviour
- Quick to make
- Low-cost materials/fabrication

Phase 2:

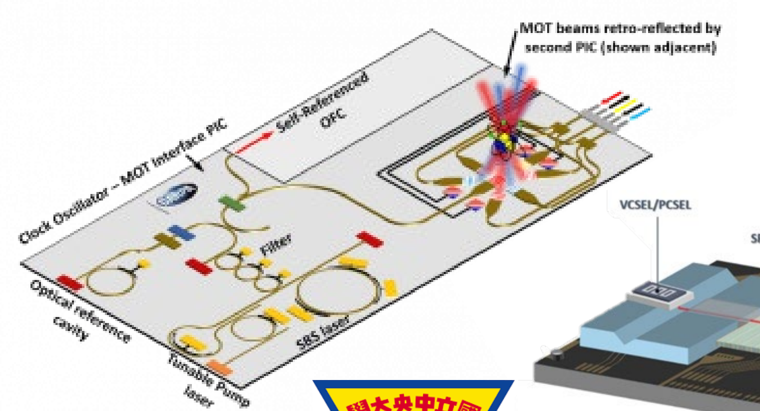
Nanofibers with FBG cavities:

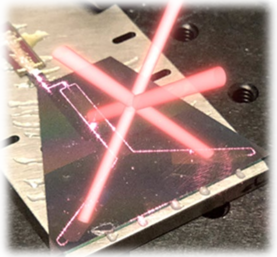
- State of the art cooperativity
- State of the art light/matter interface

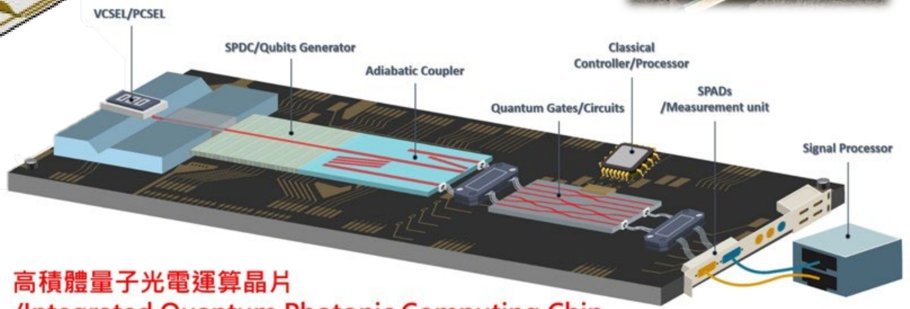


Phase 3: Hybrid atom-nanophotonic chips

could enable creating vastly more complex quantum systems and processors that can be integrated with atom quantum processors



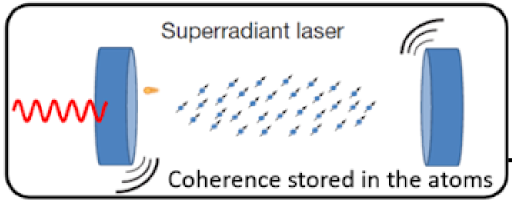




@ NCU, Taiwan

Quantum Photonic Chips Will Continue the Glory of Taiwan's Semiconductor Chips, Yen-Hung Chen, https://trh.gase.most.ntnu.edu.tw/en/article/content/264

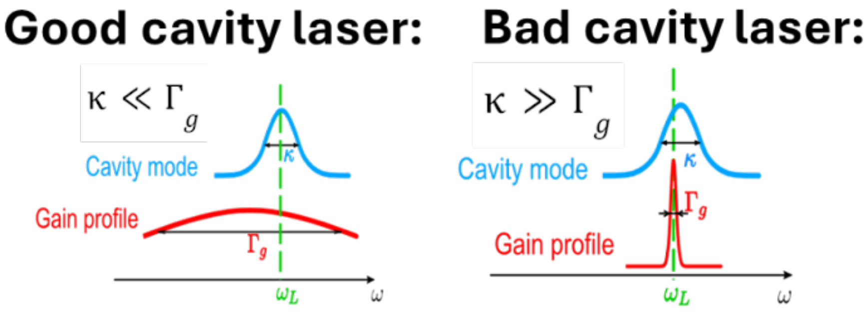
Looking to seed collaborations with photonic chip groups.

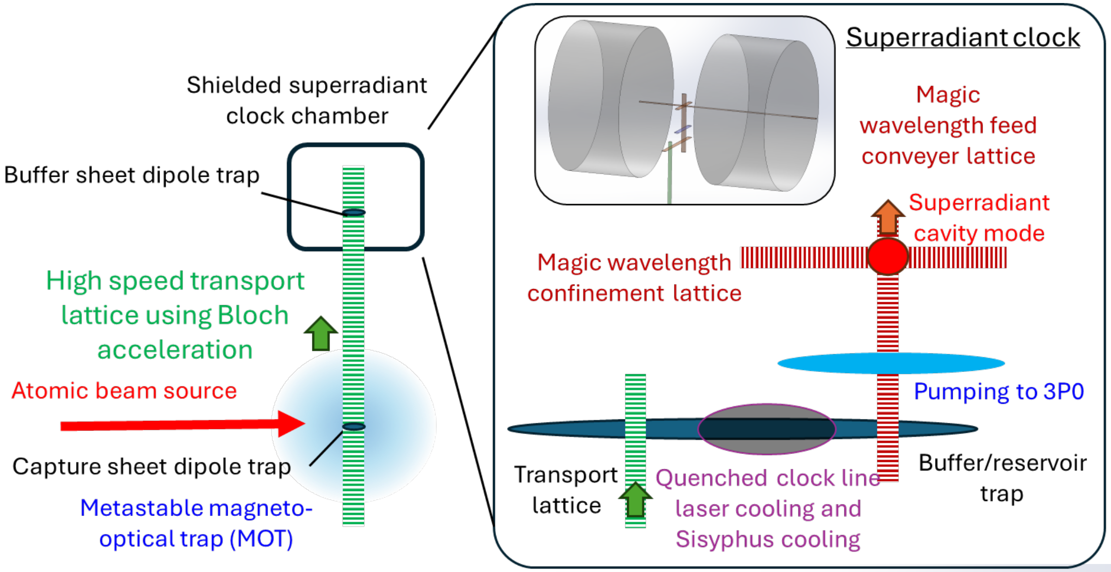


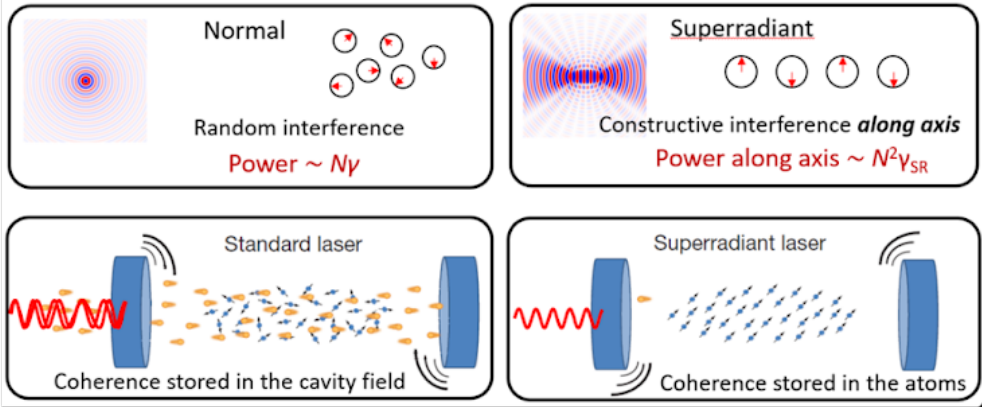
Superradiant atomic clocks

October 2025 - Current Projects

Developing a new kind of atomic clock:
A superradiant laser on a **mHz** transition.

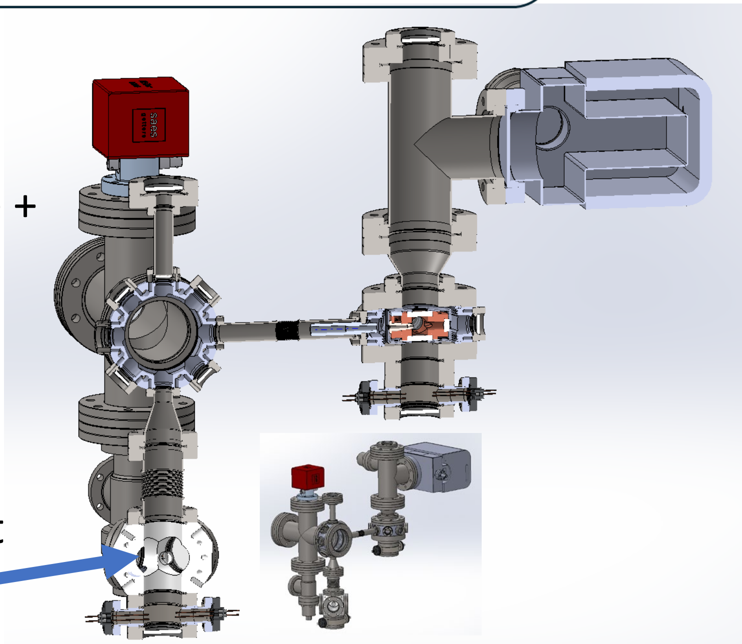






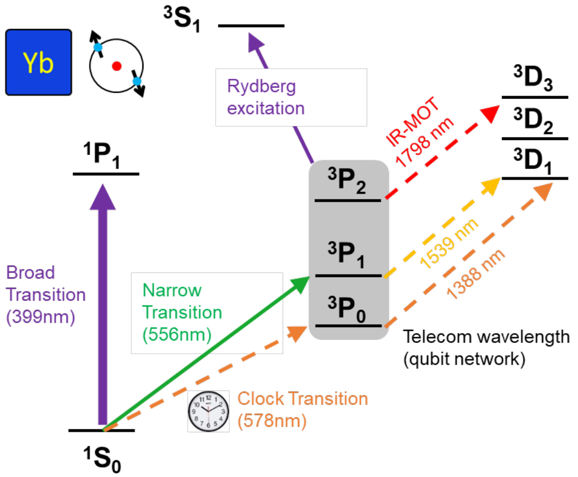
A) "Hot MOT source + mHz superradiant clock platform

B) mHz superradiant clock cavity

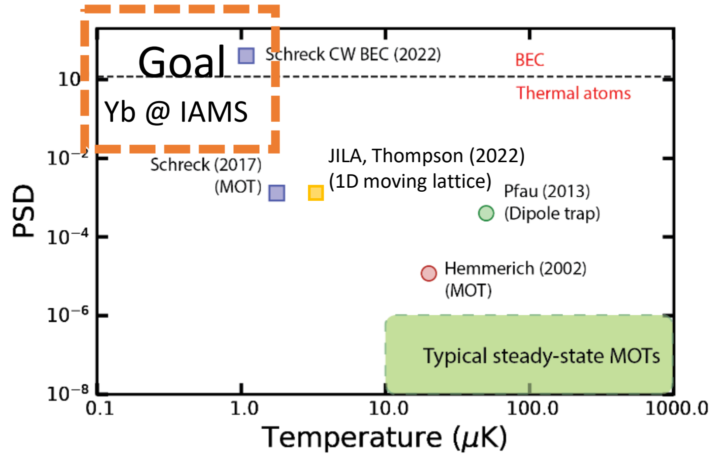


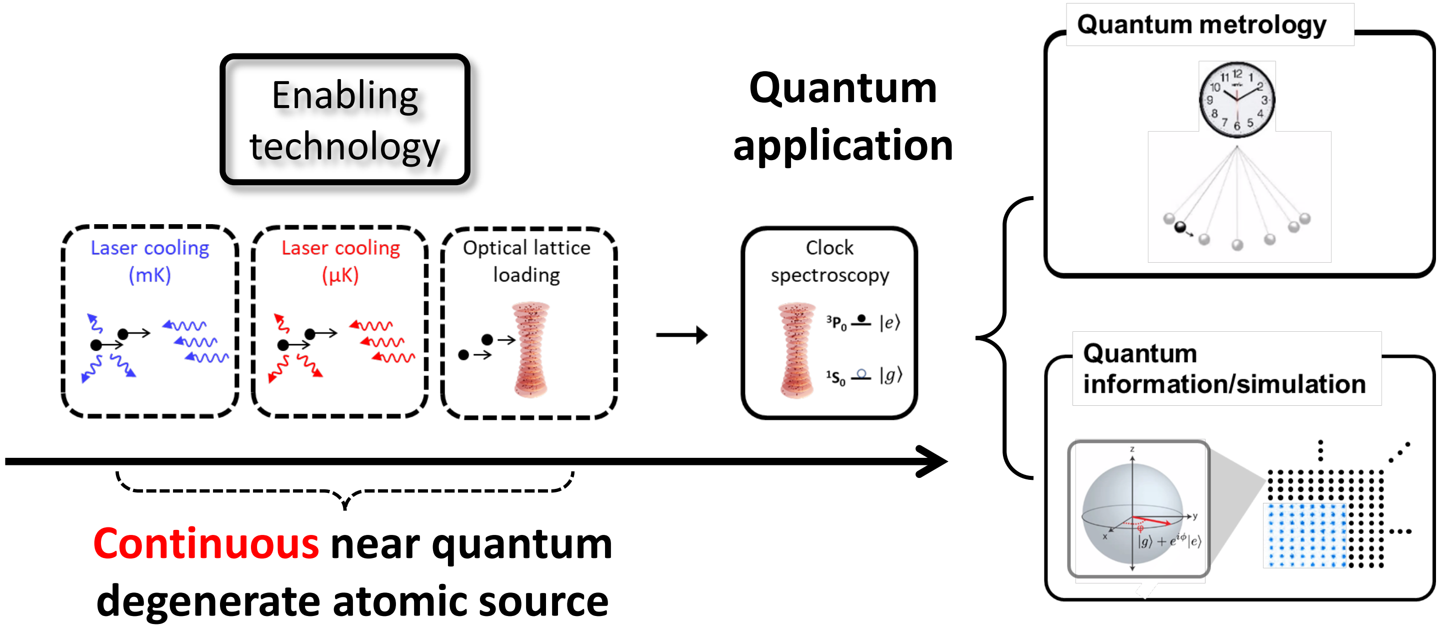
Shayne Bennetts, Lab 110, IAMS
s.p.bennetts@g.iams.sinica.edu.tw
https://iamsquantum.github.io/

Towards Continuous Yb Optical clock and large array quantum registers



Quantum Metrology Laboratory
Chun-Chia Chen (陳俊嘉)
chunchia.chen@g.iams.sinica.edu.tw

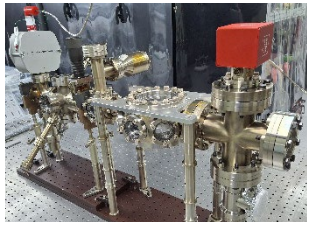


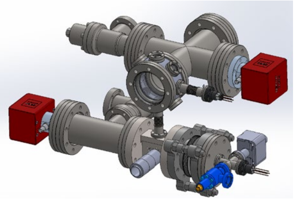


Visit ● Collaborate ● Join

IAMS Ytterbium Lab

Platforms

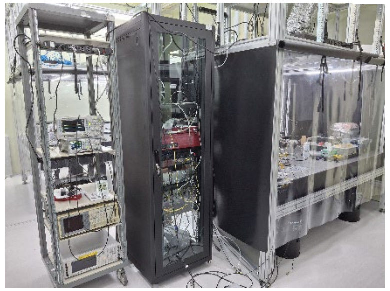

“Pathfinder”


“Nanofiber”

Join us

Projects available
Email us, visit us

Laser Systems



Lab Infrastructure

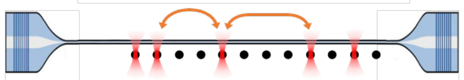



Research

Hybrid Atom-Nanophotonic Quantum Technology

Shayne Bennetts
s.p.bennetts@g.iams.edu.tw

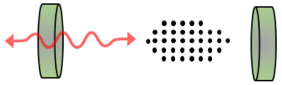
Hybrid atom-nanophotonics



- Any-to-any interaction and gates
- Individual tweezer atom control
- Arbitrary defect free arrays
- Telecom wavelength operation
- Scalability

Tapered nanofiber cavities

Cavity photon – atom interfaces

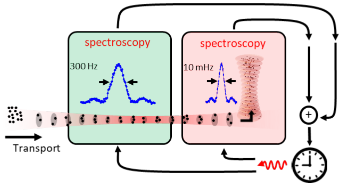


Collective atom-cavity light-matter interfaces

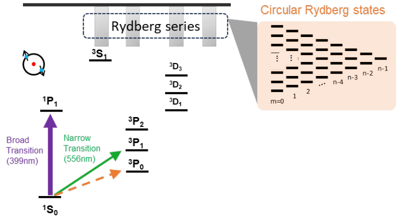
Quantum Metrology Laboratory

Chun-Chia Chen (陳俊嘉)
chunchia.chen@g.iams.sinica.edu.tw

Continuous Passive optical clocks



Ytterbium Rydberg spectroscopy



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